

SingleRAN

Automatic OMCH Establishment Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions of OMCH networking. For details, see [4.4.1 OMCH Networking](#).

Revised other descriptions in this document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue does not introduce any changes to SRAN21.1 02 (2025-04-27).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RA T	Feature ID	Feature Name	Chapter/Section
UM TS	WRFD-031100	BOOTP	5 ATM-based Automatic OMCH Establishment for Base Stations (UMTS)
UM TS	WRFD-031101	NodeB Self-discovery Based on IP Mode	4 IP-based Automatic OMCH Establishment for Base Stations
LTE FD D	LBFD-002035	Self-configuration	4 IP-based Automatic OMCH Establishment for Base Stations
LTE TD D	TDLBFD-002036	Self-configuration	4 IP-based Automatic OMCH Establishment for Base Stations
NB- IoT	MLBFD-12000241	Self-configuration	4 IP-based Automatic OMCH Establishment for Base Stations

3 Overview

3.1 Introduction

Operation and maintenance channels (OMCHs) are established between base stations and the operation and maintenance center (OMC, either the MAE or BSC). OMCHs are used to transmit operation and maintenance information about base stations and are classified as follows:

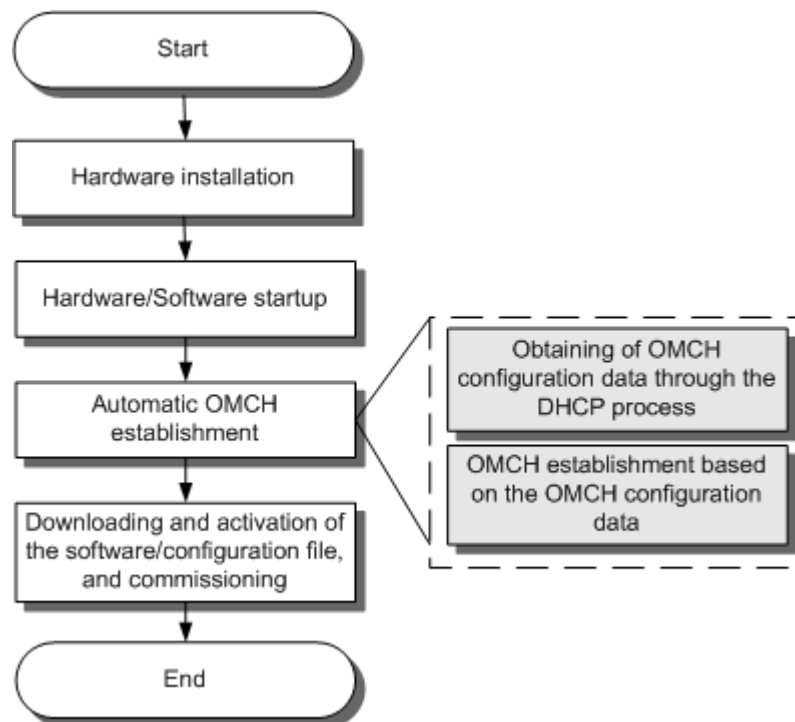
- OMCHs between the eGBTS, NodeB, eNodeB, gNodeB, co-MPT multimode base station and the MAE
- OMCH between the NodeB and the MAE on an ATM-based network
- OMCH between the GBTS and the BSC

NOTE

One end of an OMCH is located at the main control board of a base station. Depending on the configuration of the main control board, multimode base stations are classified into co-MPT multimode and separate-MPT multimode base stations. For co-MPT multimode base stations, all RATs share one main control board and one OMCH. For separate-MPT multimode base stations, each RAT has individual main control board and OMCH.

The Automatic OMCH Establishment feature enables a powered-on base station, which is configured with hardware but no transmission information, to obtain OMCH configuration information. This information is collected through the transport network and is used to automatically establish an OMCH to the MAE or BSC. This feature applies to base station deployment by PnP. [Figure 3-1](#) shows the automatic OMCH establishment phase during deployment by PnP.

Figure 3-1 Automatic OMCH establishment phase during base station deployment by PnP



A base station must obtain the following transmission configuration data to automatically establish an OMCH:

- Basic transmission configuration information, including the OM IP address, OM virtual local area network (VLAN) ID, interface IP address, interface IP address mask, IP address of the next-hop gateway, IP address of the MAE/BSC, and IP address mask of the MAE/BSC.
- Security-related information, including the Certificate Authority (CA) name, transmission protocol (HTTP or HTTPS) used by the CA, CA IP address, CA port number, CA path, IP address of the security gateway (SeGW), and name of the SeGW. The operator's CA information is required only when the base station uses digital certificates issued by the operator's CA to perform identity authentication with other devices.

For details about how the base station obtains the preceding information, see [4.2 Base Station Obtaining Transmission Configuration Information](#).

After the OMCH is established, the base station can automatically download software and configuration file/configuration data from the MAE/BSC, and activate the software and configuration file/configuration data. After being commissioned, the base station enters the working state. For details, see *3900 & 5900 Series Base Station Commissioning Guide*.

With the Automatic OMCH Establishment feature, a base station can establish OMCHs by network communication (not requiring local end operations). This enables remote base station deployment by PnP, thereby reducing site visits and deployment cost and time.

In IPsec networking, if two UMPT boards work in active/standby mode or one UMPT board functions as the transmission extension board, secure base station deployment by PnP is not supported.

3.2 Application Networking Scenarios

GBTs support automatic OMCH establishment in TDM- or IPv4-based networking. NodeBs support automatic OMCH establishment in ATM- or IPv4-based networking. eGBTs support automatic OMCH establishment in IPv4-based networking. eNodeBs and gNodeBs support automatic OMCH establishment in IPv4/IPv6-based networking. [Table 3-1](#) describes the application networking scenarios for the Automatic OMCH Establishment feature.

Table 3-1 Networking scenarios for the Automatic OMCH Establishment feature

Networking Scenario		Description
Non-IPsec in IP networking		No IP packet is secured by IPsec, including Dynamic Host Configuration Protocol (DHCP) packets, OM data, service data, signaling messages, and clock packets.
IPsec in IPv4 networking	Scenario 1	DHCP packets, OM data, and all or part of other data are secured by IPsec.
	Scenario 2	DHCP packets are not secured by IPsec, but OM data and all or part of other data are secured by IPsec.
	Scenario 3	OM data is not secured by IPsec, but service data, signaling messages, and all or part of other data are secured by IPsec.
ATM		The OMCH (UMTS) between the NodeB and MAE is carried over ATM.
TDM		The OMCH (GSM) between the GBTS and BSC uses TDM transmission. The OMCH is carried over E1 or T1 links.

 NOTE

In this document, the IPsec or non-IPsec networking indicates that the IP layer communication between the base station and other devices is secured or not secured by IPsec.

4 IP-based Automatic OMCH Establishment for Base Stations

4.1 OMCH Protocol Stacks

- The following base stations support IPv4 transmission: eGBTS, NodeB, eNodeB, gNodeB, co-MPT multimode base station, and GBTS.
- The following base stations support IPv6 transmission: eNodeB, gNodeB, and LTE/NR co-MPT multimode base station.
- DHCPv6 deployment is supported only when the main control board version is SRAN15.1 or later.

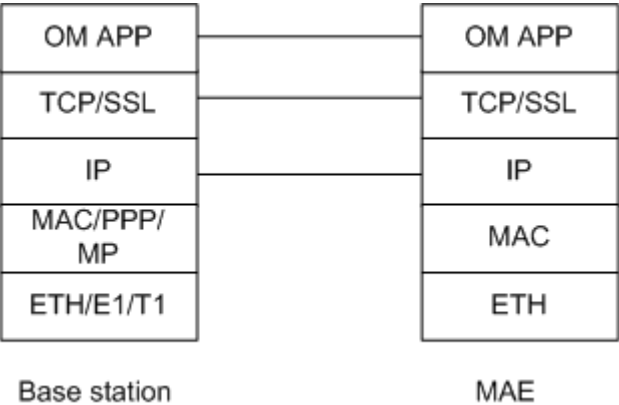
The OMCH between the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station and the MAE is carried over the Transmission Control Protocol (TCP). The OMCH between the GBTS and the BSC is carried over the User Datagram Protocol (UDP).

4.1.1 IPv4 Transmission

4.1.1.1 Non-IPsec Networking Scenario

Figure 4-1 shows the protocol stack for an OMCH between the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station and the MAE.

Figure 4-1 Protocol stack for an OMCH between the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station and the MAE



As shown in [Figure 4-1](#), an OMCH between the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station and the MAE is carried over TCP and SSL.

The eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station listens to the TCP connection establishment request with a specific TCP port number from the MAE, and establishes the TCP connection to the MAE as requested. After the TCP connection is established, the MAE initiates an OMCH establishment request to the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station.

The MAE can optionally use SSL to perform encryption and authentication for OMCHs and enable the establishment of SSL-based OMCHs. SSL uses the PKI, with which the communication between the base station and the MAE is protected against eavesdropping and confidentiality and reliability are guaranteed. For details about SSL, see *SSL Feature Parameter Description* for SingleRAN.

NOTE

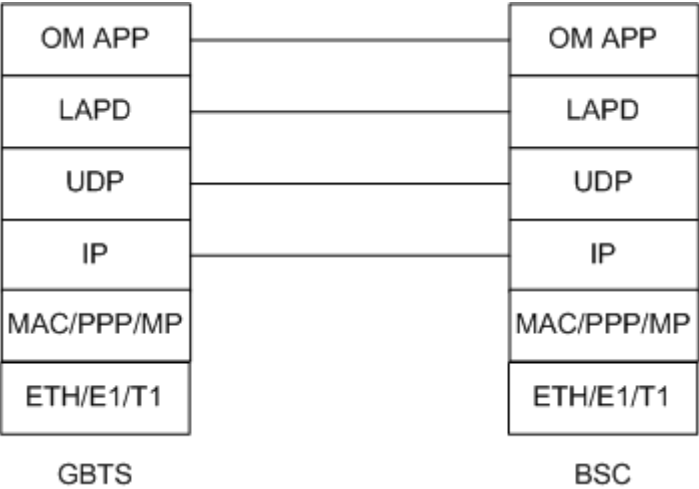
If the base stations use SSL to protect the OMCH and use initial certificates for authentication with the MAE, you are advised to contact Huawei engineers when the OMCH SSL link setup fails in either of the following scenarios:

- The boards where the base station certificates should be deployed have no certificate. For example, some UMPTb main control boards may not have deployed certificates.
- The certificate is faulty on a board where a base station certificate is deployed. For example, the certificate is damaged, revoked, or cannot be loaded.

Note: The preceding fault scenarios may easily cause OMCH SSL link setup failures in reused PnP-based site deployment.

[Figure 4-2](#) shows the protocol stack for an OMCH between the GBTS and the BSC.

Figure 4-2 Protocol stack for an OMCH (GSM) between the GBTS and the BSC



As shown in [Figure 4-2](#), an OMCH between the GBTS and the BSC is carried over UDP. The GBTS listens to the UDP connection establishment request with a specific UDP port number from the BSC, and establishes the UDP connection to the BSC as requested. After the UDP connection is established, the BSC initiates an OMCH establishment request to the GBTS.

NOTE

For the deployment policy of the DHCP server, see [4.2.4.2.3 DHCPv4 Client and DHCPv4 Server](#) and [4.2.4.3.3 DHCPv6 Client and DHCPv6 Server](#).

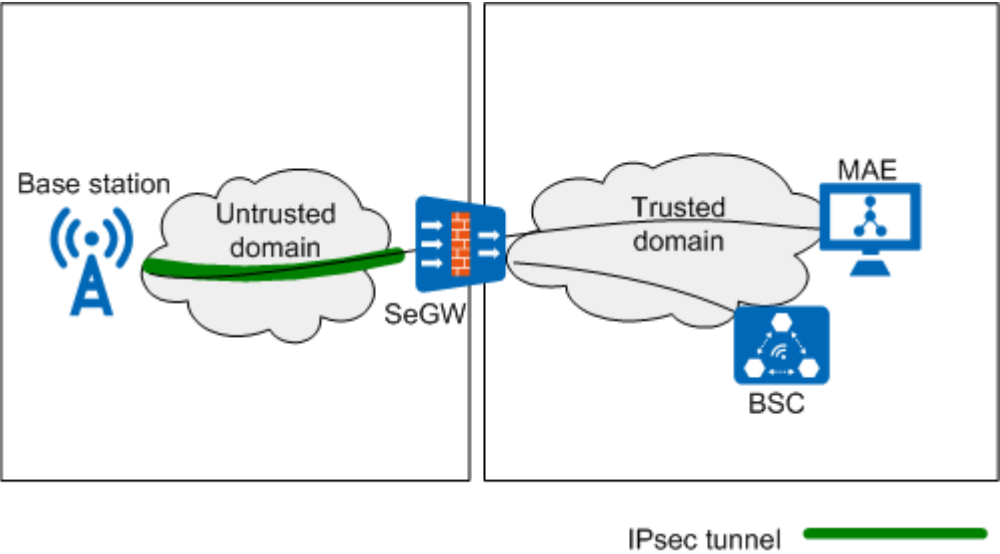
During the OMCH establishment, the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station listens to a specific TCP port number, and the GBTS listens to a UDP port number. For details, see *3900 & 5900 Series Base Station Communication Matrix*. The packets with these port numbers must be allowed to pass through the firewall between the base station and the DHCP server, MAE, or BSC.

After establishing an OMCH to the MAE, the base station uses File Transmission Protocol (FTP) to download the software and configuration file from the FTP server. FTP runs over TCP/IP, and the transport layer can be optionally secured using SSL. For details about FTP, see RFC 959. After establishing an OMCH to the BSC, the GBTS uses the proprietary protocol that runs over UDP to download the software and configuration file from the BSC. The FTP protocol has security risks. You are advised to use SSL at the transport layer.

4.1.1.2 IPsec Networking Scenario

In IPsec networking scenarios, OMCH data can be secured or not secured by IPsec. [Figure 4-3](#) shows the networking scenario in which IPsec secures OMCH data.

Figure 4-3 Networking scenario in which IPsec secures OMCH data



As shown in [Figure 4-3](#), the network is divided into the trusted and untrusted domains, which are separated by the SeGW. Devices in the untrusted domain cannot access the devices in the trusted domain. After a base station starts, an IPsec tunnel is established to the SeGW. Packets from the base station are sent over the IPsec tunnel to the untrusted domain and then forwarded by the SeGW to the MAE or BSC in the trusted domain.

[Figure 4-4](#) shows the protocol stack for an OMCH between the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station and the MAE in IPsec networking scenarios, and [Figure 4-5](#) shows the protocol stack for an OMCH between the GBTS and the BSC.

Figure 4-4 Protocol stack for an OMCH between the eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station and the MAE (IPsec networking)

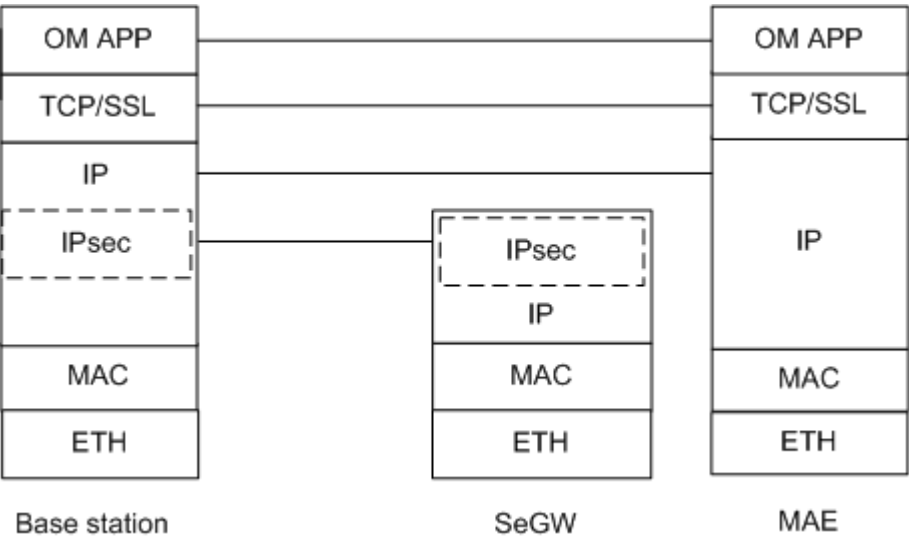
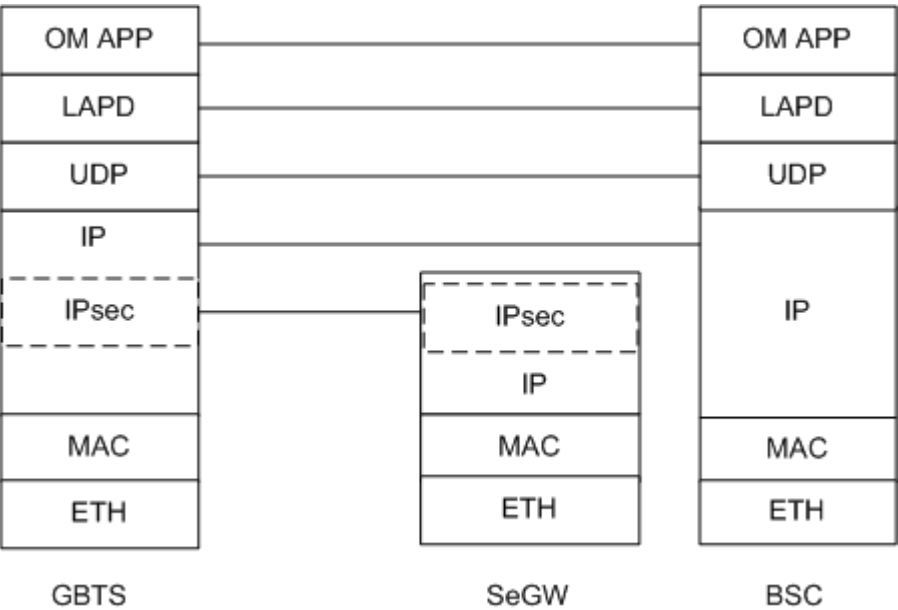


Figure 4-5 Protocol stack for an OMCH between the GBTS and the BSC (IPsec networking)



NOTE

- The protocol stacks shown in [Figure 4-4](#) and [Figure 4-5](#) are supported only when IPsec is used. Whether a base station supports IPsec depends on the base station model and the software and hardware of the main control board.

In IPsec networking scenarios, IPsec secures base station data. IPsec is a security architecture defined by the Internet Engineering Task Force (IETF) and applicable to the IP layer. IPsec secures data communication by identity authentication, data encryption, data integrity, and address encryption. During automatic OMCH establishment, the base station establishes an IPsec tunnel to the SeGW and then an OMCH secured by the IPsec tunnel.

The base station uses two types of IP addresses:

- IP addresses that can be used to access an untrusted domain
Interface IP addresses for the base station to communicate with the SeGW in an untrusted domain
- IP addresses that can be used to access a trusted domain
IP addresses for the base station to communicate with the peer end such as the MAE, BSC, or MAE DHCP server in the trusted domain

During base station deployment, NEs in the trusted and untrusted domains may communicate with one another. For example, a base station uses an interface IP address in the untrusted domain to communicate with the DHCP server in the trusted domain. Alternatively, the DHCP relay in the untrusted domain uses an IP address in the untrusted domain to communicate with the DHCP server in the trusted domain. For details, see [4.3.3 Automatic OMCH Establishment in IPsec Networking Scenario 1 \(Using MAE DHCP Server Data\)](#) and [4.3.4 Automatic OMCH Establishment in IPsec Networking Scenario 2](#).

The base station uses the interface IP address to access the untrusted domain. Unless otherwise specified, the base station uses the logical IP address to access the trusted domain.

When using IPsec to secure data and digital certificates to perform identity authentication, an operator must deploy the PKI. During automatic OMCH establishment, the base station interworks with the operator's PKI using the Certificate Management Protocol (CMP) and obtains the operator-issued device certificate and CA root certificate. The base station then establishes an IPsec tunnel to the SeGW as well as the OMCH to which the new IPsec tunnel provides security. For details about IPsec tunnels, see *IPsec Feature Parameter Description* for SingleRAN. For details about digital certificate management, see *PKI Feature Parameter Description* for SingleRAN.

When the operator uses IPsec to secure data and the pre-shared key (PSK) for identity authentication, the base station fails to automatically establish an OMCH. In this case, it is required to use other alternative methods to deploy the base station.

The MAE can optionally use SSL to perform encryption and authentication for OMCHs and enable the establishment of SSL-based OMCHs. SSL uses the PKI, with which the communication between the base station and the MAE is protected against eavesdropping and confidentiality and reliability are guaranteed. For details about SSL, see *SSL Feature Parameter Description* for SingleRAN.

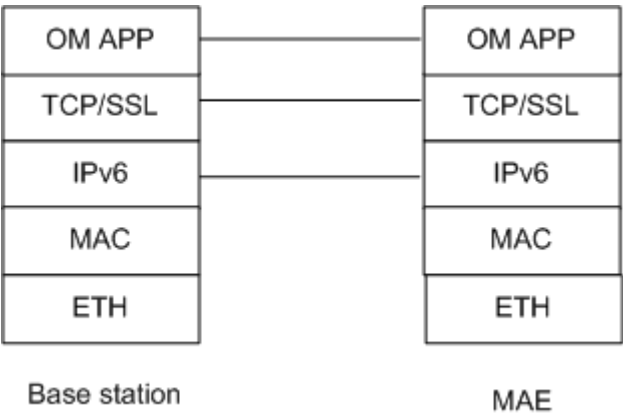
4.1.2 IPv6 Transmission

Currently, IPv6 transmission supports only automatic OMCH establishment in non-IPsec networking scenarios.

4.1.2.1 Non-IPsec Networking Scenario

Figure 4-6 shows the IPv6 protocol stack for an OMCH between the eNodeB, gNodeB, or co-MPT multimode base station and the MAE.

Figure 4-6 IPv6 protocol stack for an OMCH between the eNodeB, gNodeB, or co-MPT multimode base station and the MAE



The IPv6 protocol stack is the same as the IPv4 protocol stack. The OMCH between the eNodeB, gNodeB, or co-MPT multimode base station and the MAE is

carried over TCP and SSL. The mechanism for automatic OMCH establishment in IPv6 networking is the same as that in IPv4 networking.

The eNodeB, gNodeB, and co-MPT multimode base station support only Ethernet transmission in IPv6 networking.

4.2 Base Station Obtaining Transmission Configuration Information

4.2.1 Transmission Mode of the OMCH

A base station has two types of transmission ports: E1/T1 ports and Ethernet ports. E1/T1 ports support TDM, ATM, and IP over E1/T1 transmission modes, and Ethernet ports support IP transmission mode. No transmission mode is configured on the base station before the OMCH is established. The base station tries different transmission modes over the transmission ports until the OMCH is successfully established.

An eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station tries transmission modes in the following sequence:

1. IP over FE/GE
2. ATM
3. IP over E1/T1

A GBTS tries transmission modes in the following sequence:

1. TDM
2. IP over E1/T1
3. IP over FE/GE

4.2.2 Physical Layer Detection

A base station negotiates the duplex mode and data rate for an Ethernet port on the physical layer with a peer transmission device. The peer transmission device can work in auto-negotiation or full duplex mode.

If an E1/T1 port is available on the physical layer, an eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station attempts to set the working mode of a detection port to E1/T1 mode, and users can set the working mode of a detection port to E1/T1 mode for a GBTS by using the related DIP switch.

4.2.3 Data Link Layer Detection

IP over FE/GE Transmission

A base station obtains the VLANs used by the data link layer through the VLAN acquisition process. For details, see [4.2.7 Obtaining VLAN Information for DHCP Packets](#).

IP over E1/T1 Transmission

Physical layer detection shows that a base station works in E1 or T1 mode. The base station supports PPP/MLPPP (also called MP) detection on E1/T1 timeslot combinations. [Table 4-1](#) and [Table 4-2](#) describe the E1 and T1 timeslot combinations, respectively. PPP is short for Point-to-Point Protocol and MLPPP for Multi-Link Point-to-Point Protocol.

Table 4-1 E1 timeslot combinations

Serial Number	31 Timeslot<----->0 Timeslot	Hexadecimal Digit
1	1111111111111111111111111111110	0xFFFFFFFF
2	0000000000000000111111111111110	0x0000FFFF
3	0000000000000000111111111111110	0x0001FFFF
4	0000000000000000111111111111110	0x0007FFFF
5	0000000000000000111111111111110	0x0003FFFF
6	0000000000001111111111111111110	0x001FFFFF
7	000000000000000000000000111111110	0x00000FFF
8	0000000001111111111111111111110	0x007FFFFF
9	000000000000000000000000111111110	0x000003FE
10	0000000111111111111111111111110	0x01FFFFFE
11	0000011111111111111111111111110	0x07FFFFFE
12	0001111111111111111111111111110	0x1FFFFFE
13	0111111111111111111111111111110	0x7FFFFFE
14	000000000000000000000000111111110	0x000000FE
15	000000000000000000000000111111110	0x0000003E
16	000000000000000111111111111111110	0x0003FFFF
17	000000000000000011111111111111110	0x00007FFF
18	000000000000111111111111111111110	0x000FFFFF
19	00000000000000000000000111111111110	0x00001FFF
20	000000000011111111111111111111110	0x003FFFFF
21	00000000000000000000000111111111110	0x000007FE
22	000000001111111111111111111111110	0x00FFFFFF
23	000000111111111111111111111111110	0x03FFFFFF
24	000011111111111111111111111111110	0x0FFFFFFF

Serial Number	31 Timeslot<----->0 Timeslot	Hexadecimal Digit
25	00111111111111111111111111111110	0x3FFFFFFE
26	00000000000000000000000011111110	0x000001FE
27	00000000000000000000000001111110	0x0000007E

Table 4-2 T1 timeslot combinations

Serial Number	23 Timeslot<----->0 Timeslot	Hexadecimal Digit
1	111111111111111111111111	0x00FFFFFF
2	000000001111111111111111	0x00007FFF
3	000000011111111111111111	0x0001FFFF
4	000000000001111111111111	0x00001FFF
5	000001111111111111111111	0x0007FFFF
6	000000000000011111111111	0x000007FF
7	000111111111111111111111	0x001FFFFF
8	000000000000000111111111	0x000001FF
9	011111111111111111111111	0x007FFFFF
10	000000000000000001111111	0x0000007F
11	000000000000000000011111	0x0000001F
12	000000001111111111111111	0x0000FFFF
13	000000000011111111111111	0x00003FFF
14	000000111111111111111111	0x0003FFFF
15	000000000000111111111111	0x00000FFF
16	000011111111111111111111	0x000FFFFF
17	000000000000001111111111	0x000003FF
18	001111111111111111111111	0x003FFFFF
19	000000000000000011111111	0x000000FF
20	000000000000000001111111	0x0000003F

 NOTE

In [Table 4-1](#) and [Table 4-2](#), 1 indicates that the timeslot is occupied and 0 indicates that the timeslot is not occupied. Timeslot combinations that are not listed in the tables cannot be used for PnP deployment.

If a base station works in IP over E1/T1 mode, the peer transmission device must be configured as follows:

- PPP/MP detection is configured as non-authentication.
- The peer IP address is configured for PPP/MLPPP detection.

If the peer transmission device is not functioning as a DHCP server, the DHCP relay agent function must be enabled on the interface for PPP/MLPPP detection on the peer transmission device.

4.2.4 DHCP Overview

4.2.4.1 Introduction

Before an OMCH is established, a base station is not configured with any data and cannot perform end-to-end communication with other devices at the IP layer. The base station implements this communication by obtaining the following information:

- OMCH configuration data such as the OM IP address, VLAN ID, interface IP address, and IP address of the next-hop gateway.
- During base station deployment by PnP, if the base station needs to use digital certificates issued by the operator's CA to perform identity authentication with other devices, it also needs to obtain the operator's CA information.
- In IPsec networking scenarios, the base station must obtain SeGW information.

The base station uses DHCP to obtain the preceding OMCH information, operator's CA information, and SeGW information. DHCP works in client or server mode. The DHCP procedure involves the following logical NEs:

- DHCP client: a host that uses DHCP to obtain configuration parameters
- DHCP server: a host that allocates and distributes configuration parameters to a DHCP client
- DHCP relay agent: an NE that transmits DHCP packets between a DHCP server and a DHCP client. A DHCP relay agent must be deployed between a DHCP server and a DHCP client that are in different broadcast domains.

After a DHCP client accesses the network, it actively exchanges DHCP packets with its DHCP server to obtain configuration parameters. Before the OMCH is automatically set up, no data is configured on the base station. Therefore, it is uncertain whether the OMCH uses IPv4 or IPv6 transmission. When functioning as a DHCP client, the base station initiates the DHCPv4 or DHCPv6 process to attempt to establish an OMCH. If the OMCH of the base station uses IPv4 transmission, the data required for automatic OMCH establishment is obtained through the DHCPv4 process. If the OMCH of the base station uses IPv6 transmission, the data required for automatic OMCH establishment is obtained

through the DHCPv6 process. DHCPv4 and DHCPv6 are different protocols. They both use UDP packets but use different UDP port numbers. During the interaction, the DHCP entity listens to different UDP port numbers when different protocol stacks are used, as described in [Table 4-3](#).

Table 4-3 Listening port numbers for different protocol stacks of the DHCP entity

DHCP Entity	Protocol Stack	Listening Destination Port
DHCP client	DHCPv4	UDP port 68
	DHCPv6	UDP port 546
DHCP server/DHCP relay	DHCPv4	UDP port 67
	DHCPv6	UDP port 547

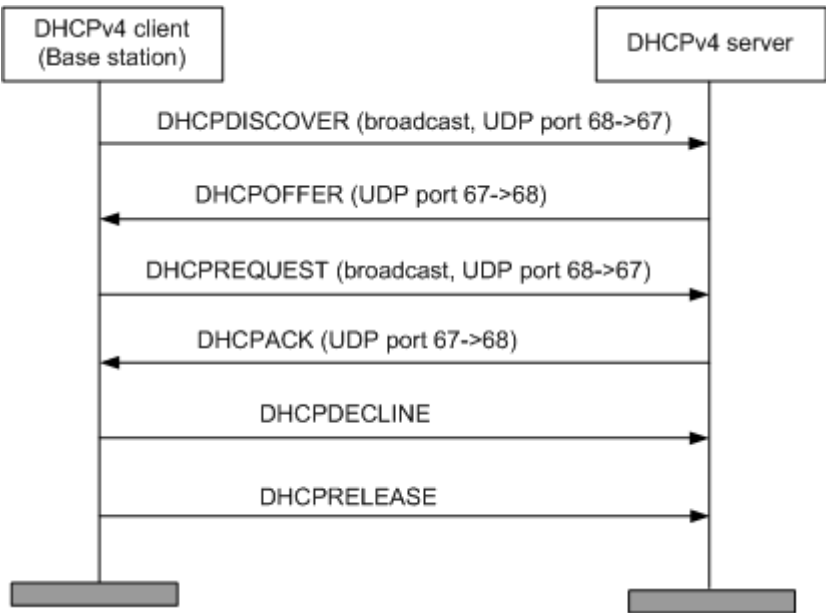
4.2.4.2 DHCPv4

4.2.4.2.1 DHCPv4 Working Principles

Not Involving the DHCP Relay Agent

When a DHCPv4 client and a DHCPv4 server are in the same broadcast domain, broadcast packets can be received by each other. [Figure 4-7](#) shows the interworking between the DHCPv4 client and DHCPv4 server that are in the same broadcast domain.

Figure 4-7 DHCPv4 interworking (without a DHCP relay agent)



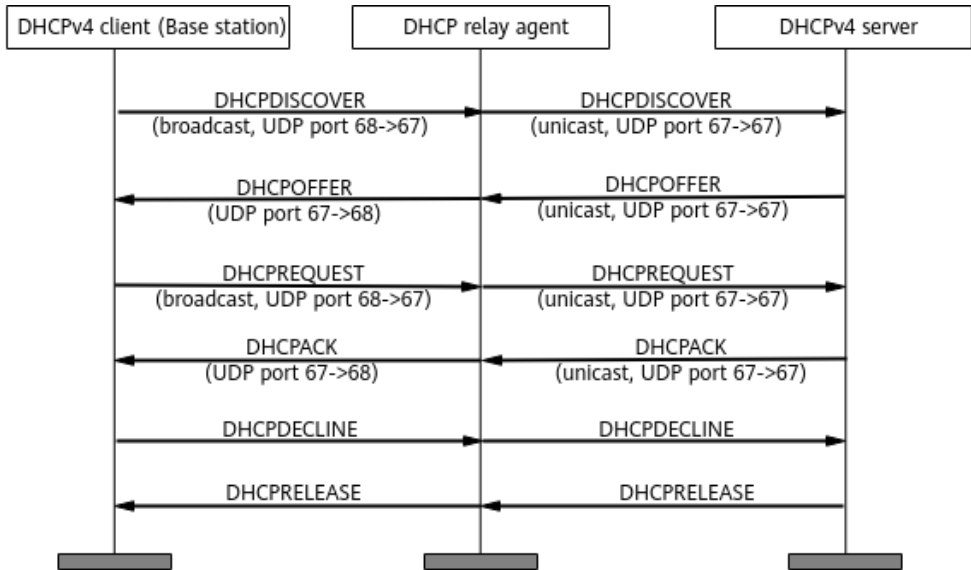
1. After the DHCPv4 client starts, a DHCPDISCOVER packet is broadcast to search for an available DHCPv4 server. The DHCPDISCOVER packet carries the identification information about the DHCPv4 client.

2. After receiving the DHCPDISCOVER packet, each DHCPv4 server selects an available IP address from the IP address pool, reserves the IP address for the DHCPv4 client, and sends parameters such as the IP address to the DHCPv4 client through a DHCPOFFER packet.
3. After receiving the DHCPOFFER packets, the DHCPv4 client selects one of them and broadcasts the result in a DHCPREQUEST packet. The DHCPv4 server that is not selected reclaims the IP address reserved for the DHCPv4 client and puts the IP address back to the IP address pool.
4. The selected DHCPv4 server sends a DHCPACK packet to the DHCPv4 client to allocate parameters such as an IP address.
5. If the assigned parameters cannot be used, for example, an assigned IP address has been used by other DHCPv4 clients, then the DHCPv4 client sends a DHCPDECLINE packet to notify the DHCPv4 server.
6. If the DHCPv4 client no longer requires the assigned parameters, the DHCPv4 client sends a DHCPRELEASE packet to notify the DHCPv4 server so that the DHCPv4 server can assign these parameters to other DHCPv4 clients.

Involving the DHCP Relay Agent

When the DHCPv4 client and DHCPv4 server are not in the same broadcast domain, broadcast packets cannot be received by each other. In this case, the DHCPv4 relay agent function must be enabled in the broadcast domain of the DHCPv4 client to ensure the communication between the DHCPv4 client and DHCPv4 server. In general, the DHCPv4 relay agent function is enabled on the gateway. When the DHCP relay agent function is enabled, the IP address of the corresponding DHCPv4 server needs to be configured to ensure that the DHCP relay agent can correctly forward the DHCP packets sent by the base station to the DHCPv4 server. [Figure 4-8](#) shows the working principles of DHCPv4 in this scenario.

Figure 4-8 DHCPv4 interworking (with a DHCP relay agent)



4.2.4.2.2 DHCPv4 Packet Format

Figure 4-9 shows an example of a DHCPv4 packet format.

Figure 4-9 DHCPv4 packet format

	Byte 0	Byte 1	Byte 2	Byte 3	
IP	Version	Header Length	TOS	Total Length	
	Identification		Flag (3 bits)	Fragment Offset (13 bits)	
	TTL	UDP	Header Checksum		
	Source IP@				
	Destination IP@				
UDP	Source Port		68/67		
	Total Length		CRC		
DHCP	op	htype	Hlen	Hop	
	xid				
	secs		Flag		
	ciaddr				
	yiaddr				
	siaddr				
	giaddr				
	chaddr				
	sname				
	file				
	Magic Cookie	Magic Cookie			
	Option 1	Code = 1	Length = 4	Subnet Mask	
Option 43 (Suboption)	Code = 43	Length = n	Subcode 1	Length 1	
	...				
	Subcode n	Length n			
	...				
Option n	Code...	Length	...		
...	...				

NOTE

The actual length and sequence of each field in a DHCPv4 packet in software implementation may be different from those shown in Figure 4-9.

The DHCPv4 header contains the DHCPv4 control and configuration information. In the DHCPv4 header, the fields related to automatic OMCH establishment are as follows:

- yiaddr
This field carries the interface IP address of the base station.
- giaddr
This field carries the IP address of the DHCPv4 relay agent.

- Option fields
These fields are encoded in code-length-value (CLV) format and consist of multiple subcodes. Among these fields, Option 43 carries Huawei proprietary information elements (IEs) and most configuration information of the base station. For example, subcode 1 in Option 43 carries the electronic serial number (ESN) of the Huawei base station. For details about subcodes in Option 43, see [Table 4-8](#).
Since Option 43 has a limited length, Option 224 is also used to carry Huawei proprietary IEs in SRAN8.0 and later versions.

For details about DHCPv4, see section "Dynamic Host Configuration Protocol (DHCP)" in RFC 2131 and "DHCP Options and BOOTP Vendor Extensions" in RFC 2132.

4.2.4.2.3 DHCPv4 Client and DHCPv4 Server

In this document, base stations act as DHCPv4 clients. [Table 4-4](#) describes the deployment positions of DHCPv4 servers.

Table 4-4 Deployment positions of DHCPv4 servers

Base Station Type		DHCPv4 Server in Non-IPsec Networking	DHCPv4 Server in IPsec Networking
Single-mode	GBTS	BSC	In the trusted domain: MAE DHCPv4 server In the untrusted domain: public DHCPv4 server
	eGBTS/eNodeB/gNodeB/NodeB	MAE	
Multimode	Co-MPT multimode base station	MAE	
	Separate-MPT multimode base station	The DHCP server for each RAT is same as that for a single-mode base station.	

NOTE

- The DHCPv4 server and the MAE are different logical communication entities, although they may be deployed on the same hardware. This document distinguishes between the DHCPv4 server and the MAE.
- It is recommended that the DHCPv4 server be deployed on the MAE for base stations other than GBTSs that are not protected by IPsec.
- If the DHCPv4 server is deployed on the MAE, the base station cannot be on the same L2 network as the MAE. For security reasons, the MAE's operating system can process only DHCP unicast packets, not DHCP broadcast packets.

From SRAN8.0 onwards, if single-mode or separate-MPT multimode base stations evolve to co-MPT multimode base stations, corresponding DHCPv4 servers must be migrated to the MAE. If the MAE is upgraded to SRAN18.1 or later, the DHCPv4 server of the NodeB must be migrated to the MAE.

When the base station is not on the same L2 network as the DHCPv4 server, a DHCP relay agent must be deployed. Pay attention to the following when deploying a DHCP relay agent:

- When a next-hop gateway of the base station is deployed on the transport network, the DHCP relay agent function must be enabled on the next-hop gateway. The MAE DHCPv4 server IP address must also be configured on the next-hop gateway of the base station.
 - If the Virtual Router Redundancy Protocol (VRRP) is deployed on the next-hop gateway, configure the VRRP's virtual IP address as the IP address of the DHCP relay agent. When the active router is faulty, the standby router can act as the DHCP relay agent.
 - If the base station is a GBTS, run the **SET BTSIP** command. In this step, set **BTSIP** to **ON** and **NEXTHOP** to the IP address of the base station's next-hop gateway.
- When the base station is on the same L2 network as the base station controller, DHCP packets pass through the base station controller, and the MAE serves as the DHCPv4 server for the base station (for example, eGBTS or NodeB), then this base station controller can act as the DHCP relay agent. If the DHCP relay agent function is enabled on a certain port of the base station controller, this port serves as the DHCP relay agent for all eGBTSs and NodeBs connected to this port. The **ADD DHCPRLY** command can be used to enable the DHCP relay agent function on a port of the base station controller. This command contains the following parameters:
 - **DHCPRLYID** indicates the identity of a DHCP relay agent.
 - **DHCPRLYGATEWAYIP** indicates the interface IP address of the base station controller.
 - **DHCPPID** is used to enable or disable the DHCP relay agent function only on BSC6900s. The base station controller serves as the DHCPv4 server for the base station by default. The **OTHERSWITCH** option of the **DHCPPID** parameter can be selected to enable the DHCP relay agent function for the base station.

MML command examples are as follows:

```
//Enabling the DHCP relay agent function on the base station controller when the MAE that manages this base station controller is the DHCP server for the base station
ADD DHCPRLY: DHCPRLYID=1, DHCPRLYGATEWAYIP="10.1.1.1", DHCPPID=OTHERSWITCH-1,
DHCPSPRSEMSIP=Yes;
```

Information such as the MAE IP address and route must be configured on the base station controller side. For details, see the section about configuring Abis interface operation and maintenance channels for eGBTS in BSC6900/BSC6910 GSM initial configuration guide. Also, refer to the section about configuring lub interface operation and maintenance channels in BSC6900/BSC6910 UMTS initial configuration guide.

NOTE

Whether the base station controller can serve as the DHCP server or DHCP relay agent depends on the base station type.

- For GBTSs, the base station controller can only serve as the DHCP server.
- For other types of base stations, such as the eGBTS, NodeB, and co-MPT multimode base station, the base station controller can only serve as the DHCP relay agent.
- When base stations are cascaded or backplane co-transmission is applied, an upper-level base station serves as the next-hop gateway for the lower-level

base station. In this case, the DHCP relay agent function must be enabled and the DHCPv4 server IP address of the lower-level base station must be configured on the upper-level base station.

- If the upper-level base station is an eGBTS, NodeB, eNodeB, gNodeB, or co-MPT multimode base station, run the **SET DHCPRELAYSCHWITCH** command with **ES (LTE eNodeB, 5G gNodeB)** set to **ENABLE** to enable the DHCP relay agent function. Then, run the **ADD DHCPVSRIP** command with **DHCPVSRIP (LTE eNodeB, 5G gNodeB)** set to the DHCPv4 server IP address of the lower-level base station. A maximum of four DHCPv4 server IP addresses can be configured. MML command examples are as follows:

```
//Enabling the DHCP relay agent function on the upper-level base station
SET DHCPRELAYSCHWITCH: ES=ENABLE;
//Setting the DHCP server IP address to 10.19.19.11. Each DHCP broadcast packet will be
forwarded to all DHCP servers.
ADD DHCPVSRIP: DHCPVSRIP="10.19.19.11";
```

NOTE

In backplane co-transmission, when the IP transmission of the upper transmission port is used as the DHCP relay agent IP address and the source-based route is configured for the base station, the DHCP relay agent IP address must be manually configured. If multiple DHCP server IP addresses specified by **DHCPVSRIP (LTE eNodeB, 5G gNodeB)** are configured, each must be configured with an IP address of the DHCP relay agent.

The following MML command example assumes that the IP address of the DHCP relay agent is 10.1.1.1:

```
ADD DHCPVSRIP: DHCPVSRIP="10.19.19.11", DHCPRELAYSCHW="ENABLE",
DHCPRELAYSCHW="10.1.1.1";
```

For details about the application scenarios of source-based IP routing, see *IPv4 Transmission*.

- If the upper-level base station is a GBTS, run the **ADD BTSDHCPVSRIP** command with **DHCPVSRIP** set to the IP address of the lower-level base station's DHCPv4 server. MML command examples are as follows:

```
ADD BTSDHCPVSRIP: IDTYPE=BYID, BTSID=20, DHCPVSRIP="10.100.10.10";
```

In base station cascading scenarios, manual configuration of DHCP relay agent IP addresses is not supported, and the upper-level base station will use its OM IP address and lower port IP address as the DHCP relay agent IP address.

In backplane co-transmission scenarios, manual configuration of DHCP relay agent IP addresses is supported. If no DHCP relay agent IP address is manually configured, the upper-level base station will use its OM IP address and upper transmission port interface IP address as the DHCP relay agent IP address. The upper transmission port interface IP address is on the same network as the next-hop IP address of the DHCP server IP address.

For details about configuration requirements, see [4.3.2.3 Configuration Requirements for the DHCP Server](#).

- A base station can serve as the DHCP relay agent for other base stations in the same L2 network. In this case, the DHCP relay agent function must be enabled and the DHCP server IP addresses of the other base stations must be configured on the base station in question. The enabling and configuring methods for this base station are the same as those for an upper-level base station.
- When base station cascading is used, the number of base station nodes on the chain topology cannot exceed four. This is because some DHCP relay

agents in the transport network require that the number of relay agents cannot exceed four. Otherwise, DHCP packets will be discarded.

4.2.4.3 DHCPv6

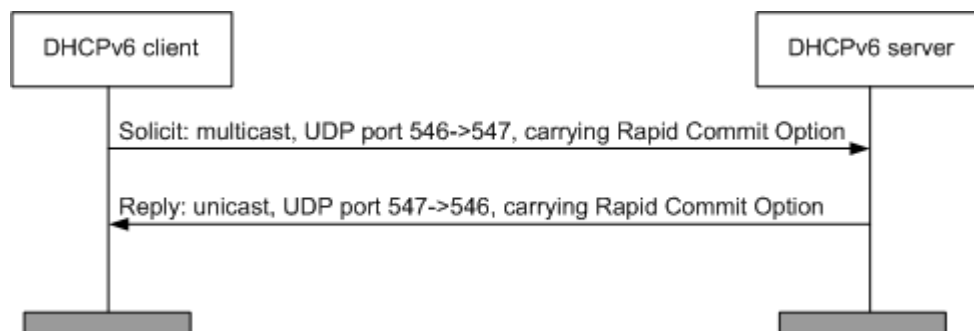
4.2.4.3.1 DHCPv6 Working Principles

If the DHCPv6 client and the DHCPv6 server are on the same link, that is, the DHCPv6 server can receive the multicast packets sent by the DHCPv6 client, RFC 3315 supports two types of DHCPv6 processes, respectively with two messages and four messages.

Not Involving the DHCPv6 Relay Agent (Two Messages)

Figure 4-10 shows the DHCPv6 process when the Solicit message sent by a DHCPv6 client carries the Rapid Commit option and the DHCPv6 server supports this option. **Figure 4-11** shows the DHCPv6 process in other cases.

Figure 4-10 DHCPv6 process with two messages (not involving the DHCPv6 relay agent)

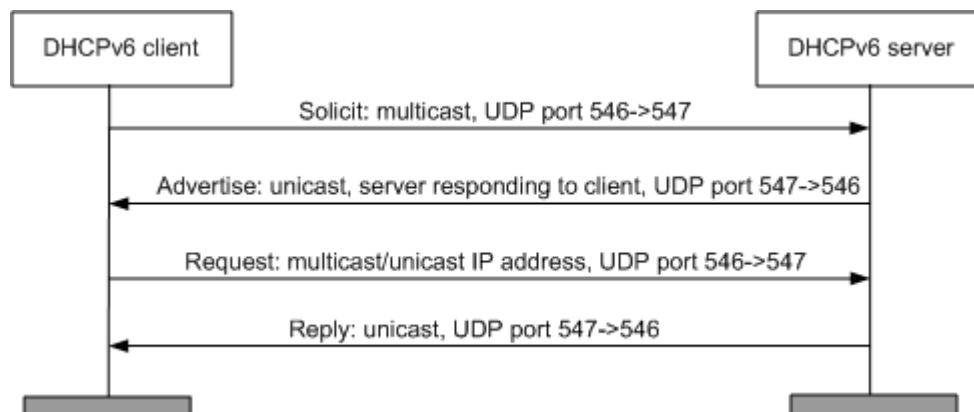


1. After the DHCPv6 client starts, it sends a Solicit message, of which the destination IP address is the multicast address ff02::1:2 and the source IP address is the link-local address. The message carries information such as the DHCPv6 client ID, Rapid Commit option, and IP address request.
2. If the Solicit message received by the DHCPv6 server carries the Rapid Commit option and this option is supported, the DHCPv6 server returns a Reply message that carries the DHCPv6 client option, DHCPv6 server option, Rapid Commit option, and IP address. If the Rapid Commit option is not supported, see **Figure 4-11**.
3. After receiving the Reply message, the DHCPv6 client obtains information such as the IP address carried in the message.

Not Involving the DHCPv6 Relay Agent (Four Messages)

Figure 4-11 shows the DHCPv6 process when the Solicit message sent by the DHCPv6 client does not carry the Rapid Commit option.

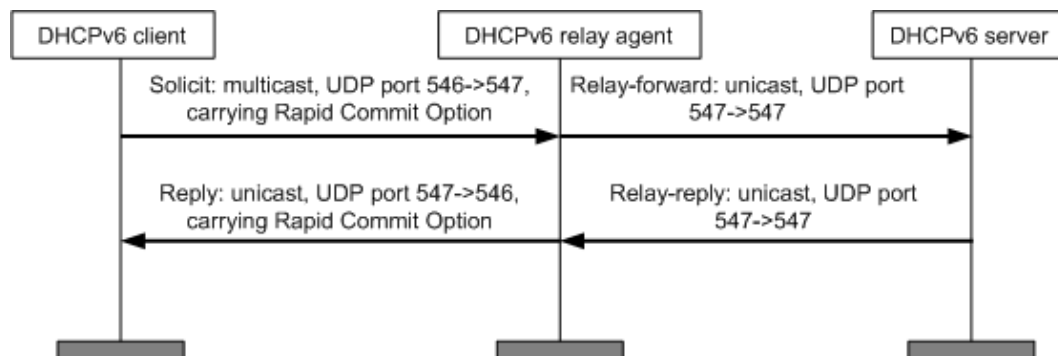
Figure 4-11 DHCPv6 process with four messages (not involving the DHCPv6 relay agent)



1. After the DHCPv6 client starts, it sends a Solicit message, of which the destination IP address is the multicast address ff02::1:2. The message carries information such as the DHCPv6 client ID, Rapid Commit option, and IP address request.
2. If the Solicit message received by the DHCPv6 server does not carry the Rapid Commit option or the DHCPv6 server does not support the option, the DHCPv6 server responds with an Advertise message carrying the DHCPv6 client option and DHCPv6 server option.
3. After receiving the Advertise message, the DHCPv6 client selects a DHCPv6 server to respond to the Request message.
4. After receiving the Request message, the DHCPv6 server returns a Reply message carrying the DHCPv6 client option, DHCPv6 server option, and IP address.
5. After receiving the Reply message, the DHCPv6 client obtains information such as the IP address carried in the message.

Involving the DHCPv6 Relay Agent (Two Messages)

If the DHCPv6 client and the DHCPv6 server are not on the same link, the DHCPv6 relay agent function must be enabled on the next-hop gateway of the DHCPv6 client. To enable this function, a unicast IPv6 address must be configured for the DHCPv6 server. [Figure 4-12](#) shows an example of the DHCPv6 process with two messages when the DHCPv6 relay agent is involved.

Figure 4-12 DHCPv6 process with two messages (involving the DHCPv6 relay agent)

- The DHCPv6 client sends a Solicit message. The DHCPv6 relay agent encapsulates this message in the Relay Message option of the Relay-forward message and forwards it to the DHCPv6 server.
- After receiving the Relay-forward message, the DHCPv6 server encapsulates a Reply message in the Relay Message option of the Relay-reply message and sends it to the DHCPv6 relay agent.
- After receiving the Relay-reply message, the DHCPv6 relay agent obtains the content of the Relay Message option, and then includes the peer-address as the destination IP address of the packet in the Relay-reply message.
- After receiving the Reply message, the DHCPv6 client obtains information such as the IP address carried in the message.

Involving the DHCPv6 Relay Agent (Four Messages)

When the DHCPv6 relay agent is involved, the DHCPv6 process with four messages is similar to the DHCPv6 process with two messages. All messages sent from the DHCPv6 client to the DHCPv6 server are encapsulated in the Relay Message option of the Relay-forward message by the DHCPv6 relay agent. The messages sent by the DHCPv6 server to the DHCPv6 client are encapsulated in the Relay Message option of the Relay-reply message.

NOTE

- If both the base station and the OSS use SRAN18.1 or a later version, the OMCH self-setup uses the DHCPv6 process with four messages.
- If the base station or OSS uses a version earlier than SRAN18.1, the OMCH self-setup uses the DHCPv6 process with two messages.

4.2.4.3.2 DHCPv6 Packet Format

DHCPv6 packets are encapsulated using UDP packets. The value of Next Header is 17, indicating UDP packets, as shown in [Figure 4-13](#). If the length of a DHCPv6 packet is greater than 1280 bytes (minimum MTU of an IPv6 packet), the packet is fragmented and carries the fragment extension header, as shown in [Figure 4-14](#).

Figure 4-13 DHCPv6 packet format (excluding the extension header)

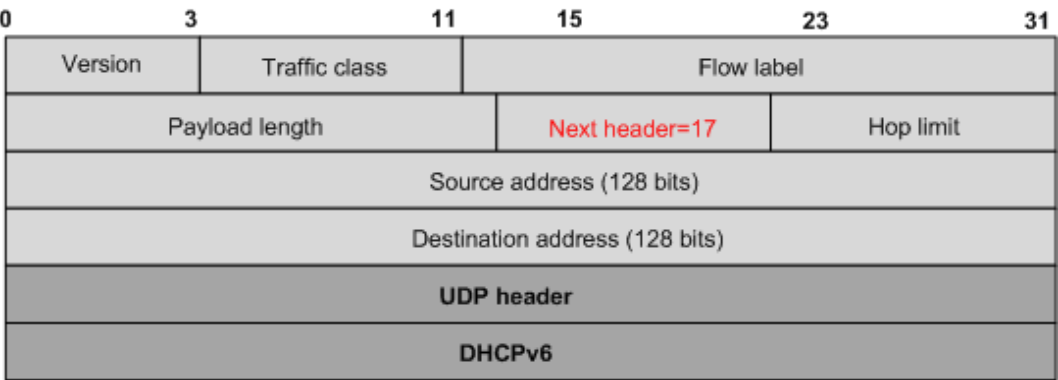
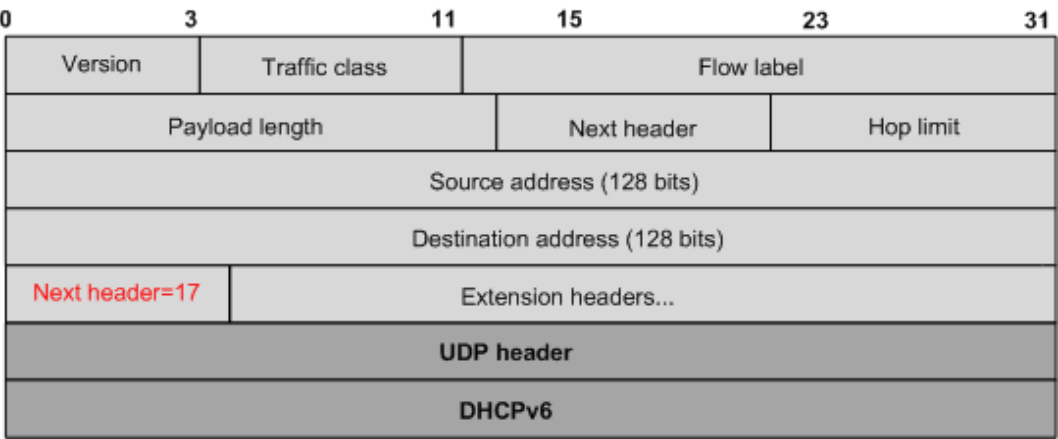
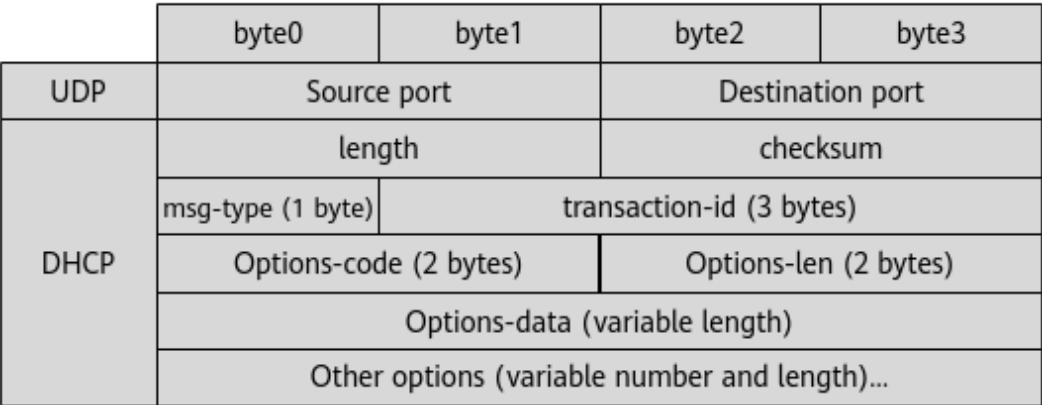


Figure 4-14 DHCPv6 packet format (including the extension header)



The format of DHCPv6 packets between the DHCPv6 client and the DHCPv6 server is different from that of DHCPv6 packets between the DHCPv6 relay agent and the DHCPv6 server, as shown in [Figure 4-15](#) and [Figure 4-16](#).

Figure 4-15 Format of DHCPv6 packets between the DHCPv6 client and DHCPv6 server



- msg-type: This field indicates the ID of a DHCPv6 message.
- transaction-id: This field indicates the ID used for message exchange and occupies 3 bytes. It is similar to xid of a DHCPv4 packet.
- Options: These fields indicate options and have variable lengths. The options include options-code, options-len, and options-data. Multiple options can be carried. Options-code 17 is used to define vendor-defined information.

There are two types of messages transmitted between the DHCPv6 relay agent and the DHCPv6 server: RELAY-FORW message sent from the DHCPv6 relay agent to the DHCPv6 server and RELAY-REPL message from the DHCPv6 server to the DHCPv6 relay agent. These messages have the same packet structure shown in [Figure 4-16](#).

Figure 4-16 Format of DHCPv6 packets between the DHCPv6 relay agent and DHCPv6 server

	byte0	byte1	byte2	byte3
UDP	Source port		Destination port	
	length		checksum	
DHCP	msg-type (1 byte)	hop-count (1 byte)	link-address (16 bytes)	
	...			
	...		peer-address (16 bytes)	
	...			
	...		options (variable number and length)	
	...			

In the RELAY-FORW message:

- msg-type: This field indicates the ID of the RELAY-FORW message.
- hop-count: This field indicates the number of times that messages are forwarded by the DHCPv6 relay agent.
- link-address: IP address of the DHCPv6 relay agent, which can be either a global address or 0 (0 indicates multi-level DHCPv6 relay agent forwarding).
- peer-address: This field indicates the source IP address of the message forwarded by this message, that is, the source address of the message received from the DHCPv6 client or DHCPv6 relay agent.
- options: These fields indicate all the options contained in this message, including the Relay Agent Option (mandatory) and the Interface-Id Option. The Relay Agent Option contains the messages received from the DHCPv6 client or DHCPv6 relay agent.

In the RELAY-REPL message:

- msg-type: This field value is RELAY-REPL.
- hop-count: This field value is copied from the RELAY-FORW message.
- link-address: This field value is copied from the RELAY-FORW message.
- peer-address: This field value is copied from the RELAY-FORW message.
- options: These fields indicate all the options contained in this message, which must include the Relay Agent Option. The Relay Agent Option contains the messages received from the DHCPv6 server or DHCPv6 relay agent.

For details about DHCPv6, see RFC 3315 Dynamic Host Configuration Protocol for IPv6(DHCPv6).

4.2.4.3.3 DHCPv6 Client and DHCPv6 Server

When the OMCH uses IPv6 transmission, base stations act as DHCPv6 clients. The DHCPv6 server is deployed on the MAE in non-IPsec networking.

When the base station and the DHCPv6 server are located on different L2 networks, the DHCPv6 relay agent must be deployed on the next-hop gateway of the base station. The following precautions must be noted:

- The DHCPv6 relay agent function is enabled on the next-hop gateway of the base station, and the DHCPv6 server IP address is the IPv6 address of the DHCPv6 server built in the MAE.
- If the Virtual Router Redundancy Protocol (VRRP) is deployed on the next-hop gateway, the IP address of the DHCPv6 relay agent is used as the virtual IPv6 address of the VRRP. When the active router is faulty, the standby router can act as the DHCPv6 relay agent.

NOTE

- The DHCPv6 server and the MAE are different logical communication entities, although they may be deployed on the same hardware. This document distinguishes between the DHCPv6 server and the MAE.
- When the MAE has a built-in DHCPv6 server, the base station and MAE cannot be located on the same L2 network, which also applies to DHCPv4. For security reasons, the MAE's operating system can process only DHCPv6 unicast packets, not DHCPv6 multicast packets.

4.2.4.3.4 DHCPv6 Relay Agent

When base stations are cascaded, backplane-based co-transmission is used, or panel-based co-transmission is used, the DHCPv6 relay agent function must be enabled on the lower transmission port of the upper-level base station and the IP address of the DHCPv6 server must be configured. The downlink interface is a VLAN interface used to forward OM channel data of a lower-level base station. The DHCPv6 relay agent IP address does not need to be manually configured. The upper-level base station attempts to use its OM IPv6 address and the IPv6 address of the downlink interface as the DHCPv6 relay agent IP address for communication. On the upper-level base station, run the **ADD INTERFACEPARA6** command to enable the DHCPv6 relay agent function and run the **ADD DHCPV6SVRIP6** command to set the IP address of the DHCPv6 server.

The DHCPv6 relay agent function of the base station must meet the following requirements:

- Only single-level DHCPv6 relay is supported. In the chain topology, multiple levels of base stations are deployed on one transmission link. All upper-level base stations need to enable the DHCPv6 relay agent function to forward DHCPv6 packets to adjacent lower-level base stations. The upper-level base stations convert multicast DHCPv6 packets received from DHCPv6 clients to unicast DHCPv6 packets and send the unicast DHCPv6 packets to the DHCPv6 server.
- This function does not apply to dual-MPT or BBU interconnection scenarios.
- A base station supports a maximum of four DHCPv6 server IP addresses.
- In the case of co-transmission through backplane interconnection, if the route from the upper-level base station to the DHCPv6 server of the lower-level base station is configured as a source-based route and the transmission between the OM IP address of the upper-level base station and the DHCPv6 server of the lower-level base station fails, you need to manually enable and specify the DHCPv6 relay IP address by running the **ADD DHCPsvRIP6/MOD DHCPsvRIP6** command.
- **DHCPsvRIP6** can be configured only in the same VRF.

4.2.5 DHCP Procedure

4.2.5.1 Base Station Identification

Upon receiving a DHCP packet from a base station, the DHCP server finds and sends related configuration information to the base station based on the base station ID contained in the DHCP packet. In SRAN8.0 and later versions, the MAE with a built-in DHCPv4 server uses the combination of the ESN and slot number or the combination of the deployment identifier (DID), subrack topology, and slot number as the base station ID. The base station controller and the MAE in versions earlier than SRAN8.0 use the combination of the ESN and NE type or the combination of the DID and NE type as the base station ID. In SRAN15.1 and later versions, the MAE with a built-in DHCPv6 server uses the combination of the ESN and slot number as the base station ID.

- ESN identifies the BBU backplane of the base station. Each backplane has a unique ESN. The ESN is automatically reported by the base station.
- DID is the site identifier planned by the operator. DID is scanned into the base station using a barcode scanner connected to the USB port of the main control board during base station deployment. After the DID is scanned into the base station, the DID is broadcast in the BBU where the main control board is located and other interconnected BBUs. All main control boards record the DID and use it as the base station ID in the DHCP process.
- Subrack topology identifies the interconnection relationship between BBU subracks. The combination of the DID and subrack topology uniquely identifies a BBU subrack.
- Slot number identifies the number of the slot that houses the main control board. The slot number is used to differentiate main control boards in a BBU subrack. If the base station is configured with active and standby main control boards, the slot number is that of the active main control board. The slot number is automatically reported by the base station.
- NE type indicates the RAT of the base station, which can be GSM, UMTS, LTE, or NR.

When a commissioning task by PnP is created, the ESN must be specified if the combination of ESN and slot number is used as the base station ID. The DID must be included in the base station configuration file if the combination of DID, subrack topology, and slot number is used as the base station ID.

When the base station ID information such as the ESN is entered, the MAE automatically delivers the ID information to the DHCPv4 or DHCPv6 server built in the MAE based on the IP transmission mode of the OMCH. If the bearer network is a dual-stack network, the MAE may receive DHCPv4 and DHCPv6 packets sent by the base station. The MAE searches for the base station ID in the DHCP server based on the base station ID in the DHCP packets, and responds to the DHCPv4 or DHCPv6 packets. Only one DHCP server responds to the DHCP request from the base station.

NOTE

In some networking scenarios, such as IPsec networking scenario 1, it is not recommended that the public DHCP server deliver the transmission configuration based on the base station ID.

A combination of DID, subrack topology, and slot number can be used as the base station ID only if the transmission port of the base station is an Ethernet port. This also requires that the DHCP server of the base station be deployed on the MAE.

In SRAN15.1 and later versions, automatic OMCH establishment in IPv6 transmission is supported but the combination of DID, subrack topology, and slot number cannot be used as the base station ID.

4.2.5.2 Obtaining Configuration Information in Non-IPsec Networking Scenarios

- A DHCP client and a DHCP server on the same L2 network can directly communicate with each other. The L2 network is a subnet in which broadcast IP packets can be exchanged and forwarded by Media Access Control (MAC) addresses and VLAN IDs. An example is Ethernet or Ethernet VLAN.
Figure 4-7 shows the process for a base station to obtain configuration information from a DHCP server when no DHCP relay agent is deployed. After the base station is powered on, a DHCPDISCOVER packet with the base station ID is broadcast. The DHCP server then sends configuration information to the base station based on the base station ID.
- If a DHCP server is not deployed on the same L2 network as a DHCP client, a DHCP relay agent must be deployed on the next-hop gateway of the base station to forward DHCP packets. In this case, the DHCP relay agent must be located on the same L2 network as that of the DHCP client, and the DHCP server must be located on the L3 network. The L3 network refers to the network that forwards packets based on the IP address.

Figure 4-8 shows the process for a base station to obtain configuration information when a DHCPv4 relay agent is deployed in an IPv4 transmission network. The DHCPv4 relay agent converts DHCPv4 packets broadcast by the base station into unicast packets, and sends them to the corresponding DHCPv4 server. When receiving the DHCPv4 request, the DHCPv4 server sends the DHCPv4 unicast packets to the DHCPv4 relay agent. At last, the DHCPv4 relay agent broadcasts the packets on the L2 network.

Figure 4-12 shows the process for a base station to obtain configuration information when a DHCPv6 relay agent is deployed in an IPv6 transmission network.

In the process in which the base station and the built-in DHCPv6 server of the MAE use two DHCPv6 messages to obtain IP addresses, the base station acts as the DHCPv6 client and sends packets carrying the Rapid Commit Option. The Reply message sent by the DHCPv6 server also carries the Rapid Commit Option.

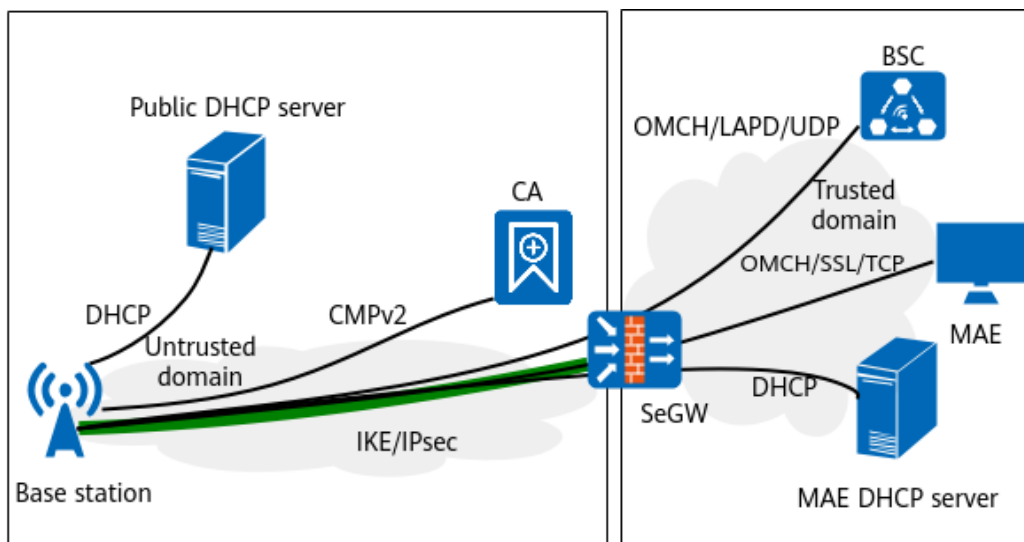
4.2.5.3 Obtaining Configuration Information in IPsec Networking Scenarios

NOTE

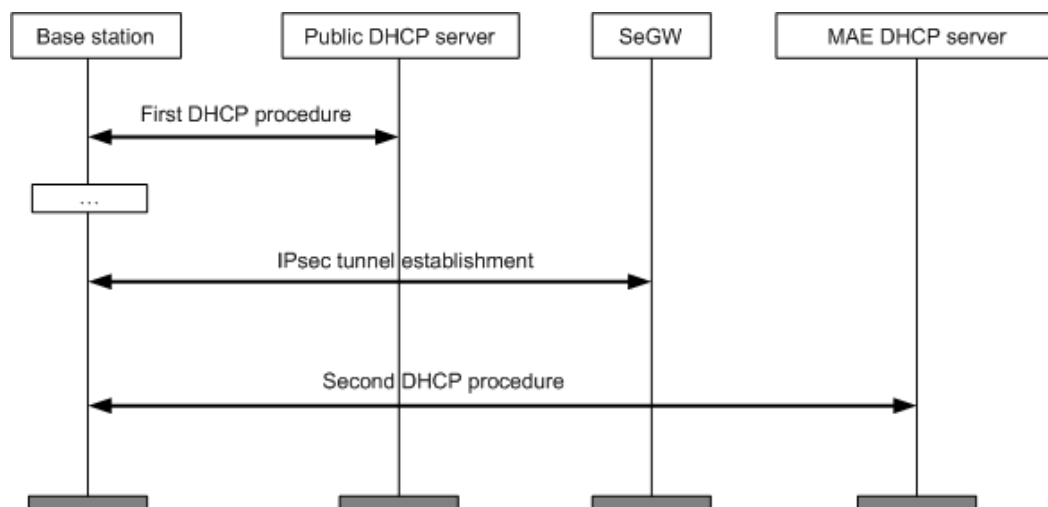
IPsec networking based on IPv6 transmission does not support automatic OMCH establishment.

In IPsec networking scenarios, the DHCP server in the trusted domain can be secured or not secured by IPsec. When the DHCP server is secured by IPsec, a public DHCP server must be deployed in the untrusted domain. [Figure 4-17](#) shows the OMCH networking in this scenario.

Figure 4-17 IPsec OMCH networking



[Figure 4-18](#) shows the two processes for the base station to obtain transmission configuration in the networking shown in [Figure 4-17](#).

Figure 4-18 Two processes for obtaining transmission configuration in IPsec networking scenarios

1. The base station exchanges DHCP packets with a public DHCP server to obtain information, such as the interface IP address for accessing the untrusted domain and the SeGW IP address. The base station must also obtain the certificate key type and CA IP address because digital certificates are required for identity authentication with the SeGW. This process is referred to as the first DHCP process.
2. The base station negotiates with the SeGW on the Internet Key Exchange (IKE) security association (SA) and IPsec SA, and then establishes an IPsec tunnel. Since digital certificates are required for identity authentication with the SeGW, the base station must generate a certificate request based on the certificate key type and apply to the CA for digital certificates that can be identified by the SeGW before establishing an IPsec tunnel.
3. The base station exchanges DHCP packets with the MAE built-in DHCP server to obtain the OM IP address used for accessing the trusted domain. This process is referred to as the second DHCP process. The second DHCP process varies depending on IPsec networking scenarios. For details, see [4.3.3.7 Obtaining Formal Transmission Configuration Information from the MAE DHCP Server](#).

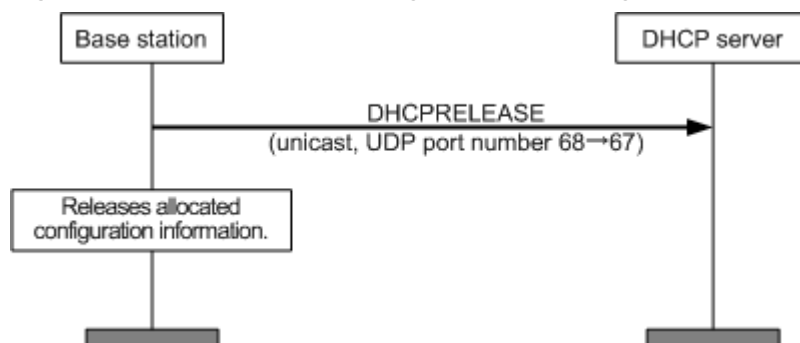
During the first DHCP process, the public DHCP server runs the general DHCP protocol. It may not support Huawei-defined DHCP Option fields and fail to identify the base station ID reported by the base station. In this case, the public DHCP server selects an IP address from the IP address pool and sends it to the base station. During the second DHCP process, the MAE built-in DHCP server sends configuration parameters to the base station based on the base station ID reported by the base station.

4.2.5.4 Releasing Allocated Configuration Information in IPsec Networking Scenarios

When a base station obtains configuration information from an MAE built-in DHCP server, but does not require configuration information allocated by a public DHCP server, the base station sends a DHCPRELEASE message to the public DHCP server. After receiving the DHCPRELEASE message, the public DHCP server can

redistribute allocated configuration information to other NEs. [Figure 4-19](#) shows the process of releasing allocated configuration information.

Figure 4-19 Process of releasing allocated configuration information



NOTE

In addition to the preceding process, DHCP also supports the process of updating configuration information. However, base stations in the current version do not support the process of updating configuration information.

4.2.6 Automatic DHCP Data Synchronization

The principles of automatic data synchronization are the same for the DHCPv4 server and the DHCPv6 server.

Ensure that the correct DHCP data of a base station is available on the MAE DHCP server before using the Automatic OMCH Establishment feature. Any manual modifications to a base station's transmission configuration data may change its DHCP data on the MAE. In earlier versions, users had to manually ensure that the DHCP data on the MAE DHCP server was correct before the startup of the next automatic OMCH establishment. As a manual data check is a complicated and error-prone process, the automatic DHCP data synchronization function is introduced.

After the base station is deployed, the system automatically synchronizes manual modifications to the transmission configuration data in the base station configuration file with the MAE DHCP server. This ensures the configuration information consistency between the MAE DHCP server and the base station. For manual modifications on a single base station, the system starts data synchronization (completed within 5 minutes), which begins 10 minutes after the last manual data modification. For manual modifications on multiple base stations, the system starts data synchronization for every 200 base stations as a batch, with each batch completed within less than or equal to 30 minutes. If DHCP data is to be modified, batch modification is supported for base stations of which OMCHs have not been established, and manual modification on the MAE GUI is required for base stations of which OMCHs work properly.

However, the automatic DHCP data synchronization function does not support automatic synchronization of the NE name, NE type, ESN, and working mode fields because they identify a specific NE. In addition, this function does not support automatic synchronization of the Security Gateway Emergency Bypass, ACLRule Destination IP Address, and ACLRule Destination Subnet Mask fields because they must be manually configured. Automatic DHCP data synchronization

supports synchronization of other information on the MAE DHCP server. Ensure that the related NE data exists in the current data area on the MAE-Deployment before starting automatic DHCP data synchronization.

4.2.7 Obtaining VLAN Information for DHCP Packets

Packets sent by a base station on a VLAN-based network must carry the VLAN ID. Before an OMCH is established, that is, before the base station sends the first DHCP packet, the base station must learn VLAN information. After the base station is started, the VLAN learning function is enabled. For IPv4 transmission and IPv6 transmission on the base station, the VLAN learning functions are independent of each other, and the VLAN obtaining processes are different.

4.2.7.1 Obtaining VLAN Information in IPv4 Transmission

On an IPv4 transmission network, after the base station starts, it receives an ARP packet. The base station acquires the IPv4 VLAN information by parsing the received ARP packet carrying the VLAN ID. After acquiring the VLAN information, the base station sends a DHCPv4 packet carrying the VLAN ID, and communicates with the DHCPv4 server to obtain the transmission configuration. The process is as follows:

1. Once the DHCP function is enabled on the base station, the base station starts the VLAN acquisition process in IPv4 transmission. The base station then acquires VLAN IDs from all received ARP packets and records these VLAN IDs in a PnP VLAN-ID table.
The base station sends DHCPv4 packets without VLAN IDs or with VLAN ID being either 0 or 1.
2. The base station waits 20s. If the base station receives a DHCPOFFER packet within 20s, it exits the DHCPv4 process and enters the subsequent PnP deployment process. Otherwise, the base station goes to the next step.
3. The base station checks the PnP VLAN-ID table and sends DHCP packets using all acquired VLAN IDs. If the base station receives a valid DHCPOFFER packet, it exits the DHCPv4 process and enters the subsequent PnP deployment process.
4. If the preceding steps fail:
 - If the base station has only one transmission port, the base station repeats the preceding steps on this port.
 - If the base station has multiple transmission ports, it repeats the preceding steps on other transmission ports.

Table 4-5 describes the recommended schemes for the base station in SRAN8.0 and later versions to obtain VLAN information during deployment.

Table 4-5 Schemes of obtaining VLAN information

Networking Scenario	Requirements for Network Devices	Scheme
Non-IPsec networking or IPsec securing service data but does not secure OMCH data (IPsec networking scenario 3)	N/A	Scheme 1
IPsec securing DHCP packets and OMCH data (IPsec networking scenario 1)	The SeGW initiates a request for IKE negotiation with the base station. The destination IP address of the request is the interface IP address that the base station uses to access the untrusted domain. The VLAN information in DHCP packets sent by the base station must be the same as the VLAN information in the configuration file of the base station.	
IPsec securing OMCH data but not DHCP packets (IPsec networking scenario 2).	The security policy allows the transmission of DHCP packets sent by the MAE DHCP server to the base station.	Scheme 2
IPsec securing DHCP packets and OMCH data (IPsec networking scenario 1)	The L2 network is configured with the default VLAN ID or no VLAN ID.	Scheme 3
IPsec securing DHCP packets and OMCH data (IPsec networking scenario 1)	The next-hop gateway of the base station can periodically send ping packets to the interface IP address of the base station.	Scheme 4

If a base station is deployed by PnP, the scheme of obtaining VLAN information varies depending on whether IPsec secures OMCH data and NE capability.

- If IPsec does not secure OMCH data, scheme 1 is used:
Scheme 1: The MAE or BSC actively and periodically sends OMCH establishment requests to the base station. After receiving the requests, the next-hop gateway of the base station sends ARP packets to the base station. The base station then records VLAN IDs derived from ARP packets and includes recorded VLAN IDs in DHCPv4 packets.
- If IPsec secures OMCH data, any of the following schemes is used:
 - Scheme 1
 - Scheme 2: The DHCPv4 server on the MAE periodically sends empty DHCPv4 Offer packets (containing DHCPv4 headers only) to the base station. The destination IP address is the interface IP address of the base station in the untrusted domain. This enables the next-hop gateway of

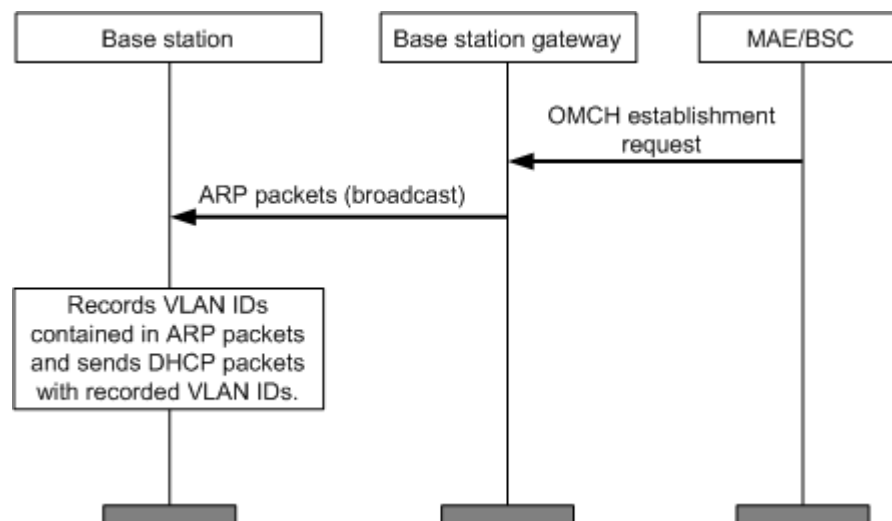
the base station to send ARP packets, from which the base station acquires VLAN information.

- Scheme 3: The base station sends DHCPv4 packets without VLAN ID, and the L2 network attaches a VLAN ID to DHCPv4 packets sent by the base station. In this case, the base station does not need to acquire VLAN information.
- Scheme 4: The gateway of the base station, or another NE periodically sends packets to the base station or an idle address of the subnet to which the base station belongs. This enables the gateway of the base station to send ARP packets, from which the base station acquires VLAN information.

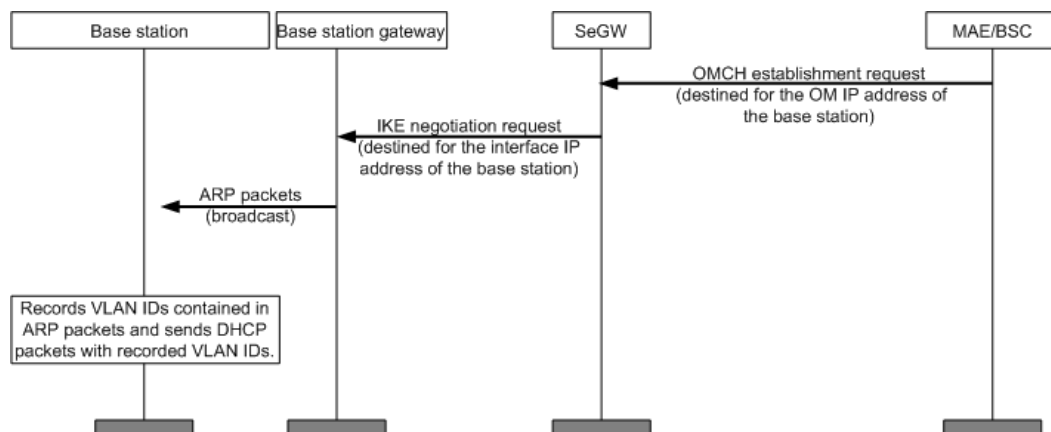
4.2.7.1.1 Scheme 1

Scheme 1 applies to scenarios 1 and 3 described in [4.2.7.1 Obtaining VLAN Information in IPv4 Transmission](#). [Figure 4-20](#) and [Figure 4-21](#) show the procedures in the two scenarios.

Figure 4-20 Scheme 1 (IPsec does not secure OMCH data)



1. The MAE/BSC sends an OMCH establishment request to the OM IP address of the base station.
2. To forward the OMCH establishment request to the correct base station, the next-hop gateway of the base station broadcasts ARP packets to obtain the MAC address mapping the destination IP address of the request. The next-hop gateway or the L2 network attaches VLAN IDs to ARP packets so that correct VLAN IDs are contained in the ARP packets received by the base station.
3. The base station parses all received ARP packets and records the VLAN IDs contained in the packets.
4. The base station sends all DHCP packets with recorded VLAN IDs. Only DHCP packets with correct VLAN IDs can reach the DHCP relay agent which is installed on the next-hop gateway of the base station.

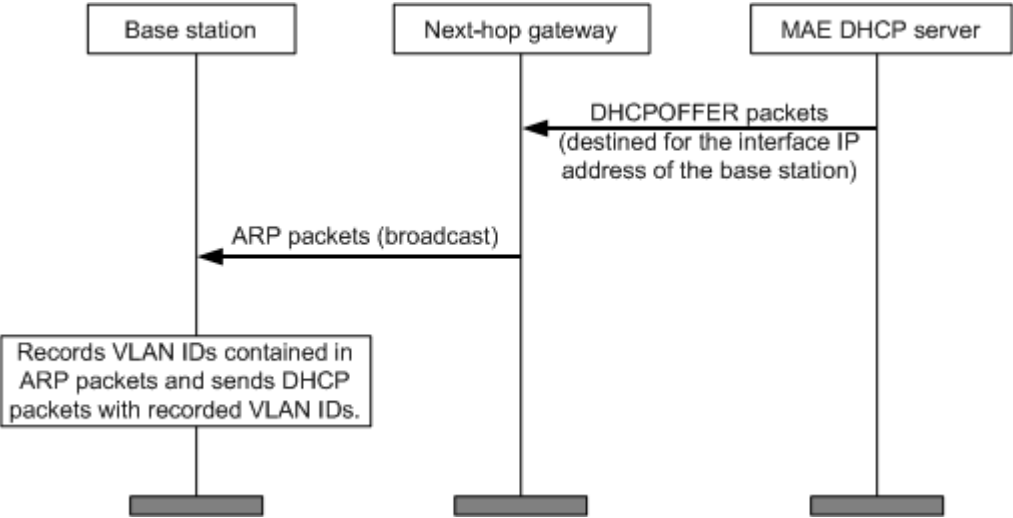
Figure 4-21 Scheme 1 (IPsec secures OMCH data)

1. The MAE/BSC sends an OMCH establishment request to the OM IP address of the base station. The request is forwarded to the SeGW.
2. The SeGW detects that the IPsec SA with the base station is not established and sends an IKE negotiation request to the interface IP address of the base station in the untrusted domain. The request is then routed to the next-hop gateway of the base station.
3. To forward the IKE negotiation request to the correct base station, the next-hop gateway of the base station broadcasts ARP packets to obtain the MAC address mapping the destination IP address of the request. The next-hop gateway or the L2 network attaches VLAN IDs to ARP packets so that correct VLAN IDs are contained in the ARP packets received by the base station.
4. The base station parses all received ARP packets and records the VLAN IDs contained in the packets. It may record the VLAN ID in an ARP packet destined for another base station.
5. The base station sends all DHCP packets with recorded VLAN IDs. Only DHCP packets with correct VLAN IDs can reach the DHCP relay agent.

4.2.7.1.2 Scheme 2

Figure 4-22 shows the procedure for a base station to obtain VLAN information in scheme 2.

Figure 4-22 Scheme 2

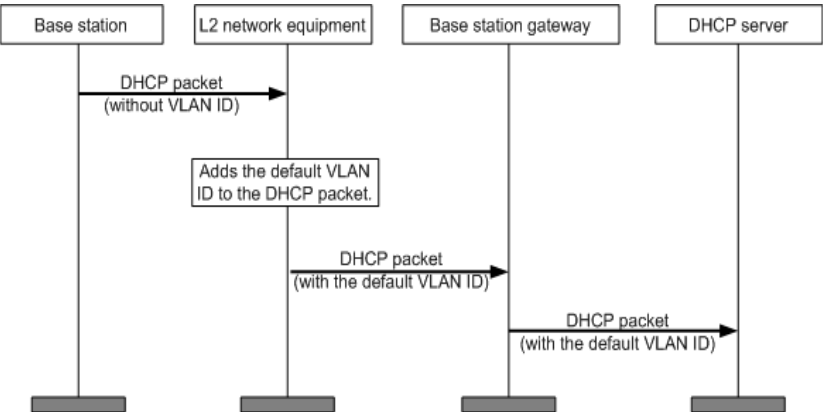


1. The MAE sends a DHCPOFFER packet with no content to the interface IP address of the base station in the untrusted domain. The packet is then forwarded to the next-hop gateway of the base station.
2. To forward the DHCPOFFER packet to the correct base station, the next-hop gateway of the base station broadcasts ARP packets to obtain the MAC address mapping the destination IP address of the request. The next-hop gateway or the L2 network attaches VLAN IDs to ARP packets so that correct VLAN IDs are contained in the ARP packets received by the base station.
3. The base station parses all received ARP packets and records the VLAN IDs contained in the packets. It may record the VLAN ID in an ARP packet destined for another base station.
4. The base station sends all DHCP packets with recorded VLAN IDs. Only DHCP packets with correct VLAN IDs can reach the DHCP relay agent.

4.2.7.1.3 Scheme 3

Figure 4-23 shows the procedure for a base station to obtain VLAN information in scheme 3.

Figure 4-23 Scheme 3

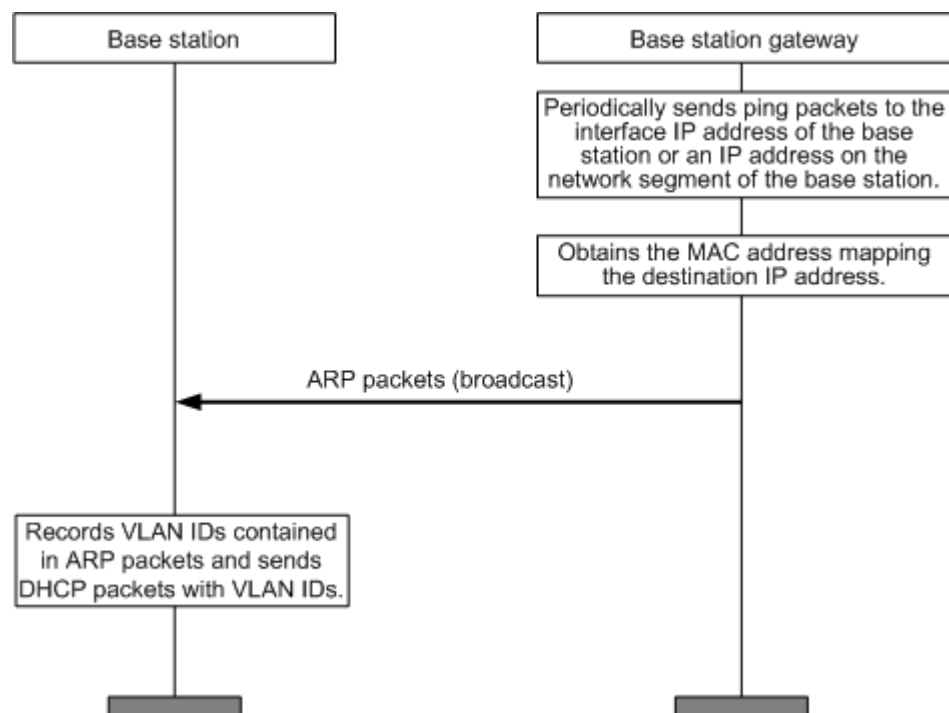


1. The base station sends a DHCP packet with no VLAN ID.
2. The L2 network between the base station and the next-hop gateway of the base station automatically attaches the default VLAN ID to the DHCP packet. The default VLAN ID is the same as the VLAN ID required for deploying the base station. With the correct VLAN ID, the DHCP packet can then be forwarded over the L2 network to the DHCP relay agent to reach the DHCP server.

4.2.7.1.4 Scheme 4

Figure 4-24 shows the procedure for a base station to obtain VLAN information in scheme 4.

Figure 4-24 Scheme 4



1. The next-hop gateway periodically sends ping packets to the interface IP address of the base station or an IP address on the network segment of the base station.
2. To forward ping packets to the correct base station, the next-hop gateway of the base station broadcasts ARP packets to obtain the MAC address of the base station mapping the destination IP address of the ping packets. The ARP packets received by the base station carry correct VLAN IDs.
3. The base station parses all received ARP packets and records the VLAN IDs contained in the packets. It may record the VLAN ID in an ARP packet destined for another base station.
4. The base station sends all DHCP packets with recorded VLAN IDs. Only DHCP packets with correct VLAN IDs can reach the DHCP relay agent.

4.2.7.1.5 Enabling and Disabling the VLAN Scanning Function

In SRAN7.0, the VLAN scanning function is provided for eNodeBs to solve the issue that eNodeBs cannot acquire VLAN IDs in secure networking scenarios. After the VLAN scanning function is enabled, the base station sends DHCPv4 packets with random VLAN IDs. This occurs if the base station does not receive a response after sending DHCPv4 packets without a VLAN ID and DHCPv4 packets with acquired VLAN IDs.

After the VLAN scanning function is enabled, some DHCP packets with invalid VLAN IDs may be broadcast. When VLANs are not isolated, these broadcast packets may have impacts on the network. Therefore, this function is disabled by default on the base stations in SRAN8.0 and later versions. For base stations upgraded from SRAN7.0 to SRAN8.0 and later versions, you are advised to run the **SET DHCP** command to locally or remotely enable or disable this function.

- Enabling the VLAN scanning function
Run the **SET DHCP** command with *SWITCH* set to **ENABLE** and *VLANSCAN* set to **ENABLE**.
- Disabling the VLAN scanning function
Run the **SET DHCP** command with *SWITCH* set to **ENABLE** and *VLANSCAN* set to **DISABLE**.

NOTE

When the OMCH and service links are disconnected, the **SET DHCP** command is used to determine whether to automatically start the DHCP process to obtain the initial configuration information or to restore the base station configuration. The *SWITCH* parameter specifies whether to enable this function. The *VLANSCAN* parameter specifies whether to enable the VLAN scanning function when the base station sends DHCP packets.

4.2.7.2 Obtaining VLAN Information in IPv6 Transmission

On an IPv6 transmission network, after the base station is started, it receives a neighbor solicitation (NS) packet or a router advertisement (RA) packet defined in the Neighbor Discovery Protocol. The base station parses the received NS or RA packet, which carries the VLAN ID, to acquire the VLAN information in IPv6 transmission. After acquiring the VLAN information, the base station sends a DHCPv6 packet carrying the VLAN ID, and communicates with the DHCPv6 server to obtain the IPv6 transmission configuration. The process is as follows:

1. The base station starts the VLAN acquisition process in IPv6 transmission. It then acquires VLAN IDs from all received NS and RA packets and records these VLAN IDs in an IPv6 VLAN-ID table. The base station also parses NS and RA packets that do not carry VLAN IDs to send DHCPv6 packets without VLAN IDs.
2. The base station sends a DHCPv6 packet carrying the acquired IPv6 VLAN ID. If the base station does not acquire the IPv6 VLAN ID, it does not send DHCPv6 packets.
3. The base station waits 20s. If the base station receives a valid DHCPv6 Reply message within 20s, it exits the DHCPv6 process and enters the subsequent PnP deployment process.
4. If the base station does not receive a DHCPv6 Reply message:

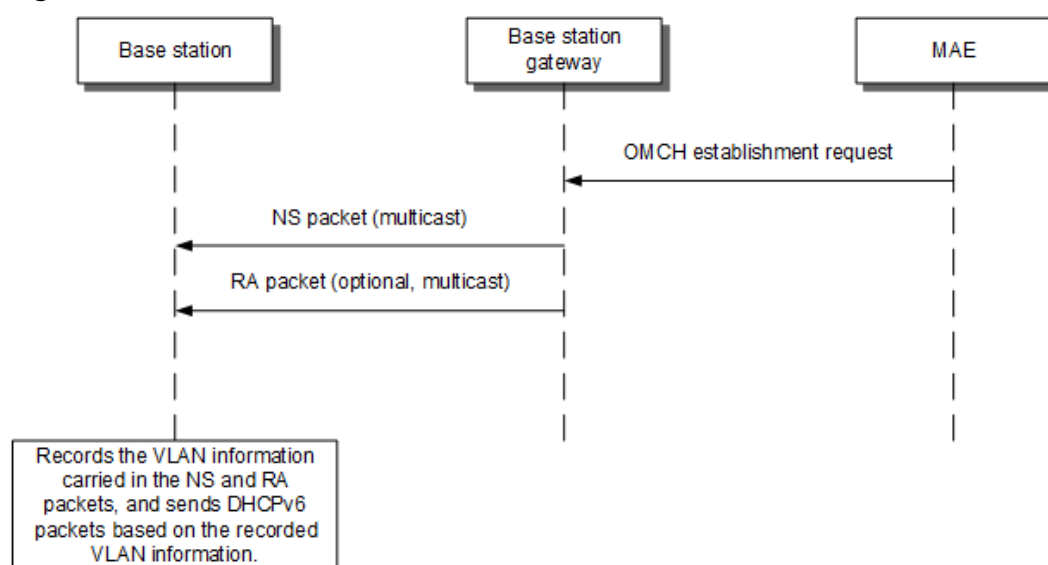
- If the base station has multiple transmission ports, it repeats the preceding steps on other transmission ports.
- If the base station only has one transmission port, it repeats the preceding steps on this transmission port.

If the bearer network is a IPv4/IPv6 dual-stack network, the base station may attempt to acquire both the IPv4 VLAN ID and the IPv6 VLAN ID.

Scheme for the Scenario Where IPsec Does Not Secure OMCH Data

Figure 4-25 shows the process for a base station to obtain VLAN information when IPsec does not secure OMCH data in IPv6 transmission

Figure 4-25 Scheme for the scenario where IPsec does not secure OMCH data



1. The MAE sends an OMCH establishment request to the OM IPv6 address of the base station.
2. To forward the OMCH establishment request to the destination IPv6 address, the next-hop gateway of the base station multicasts NS packets to obtain the MAC address mapping the destination IPv6 address of the request. The NS packets received by the base station may carry the VLAN ID or not. The VLAN ID is attached by the next-hop gateway or the L2 network.
3. The base station parses the received NS packets and records the VLAN information in the NS packets. The VLAN information may carry the VLAN ID or not.
4. If periodic delivery of multicast RA packets is enabled on the base station gateway, the base station can receive RA packets. The base station then parses the received RA packets and records the VLAN information in the RA packets. Periodic delivery of multicast RA packets may be enabled or not on the base station gateway when the OM data is not protected by IPsec.
5. The base station sends DHCPv6 packets based on the learned VLAN information. Finally, only DHCPv6 packets carrying the correct VLAN ID can reach the DHCPv6 relay agent deployed on the base station gateway.

4.2.7.3 Saving VLAN IDs

The base station supports saving the VLAN IDs for successful DHCP procedures. After receiving the last response message from the DHCP server, the base station saves the VLAN ID used for the DHCP procedure. A maximum of eight VLAN IDs can be saved in IPv4 transmission, and a maximum of six VLAN IDs can be saved in IPv6 transmission. When the maximum number of VLAN IDs that can be saved is exceeded, the new VLAN ID overwrites the earliest VLAN ID in the table.

The base station can use the saved and acquired VLAN IDs to send DHCP packets when reinitiating a DHCP procedure during or after deployment of the base station.

The saved VLAN IDs will be automatically cleared after the base station experiences a power-off reset.

4.3 Automatic OMCH Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations

4.3.1 Overview

This chapter describes the automatic OMCH establishment implemented on the single-mode base station and co-MPT multimode base station in IPsec or non-IPsec networking scenarios in IPv4 transmission and non-IPsec networking scenarios in IPv6 transmission, and outlines the requirements on network equipment. In IPv4 IPsec networking scenarios, the network is divided into the trusted and untrusted domains. Depending on NE distribution in these domains, IPsec networking scenarios are classified as follows:

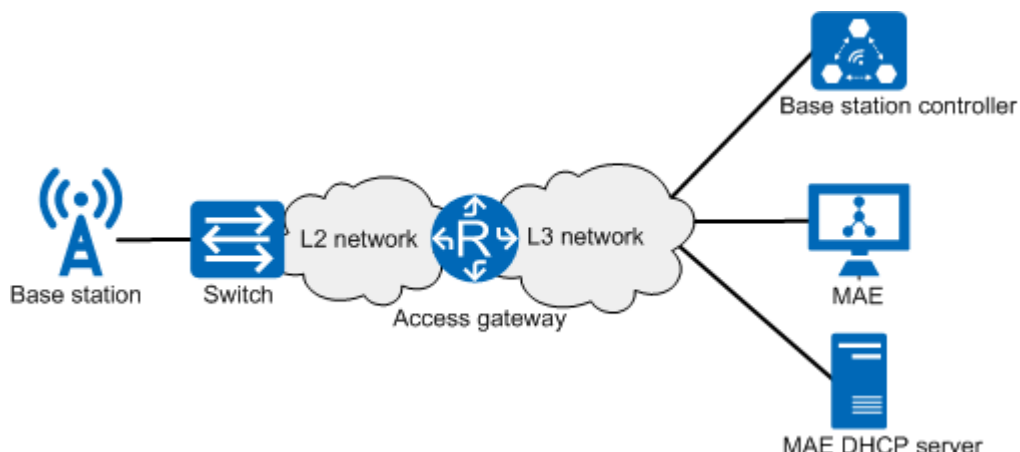
- IPsec networking scenario 1: IPsec secures DHCP packets, OM data, and all or a portion of other data.
- IPsec networking scenario 2: IPsec secures OM data and all or a portion of other data. It does not secure DHCP packets.
- IPsec networking scenario 3: IPsec secures service data, signaling data, and all or a portion of other data. It does not secure DHCP packets or OM data.

Automatic OMCH establishment may fail if the peer equipment is not ready or the configuration of the base station, transmission equipment, or peer equipment is incorrect. In this case, the base station initiates another DHCP process to obtain the configuration and then restarts automatic OMCH establishment.

4.3.2 Automatic OMCH Establishment in Non-IPsec Networking Scenarios

4.3.2.1 Introduction to OMCH Networking

[Figure 4-26](#) shows the OMCH networking.

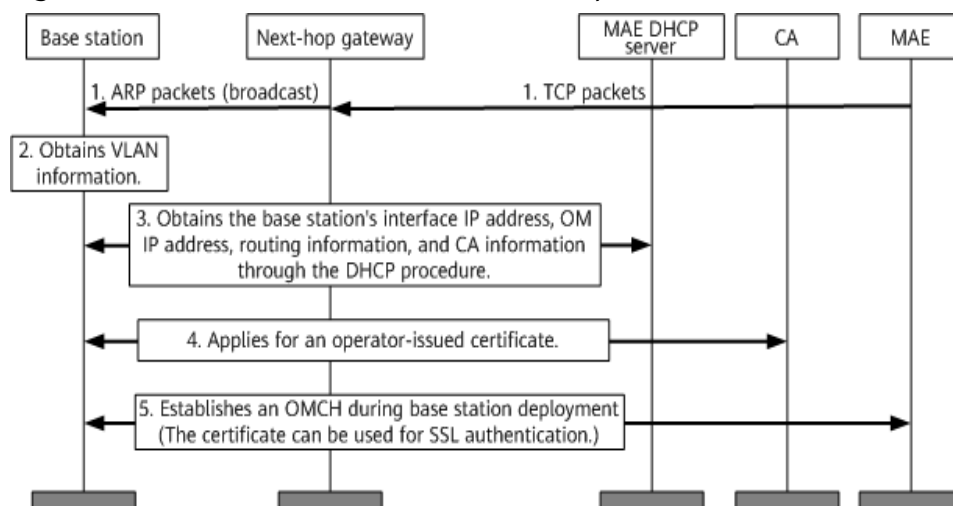
Figure 4-26 OMCH networking

This networking has the following characteristics:

- The DHCP server is not deployed on the L2 network of the base station.
- The DHCP relay agent is deployed on the next-hop gateway of the base station.
- IPsec does not secure OMCH data.
- IPv4/IPv6 OMCH networking is supported.

4.3.2.2 Automatic OMCH Establishment Process

If an OMCH uses SSL authentication, the base station must obtain an operator-issued device certificate before establishing the OMCH with the MAE. **Figure 4-27** shows the automatic OMCH establishment procedure in this scenario.

Figure 4-27 Automatic OMCH establishment process

1. After a PnP-based commissioning task is created on the MAE, the MAE periodically sends SSL-based OMCH establishment requests to the base station. If the OM IP address of the base station is an IPv4 address, the MAE sends an IPv4 OMCH establishment requests. If the OM IP address of the base station is an IPv6 address, the MAE sends an IPv6 OMCH establishment

requests. In an IPv4 OMCH establishment request packet, the source IP address is the IPv4 address of the MAE, and the destination IP address is the OM IPv4 address of the base station. In an IPv6 OMCH establishment request packet, the source IP address is the IPv6 address of the MAE, and the destination IP address is the OM IPv6 address of the base station. After the base station gateway receives the request: the IPv4 base station gateway sends an ARP broadcast packet to the base station to parse the MAC address corresponding to the interface IP address of the base station; the IPv6 base station gateway sends a multicast NS packet to the base station to parse the MAC address corresponding to the interface IP address of the base station.

2. The base station obtains VLAN information. For details, see [4.2.7 Obtaining VLAN Information for DHCP Packets](#).
3. The base station first sends DHCPv4 packets without VLAN IDs and then DHCPv4 packets with VLAN IDs. The base station sends DHCPv6 packets only after learning IPv6 VLAN information. By exchanging DHCP packets with its next-hop gateway and DHCP server, the base station obtains and then validates the OMCH configuration data.
4. Based on the CA information obtained from the DHCP server, the base station applies for an operator-issued device certificate from the CA. For details, see [4.3.2.4 Obtaining an Operator-Issued Device Certificate](#).
5. In response to the OMCH establishment requests from the MAE, the base station performs mutual authentication with the MAE using the obtained device certificate. After the authentication is successful, an OMCH is established between them.

In this scenario, the MAE DHCP server delivers configurations to the base station. The configurations include those described in [DHCPv4 Server](#) and [DHCPv6 Server](#) and CA information described in [Table 4-11](#).

NOTE

If the OMCH fails to be established, the base station automatically restarts the automatic OMCH establishment process.

4.3.2.3 Configuration Requirements for the DHCP Server

DHCPv4 Server

A route to the IP address of the DHCP relay agent must be configured on the DHCP server. In addition, the DHCP server must be preconfigured with the configuration information to be used in the DHCP process, including the related fields in the DHCP packet header, the common Option fields defined by RFC2132, and the subcodes in the Option 43 field defined by Huawei. [Table 4-6](#) describes the fields in the DHCP packet header. [Table 4-7](#) describes the common Option fields. [Table 4-8](#) describes the subcodes in the Option 43 field.

Table 4-6 Parameters to be contained in DHCP packet headers

Parameter Name	Mapping DHCP Field	Length (Byte)	Parameter Description	DHCP Packet Involved
Interface IP Address	yiaddr	4	Mandatory. Interface IP address of the base station.	DHCPOFFER DHCPACK
Relay Agent IP	giaddr	4	Optional. IP address of the DHCP relay agent deployed on the network, if any are available. Broadcast packets (Discovery and Request packets) sent by the base station do not carry this IP address, and the DHCP relay agent adds this IP address to DHCP packets to be forwarded. For details, see RFC 2131.	DHCPDISCOVERY DHCPOFFER DHCPREQUEST DHCPACK

Table 4-7 Parameters to be contained in DHCP Option fields

Parameter Name	Option	Length (Byte)	Parameter Description	DHCP Packet Involved
Subnet Mask	1	4	Mandatory. Subnet mask of a DHCP client.	DHCPOFFER DHCPACK
Router Option	3	N x 4	Mandatory. List of the IP addresses of routers deployed in a DHCP client's subnet. N indicates the number of next-hop gateways for the DHCP client.	DHCPOFFER DHCPACK

Parameter Name	Option	Length (Byte)	Parameter Description	DHCP Packet Involved
Vendor Specific Information	43	0-255	Mandatory. Vendor-specific information exchanged between a DHCP client and a DHCP server.	DHCPDISCOVER DHCPREQUEST DHCPOFFER DHCPACK
IP Address Lease Time	51	4	Mandatory. Lease time of an assigned IP address.	DHCPOFFER DHCPACK
DHCP Message Type	53	1	Mandatory. Value 1: DHCPDISCOVER Value 2: DHCPOFFER Value 3: DHCPREQUEST Value 5: DHCPACK	DHCPDISCOVER DHCPREQUEST DHCPOFFER DHCPACK
Server Identifier	54	4	Mandatory. IP address of a DHCP server.	DHCPOFFER DHCPACK REQUEST
Renewal (T1) Time Value	58	4	Optional. Interval between address assignment and transition to the RENEWING state.	DHCPOFFER DHCPACK
Rebinding (T2) Time Value	59	4	Optional. Interval between address assignment and transition to the REBINDING state.	DHCPOFFER DHCPACK
Vendor class identifier	60	0-255	Optional. Vendor type and configuration.	DHCPDISCOVER DHCPREQUEST
Client-identifier	61	0-255	Optional. Unique identifier of a DHCP client.	DHCPDISCOVER DHCPREQUEST

Table 4-8 Parameters to be contained in subcodes of Option 43

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
ESN	1	20	Mandatory. ESN of the BBU backplane. It is used by a DHCP server to determine the location and BBU subrack of the base station.	DHCPDISCOVER DHCPOFFER DHCPREQUEST DHCPACK
DHCP Server ID	50	1	Whether the DHCP packets are sent by the MAE DHCP server. The MAE DHCP server fills in this field when sending the DHCP packets. If the DHCP packets are not sent by the MAE DHCP server, this field is left blank. Mandatory when the MAE serves as the DHCP server. This field is left blank when a device other than the MAE serves as the DHCP server.	DHCPOFFER DHCPACK
MPT 1st Slot Number	251	1	Mandatory. Slot number of the first main control board.	DHCPDISCOVER DHCPOFFER DHCPREQUEST DHCPACK

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
MPT 2nd Slot Number	249	1	Slot number of the second main control board. Optional. Mandatory only if the base station is configured with active/standby or primary/secondary main control boards.	DHCPOFFER DHCPACK
DID	27	1~64	If the base station is configured with only one BBU, the DID serves the same purpose as the ESN. If the base station is configured with multiple BBUs that are interconnected, these BBUs use the same DID. Optional. DID is mandatory if it is used as the base station ID in DHCP packets.	DHCPDISCOVER DHCPOFFER DHCPREQUEST DHCPACK

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
Subrack Topo	246	1~16	Mandatory. Interconnection relationship between the BBU accommodating the main control board that sends the DHCP packets and other BBUs when these BBUs are interconnected. The DHCP server uses the combination of the DID, BBU subrack topology, and slot number to identify the configuration file of the base station.	DHCPDISCOVER DHCPOFFER DHCPREQUEST DHCPACK
OM Bearing Board	250	1	This parameter has two values, 0 and 1 . • Value 0: An OMCH is established on the panel. Use the value 0 for single-mode base stations. • 1: An OMCH is established on the backplane. Optional. The default value of this parameter is 0.	DHCPOFFER DHCPACK

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
OM Interface Type	2	1	Transmission port of the base station: Ethernet or E1. Optional. The default value is Ethernet. NOTE If the OM interface type is Ethernet, the OMCH MO in the configuration file of the base station must be bound to a route, or the PEERIP parameter must be set to the IP address of the MAE or the next-hop gateway of the base station.	DHCPOFFER DHCPACK
OM Interface Slot Number	248	1	Slot number of the main control board if the transmission port is provided by the main control board, or the slot number of the UTRP board if the transmission port is provided by the UTRP board. Optional. Mandatory if an Ethernet port is used as the transmission port. If this parameter is not specified, the base station automatically identifies the slot number.	DHCPOFFER DHCPACK

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
OMCH Interface Port Number	247	1	Number of the transmission port of the base station. Optional. Mandatory if an Ethernet port is used as the transmission port. If this parameter is not specified, the base station automatically identifies the port number.	DHCPOFFER DHCPACK
OMLOCATION	51	2	Numbers of the cabinet and subrack that house the main control board where the OMCH is located. Optional. Mandatory if an Ethernet port is used as the transmission port. If this parameter is not specified, the base station automatically identifies the cabinet number and subrack number.	DHCPOFFER DHCPACK
OM IP Address	3	4	Mandatory. Local IP address of the OMCH.	DHCPOFFER DHCPACK
OM IP Address Subnet Mask	4	4	Mandatory. Local IP address mask of the OMCH.	DHCPOFFER DHCPACK
MAE IP Address	5	4	Mandatory. IP address of the peer MAE, namely, peer IP address of the OMCH.	DHCPOFFER DHCPACK

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
MAE IP Subnet Mask	6	4	Mandatory. IP address mask of the peer MAE, namely, peer IP address mask of the OMCH.	DHCPOFFER DHCPACK
OM Vlan ID	11	2	VLAN ID of the OMCH. Optional. This parameter is mandatory if VLAN is configured on the Ethernet port of the base station.	DHCPOFFER DHCPACK
OM Vlan Priority	12	1	Optional. VLAN priority of the OMCH. This parameter is not included when a non-Ethernet port is used as the transmission port.	DHCPOFFER DHCPACK
BSC IP	13	4	IP address of the BSC. This parameter is mandatory for GSM and not required for UMTS and LTE.	DHCPOFFER DHCPACK
OM Next Hop IP Address	17	4	Mandatory. Next-hop IP address of the base station.	DHCPOFFER DHCPACK
GBTS OMCH DSCP	54	1	DSCP used by the GBTS to establish an OMCH. Optional. This parameter is applicable only to GBTSs. If this parameter is not specified, the DSCP subcode will not be delivered.	DHCPOFFER DHCPACK

Parameter Name	Subcode	Length (Byte)	Parameter Description	DHCP Packet Involved
Routing Mode	57	1	Optional. Routing mode indicator. Value 0: destination-based routing Value 1: source-based routing If this parameter is not specified, the base station uses the destination-based route by default.	DHCPOFFER DHCPACK
Connection Type	47	1	Mandatory. Type of connection between the NE and client. Value 0: ALL(All type) Value 1: SSL(Only SSL Connection)	DHCPOFFER DHCPACK
Capability ID	239	1	Mandatory. DHCP request capability. Value 0: basic capability	DHCPDISCOVER DHCPREQUEST

When creating a base station commissioning by PnP task on the MAE, deployment engineers can import configuration information listed in [Table 4-8](#) into the DHCP server.

Deployment engineers can only manually modify the configuration information for the DHCP server on the MAE GUI. Deployment may fail if the DHCP server is not configured with mandatory parameters listed in [Table 4-8](#) or optional parameters in certain scenarios.

DHCPv6 Server

A route to the IPv6 address of the DHCPv6 relay agent must be configured on the DHCPv6 server. In addition, the DHCPv6 server must be preconfigured with the configuration information to be used in the DHCPv6 process, including the standard Option fields defined by RFC3315 and the Option 17 fields defined by Huawei. [Table 4-9](#) describes the standard Option fields to be configured on the DHCP server. [Table 4-10](#) describes Huawei-defined Option 17 fields.

Table 4-9 Standard Option fields of the DHCPv6 server

Parameter Name	Option	Length (Byte)	Parameter Description	DHCP Packet Involved
Interface IPv6 Address	5	40	Mandatory. Interface IP address of the IPv6 OMCH.	ADVERTISE REPLY
Vendor-specific Information Option	17	Variable length	Mandatory. Vendor-specific information exchanged between a DHCPv6 client and a DHCPv6 server.	SOLICIT ADVERTISE REQUEST REPLY

Table 4-10 User-defined Option 17 fields of the DHCPv6 server

Parameter Name	Option-code	Length (Byte)	Parameter Description	DHCP Packet Involved
ESN	1	20	ESN of the BBU backplane. It is used by a DHCP server to determine the location and BBU subrack of the base station.	SOLICIT ADVERTISE REQUEST REPLY
OM IPv6 Address	3	16	Mandatory. Local IPv6 address of the base station OMCH.	ADVERTISE REPLY
OSS IPv6 Address	5	16	Mandatory. IPv6 address of the peer MAE, namely, peer IPv6 address of the OMCH.	ADVERTISE REPLY
OSS IPv6 Address Prefix Length	6	1	Mandatory. Prefix length of the IPv6 address of the peer MAE, namely, prefix length of the peer IPv6 address of the OMCH.	ADVERTISE REPLY
Interface IPv6 Address	7	Variable length	Mandatory. IPv6 interface address and prefix length.	ADVERTISE REPLY

Parameter Name	Option-code	Length (Byte)	Parameter Description	DHCP Packet Involved
OM Vlan ID	11	2	VLAN ID of the OMCH. This parameter is mandatory if VLAN is configured on the Ethernet port of the base station.	ADVERTISE REPLY
OM Vlan Priority	12	1	VLAN priority of the OMCH. Optional. This parameter is not included when a non-Ethernet port is used as the transmission port.	ADVERTISE REPLY
OM Next Hop IPv6 Address	17	16	Mandatory. Next-hop IPv6 address of the OMCH.	ADVERTISE REPLY
Prime MPT Cabinet Number	51	2	Mandatory. Number of the cabinet that houses the main control board where the OMCH is located.	ADVERTISE REPLY
Prime MPT Subrack Number			Mandatory. Number of the subrack that houses the main control board where the OMCH is located.	ADVERTISE REPLY
OMCH Interface Port Number	247	1	Mandatory. Transmission port number of the base station.	ADVERTISE REPLY
OM Interface Slot Number	248	1	Mandatory. Number of the slot that houses the main control board providing the transmission port.	ADVERTISE REPLY
Connection Type	47	1	Mandatory. Type of connection between the NE and client. Value 0: ALL(All type) Value 1: SSL(Only SSL Connection)	ADVERTISE REPLY
Capability ID	239	1	Mandatory. DHCP request capability. Value 0: basic capability	SOLICIT REQUEST

4.3.2.4 Obtaining an Operator-Issued Device Certificate

The base station obtains CA information from the DHCP server. For details, see [Table 4-11](#).

Table 4-11 Parameters specific to the MAE DHCP server or DHCPv6 server

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
CA information	CA URL	44	1-128	Mandatory. In IPsec networking scenarios, the base station needs to obtain operator-issued certificates from this URL. This URL must be reachable in the untrusted domain.	DHCP: - DHCPOFFER - DHCPACK DHCPv6: - ADVERTISE - REPLY
	CA Name	38	1-127	It is the CA name.	
	Certificate Key Type	45	1	Optional. It indicates the type of the certificate key. - The value 0 indicates RSA_2048. - The value 1 indicates RSA_3072. - The value 2 indicates RSA_4096. - The value 3 indicates ECDSA_P256. - The value 4 indicates ECDSA_P384. - The value 5 indicates ED25519. The default value is 0.	

After obtaining OMCH information and CA information, the base station generates a certificate request file. The base station then uses this certificate request file to apply for an operator-issued device certificate from the CA (obtained through the DHCP procedure) based on CMPv2.

During the certificate application, the CA needs to verify the base station. All UMP/UMDU/GTMUc boards are preconfigured with Huawei-issued device certificates before delivery. During the certification application, the base station provides the CA with Huawei-issued device certificates as its identity. The CA also needs to be preconfigured with the Huawei root certificate to authenticate the base station.

Before applying for a certificate, the base station has not obtained the configuration file but obtained limited configurations (such as the CA URL and CA name) from the DHCP server. Therefore, the base station uses the default parameters described in Table 4-12 to complete the certificate application. Parameters other than those listed in the table are not included during the certification application or in the certificate request files.

 NOTE

- For details about the certificate application procedure, see the "Certificate Management and Application Scenarios" section in *PKI Feature Parameter Description* for SingleRAN.
- PKI redundancy is not supported during base station deployment by PnP. The active PKI server must work properly during base station deployment by PnP.
- Huawei initial certificates deployed on the GTMUc boards in the GBTs can only be used for encrypting the connections between the GBTs and the site maintenance terminal (SMT). These certificates cannot be used to obtain operators' certificates during automatic OMCH establishment.

Table 4-12 Default parameters used for certificate application

Parameter Category	Parameter Name	Parameter Description	Description
CMPv2-related parameters	Source IP address	Source IP address used to apply for the operator-issued device certificate	This parameter is set to the local IP address of the OMCH configured for DHCP.
	CA URL During Site Deployment	URL of the CA	This parameter is set to the URL of the CA that is obtained through the DHCP procedure.
	Signature Algorithm	Signature algorithm for CMP messages	This parameter is set to SHA256 . NOTE This parameter is set to SHA1 in versions earlier than SRAN16.1 and is set to SHA256 in SRAN16.1 and later versions.

Parameter Category	Parameter Name	Parameter Description	Description
Parameters in the certificate request file	Request Type	Type of a certificate request. The request can be either a new certificate request or a certificate update request. The default type is new certificate request.	The parameter is set to NEW by default.
	Certificate Request File Format	Format of a certificate request file	This parameter is set to CRMF .
	Renew Key	Whether to generate a new key pair	This parameter is set to YES .
	Key type	Certificate key type. The value can be RSA or ECDSA .	This parameter is set based on Certificate Key Type in the DHCP configuration.
	Key size (valid when the certificate type is RSA)	Length of the RSA key in the certificate	This parameter is set based on Certificate Key Type in the DHCP configuration.
	Curve name (valid when the certificate type is ECDSA)	Curve name used by the ECC key in the certificate	This parameter is set based on Certificate Key Type in the DHCP configuration.
	Common Name	Common name of the certificate request file	This parameter is set to the ESN of the base station that applies for a certificate, which can be obtained from ESN.huawei.com.
	Key Usage	Usage of a key	KEY_AGREEMENT (key negotiation), DATA_ENCIPHERMENT (data encryption), KEY_ENCIPHERMENT (key encryption), and DIGITAL_SIGNATURE (digital signature) are selected for this parameter.

Parameter Category	Parameter Name	Parameter Description	Description
	Signature Algorithm	Signature algorithm for a certificate request file	This parameter is set to SHA256 .
	Local Name	Local name of a base station. This parameter is used to generate the DNS name of the certificate subject alternative name for identity verification in IKE negotiation.	The value of this parameter is <i>board ESN.huawei.com</i> .
	Local IP	Local IP address	This parameter is set to 0.0.0.0 . NOTE This parameter cannot be set to the IP address that the base station obtains from the DHCP server, because the obtained IP address may not be used finally.

In addition to the operator-issued device certificate, the base station also obtains the root certificate of the CA.

If the application for operator-issued digital certificates fails or the base station receives no response within about 30 seconds, the initial certificates are used to establish an OMCH.

For a newly deployed MAE of V100R022C10 or later, if the base station version is SRAN16.1 or later, the base station must use the obtained operator-issued certificate for authentication with the MAE. The initial certificate cannot be used to establish an OMCH.

 NOTE

In non-IPsec networking, the base station can apply for an operator-issued certificate through the LiteCA. For detailed configuration, see section "IP over ETH (Non-Secure Networking)" in *3900 & 5900 Series Base Station Initial Configuration Guide*.
During automatic OMCH establishment, the base station can obtain certificates from only one CA server.

4.3.2.5 Configuration Requirements for Network Equipment

Table 4-13 and **Table 4-14** describe the configuration requirements for network equipment during base station deployment by PnP (**4.3.2.1 Introduction to OMCH Networking** is used as an example).

Table 4-13 Configuration requirements for network equipment (IPv4 transmission)

Network Equipment	Requirement
L2 device	- Allows the transmission of DHCP broadcast and unicast packets and does not allow DHCP packet filtering or modification over the L2 network. - Is configured with correct VLAN information.
Next-hop gateway of the base station	- Is enabled with the DHCP relay agent function and configured with the IP address of the DHCP server. For the IP address requirements, see Table 4-45. If an NAT server is deployed before the MAE, the IP address of the MAE must be converted by the NAT server. - Is configured with a route of which the destination IP address is the DHCP server IP address. - If the base station's OM IP address is not its interface IP address, configure a route of which the destination IP address is the OM IP address of the base station. - Is configured with a route of which the destination IP address is the IP address of the CA if the OMCH uses SSL authentication.
L3 device	- Is configured with routes of which the destination IP addresses are the OM IP address of the base station and the MAE IP address, respectively. - Is configured with a route of which the destination IP address is the DHCP relay agent IP address. - Is configured with a route of which the destination IP address is the IP address of the CA if the OMCH uses SSL authentication.
MAE/BSC	Is configured with a route of which the destination IP address is the OM IP address of the base station.
DHCP server	Is configured with a route of which the destination IP address is the DHCP relay agent IP address.
FTP server	- Is configured with a route of which the destination IP address is the OM IP address of the base station. - Stores software and configuration file of the base station in a specified directory. - Provides access rights, such as the user name and password, for the base station.

Network Equipment	Requirement
CA	(Required only when the OMCH uses SSL authentication) - Is configured with the IP address that can be accessed by network equipment in the untrusted domain. - Is configured with Huawei root certificates.

Table 4-14 Configuration requirements for network equipment (IPv6 transmission)

Network Equipment	Requirement
L2 device	- Allows the transmission of DHCPv6 multicast packets, and multicast and unicast packets for address resolution in neighbor discovery, without filtering and modifying DHCPv6 packets. - Is configured with correct VLAN information.
Next-hop gateway of the base station	- Is enabled with the DHCPv6 relay agent function and configured with the IPv6 address of the DHCPv6 server. - Is configured with a route of which the destination IPv6 address is the DHCPv6 server IP address. - If the base station's OM IPv6 address is not its interface IP address, configure a route of which the destination IP address is the OM IPv6 address of the base station. - Is configured with a route of which the destination IP address is the IP address of the CA if the OMCH uses SSL authentication.
L3 device	- Is configured with routes of which the destination IP addresses are the OM IPv6 address of the base station and the MAE IPv6 address, respectively. - Is configured with a route of which the destination IP address is the IPv6 address of the DHCPv6 relay agent. - Is configured with a route of which the destination IP address is the IP address of the CA if the OMCH uses SSL authentication.
MAE	Is configured with a route of which the destination IP address is the OM IPv6 address of the base station.
DHCPv6 Server	Is configured with a route of which the destination IP address is the IP address of the DHCPv6 relay agent.

Network Equipment	Requirement
FTP server	- Is configured with a route of which the destination IP address is the OM IPv6 address of the base station. - Stores software and configuration file of the base station in a specified directory. - Provides access rights, such as the user name and password, for the base station.
CA	(Required only when the OMCH uses SSL authentication) - Is configured with the IPv6 address that can be accessed by network equipment in the untrusted domain. - Is configured with Huawei root certificates.

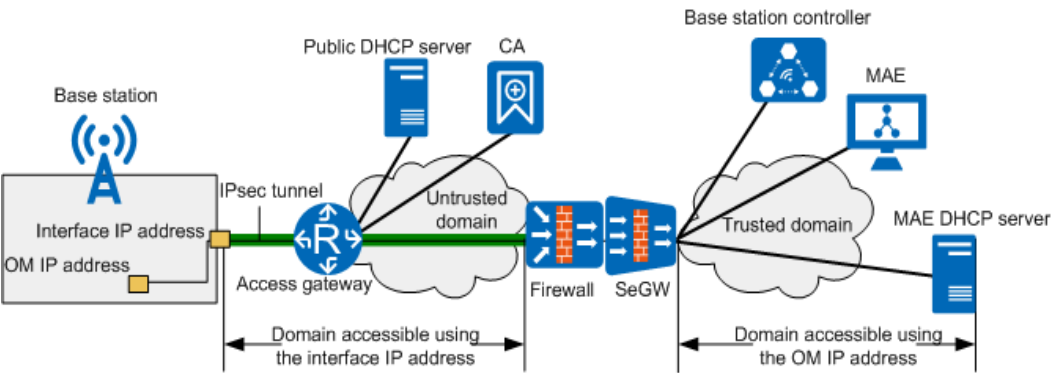
4.3.3 Automatic OMCH Establishment in IPsec Networking

Scenario 1 (Using MAE DHCP Server Data)

4.3.3.1 Introduction to OMCH Networking

Figure 4-28 shows IPsec networking scenario 1, in which IPsec secures both OMCH data and DHCP packets.

Figure 4-28 IPsec networking scenario 1



This type of networking has the following characteristics:

- A public DHCP server and an MAE DHCP server are deployed in the untrusted domain and the trusted domain, respectively. The base station obtains the transmission configuration information (from the public DHCP server) required for establishing a temporary IPsec tunnel to the SeGW and obtains the formal transmission configuration information from the MAE DHCP server.
- The base station in the untrusted domain cannot directly access NEs in the trusted domain. IP packets are encrypted over the IPsec tunnel between the base station in the untrusted domain and the SeGW before being transmitted to the MAE/BSC in the trusted domain.

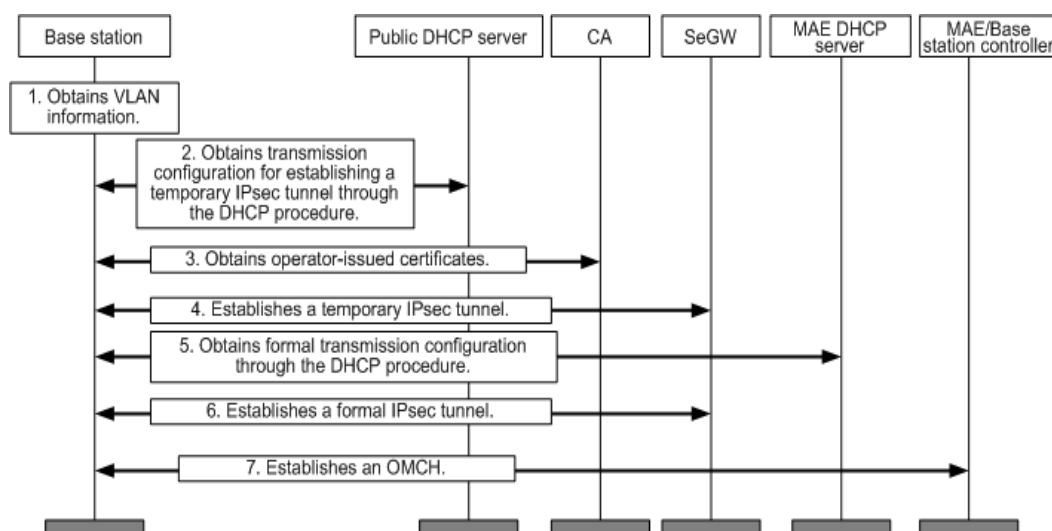
- A CA is deployed. During base station deployment, the CA is accessible through IP addresses of NEs in the untrusted domain (for example, the interface IP address of the base station).
- After the base station starts, it must apply to the CA for operator-issued digital certificates before connecting to the SeGW. After obtaining the certificates, the base station negotiates with the SeGW to establish an IPsec tunnel.
- Only IPv4 OMCH networking is supported.

4.3.3.2 Automatic OMCH Establishment Process

In IPsec networking scenario 1, the base station obtains the interface IP address from the public DHCP server for communicating with NEs located in the untrusted domain. To establish a temporary IPsec tunnel with the SeGW, the base station also needs to obtain the CA and SeGW information from the public DHCP server. After the temporary IPsec tunnel is established, the base station obtains a formal interface IP address and OMCH configuration data from the MAE DHCP server for establishing a formal IPsec tunnel with the SeGW and communicating with NEs located in the trusted domain.

Figure 4-29 shows the automatic OMCH establishment procedure in IPsec networking scenario 1.

Figure 4-29 Automatic OMCH establishment process in IPsec networking scenario 1



1. The base station obtains VLAN information. For details, see [4.2.7 Obtaining VLAN Information for DHCP Packets](#).
2. Using the DHCP procedure, the base station obtains the transmission configuration information (from the public DHCP server) used for establishing a temporary IPsec tunnel. The information includes the interface IP address of the base station, CA configuration data, SeGW configuration data, and MAE DHCP server IP address. For details about the configuration information on the public DHCP server, see [4.3.3.3 Configuration Requirements for the Public DHCP Server](#).

3. Using CMPv2, the base station applies to the CA for an operator-issued device certificate. (For details about the certificate application procedure, see [4.3.3.4 Obtaining an Operator-Issued Device Certificate](#).) The base station adds the obtained CA root certificate to the default trusted certificate list for IPsec tunnel setup and SSL authentication in subsequent procedures.
4. The base station establishes a temporary IPsec tunnel to the SeGW. For details about the security parameters used by the base station in this process, see [4.3.3.5 Establishing a Temporary IPsec Tunnel](#).
5. With protection from the temporary IPsec tunnel, the base station obtains formal transmission configuration information from the MAE DHCP server. Based on whether the internal IP address and the information about the MAE DHCP server are available, the base station uses different procedures to obtain formal transmission configuration information. For details, see [4.3.3.7 Obtaining Formal Transmission Configuration Information from the MAE DHCP Server](#).
6. The base station releases the temporary IPsec tunnel and uses formal transmission configuration information to establish a formal IPsec tunnel to the SeGW. For details, see [4.3.3.8 Establishing a Formal IPsec Tunnel](#).
7. After the formal IPsec tunnel is established, the base station waits for the OMCH establishment request from the MAE/BSC and then establishes an OMCH to the MAE/BSC. If the OMCH is not established within 10 minutes (15 minutes if the SSL authentication fails), the base station restarts the automatic OMCH establishment procedure. Since the base station has obtained the operator-issued certificate, SSL authentication is supported between the MAE and base station.

NOTE

If any steps (except step 1) fail during the automatic OMCH establishment procedure, the base station automatically restarts the procedure.

IPsec Redundancy Among Multiple SeGWs is not supported during base station deployment by PnP when multiple SeGWs are configured. The active SeGW must function properly during base station deployment by PnP.

4.3.3.3 Configuration Requirements for the Public DHCP Server

The public DHCP server must be configured with a route whose destination IP address is the IP address of the base station or whose destination network segment is the network segment of the base station. The DHCP server must also be configured with the parameters listed in [Table 4-15](#). Unless otherwise specified, these parameters are contained in subcodes of Option 43 in DHCP packets.

Table 4-15 Parameters to be configured on the public DHCP server

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	Mandatory or Optional
CA information	PKI Server IP	35	4	It is the IP address of the CA.	The combination of these parameters forms the URL of the CA. In authentication by digital certificates, if the CA URL is configured, these parameters must not be configured; otherwise, these parameters are mandatory. If digital certificate authentication is not required, you do not need to set these parameters.
	CA Protocol Type	39	1	The parameter indicates the type of the protocol used to access the CA. - The value 0 indicates HTTP. - The value 1 indicates HTTPS. When the communication between the base station and CA is protected by SSL, this parameter must be set to 1.	
	CA Port	36	2	It is the HTTP/HTTPS port number of the CA.	
	CA Path	37	1–60	If no path is required for accessing the CA, you do not need to set this parameter.	

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	Mandatory or Optional
	CA URL	44	1-128	It is the URL for accessing the digital certificate path to obtain the digital certificate. This parameter is configurable only when the base station and CA use CMPv2. The CA URL format is as follows: <i>http(s)://CA IP address.CA port number/CA path</i>	If a digital certificate is used for authentication and the PKI Server IP, CA Protocol Type, CA Port, and CA Path are configured, no CA URL can be configured. Otherwise, a CA URL must be configured. If digital certificate authentication is not required, you do not need to set these parameters.
	CA Name	38	1-127	It is the CA name.	This parameter is mandatory when the base station uses digital certificates for authentication.

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	Mandatory or Optional
	Certificate Key Type	45	1	It indicates the type of the certificate key. • The value 0 indicates RSA_2048. • The value 1 indicates RSA_3072. • The value 2 indicates RSA_4096. • The value 3 indicates ECDSA_P256. • The value 4 indicates ECDSA_P384. • The value 5 indicates ED25519.	Optional. The default value is 0 .
SeGW information	Public SeGW IP Address	18	4	It is the IP address of the public SeGW in IPsec networking scenarios. The public SeGW is used in the internal DHCP procedure and is allocated by the public DHCP server.	If the base station needs to access the MAE DHCP server through the SeGW, this parameter is mandatory.
	Public SeGW Local Name	31	1-32	It is the local name of the public SeGW.	Optional. If the SeGW is configured, the local name of the SeGW can be configured to authenticate the SeGW.

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	Mandatory or Optional
	Rsa Signature Hash Algorithm	70	1	It is the hash algorithm used by the RSA digital signature. • The value 3 indicates SHA1. • The value 14 indicates SHA256. • The value 15 indicates SHA384. This parameter is valid only when IKEPROPOSAL.AUT HMETH (LTE eNodeB, 5G gNodeB) is set to IKE_CERT_SIG and the authentication method negotiation result is RSA digital signature. This parameter does not apply to IKEv1.	Optional. This parameter is required when the SeGW does not support RFC 7427. If this parameter is not set, the base station uses the SHA1 algorithm. The SHA1 algorithm is considered insecure in the industry. If the algorithm is used, the transmitted data may be forged by attackers. The SHA256 or SHA384 algorithm is recommended.
MAE DHCP server IP address (list)	Internal DHCP Server IP Address (List)	42	N x 4	It is the IP address of the MAE DHCP server that delivers transmission configuration data to the base station. In SRAN8.0 and later versions, a maximum of eight MAE DHCP server addresses can be configured. N indicates the number of DHCP servers built into the MAE.	Optional. If this parameter is configured, the base station can send unicast DHCP packets to the DHCP server even if the SeGW cannot send any DHCP server IP address to the base station.

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	Mandatory or Optional
Transmission configuration information for the base station	Interface IP Address	-	4	Carried in the yiaddr field in DHCP packet headers	Mandatory
	Interface IP Address mask	-	4	Carried in DHCP option 1	Mandatory
	Next-hop Gateway IP Address	-	4	Carried in DHCP option 3	Mandatory
NOTE The preceding parameters are carried in the following DHCP packets: DHCP OFFER and DHCP ACK.					

All IP addresses or URLs listed in [Table 4-15](#) except **Internal DHCP Server IP Address (List)** can be used only in the untrusted domain. Particularly, NEs in the untrusted domain must have access to the CA IP address and the CA URL. If the base station cannot access the CA, any operator-issued certificates cannot be retrieved.

 NOTE

In IPsec networking scenario 1, the public DHCP server assigns an interface IP address in the IP address pool to the base station, without parsing the BS ID contained in Option 43. Therefore, the BS ID contained in DHCP packets is meaningless in such a scenario.

4.3.3.4 Obtaining an Operator-Issued Device Certificate

The base station generates a certificate request file after a temporary IP address and CA information is retrieved. The base station then uses this certificate request file to apply for an operator-issued device certificate from the specified CA (configured on the public DHCP server) based on CMPv2.

Before applying for a certificate, the base station has not obtained the configuration file but obtained limited configurations (such as the CA URL and CA name) from the public DHCP server. The default parameters for certificate application are the same as those listed in [Table 4-12](#) except for those listed in [Table 4-16](#).

Table 4-16 Default parameters used for certificate application

Parameter Category	Parameter Name	Parameter Description	Description
CMPv2-related parameters	Source IP address	Source IP address used to apply for the operator-issued device certificate	It is the interface IP address configured on the public DHCP server.
CMPv2-related parameters	CA URL During Site Deployment	URL of the CA	It is the CA URL configured on the public DHCP server or the combination of PKI Server IP , CA Protocol Type , CA Port , and CA Path configured on the public DHCP server. NOTE CA Path is optional. Whether it is required depends on the relative path of the CA in which CMPv2 services are provided for the base station.

In addition to the operator-issued device certificate, the base station also obtains the root certificate of the CA. The base station uses the operator-issued device certificate and CA root certificate to perform mutual authentication with the SeGW on the operator network. After the authentication succeeds, the base station can access the internal DHCP server and MAE in the trusted domain through the IPsec tunnel.

 NOTE

For suggestions on PKI system deployment in IPsec networking scenarios, see "PKI Architecture" in *PKI*.

During automatic OMCH establishment, the base station can obtain certificates from only one CA server.

In IPsec networking scenarios, the LiteCA cannot be used.

4.3.3.5 Establishing a Temporary IPsec Tunnel

After the base station obtains the transmission configuration information (including the interface IP address, SeGW IP address, and CA IP address) from the public DHCP server, the base station obtains digital certificates from the CA and then establishes a temporary IPsec tunnel to the SeGW. For details about how to establish an IPsec tunnel, see *IPsec* for SingleRAN. For details about how to obtain digital certificates, see *PKI* for SingleRAN. This section describes the IPsec and IKE proposal algorithms used by the base station during deployment by PnP.

IKEv1 and IKEv2 are incompatible. During base station deployment by PnP, the base station cannot predict the IKE version used by the SeGW. If the base station

has successfully negotiated an IKE version with the SeGW, the base station preferentially tries this IKE version. Otherwise, the base station tries IKEv2 and then IKEv1.

IKE SA Negotiation

During IKE SA negotiation in the normal operation of the base station, the base station supports a large number of algorithm combinations. During base station deployment by PnP, the base station supports a total of 93 IKEv2 proposal algorithm combinations (48 + 9 + 30 + 4 + 1 + 1) listed in [Table 4-17](#), [Table 4-18](#), [Table 4-19](#)/[Table 4-20](#), [Table 4-21](#), [Table 4-22](#), and [Table 4-23](#), and a total of 120 IKEv1 proposal algorithm combinations listed in [Table 4-24](#).

NOTE

The 48 IKEv2 proposal algorithm combinations are obtained as follows: Encryption Algorithm has four values, Authentication Algorithm has two values, Diffie-Hellman Group has three values, and PRF Algorithm has two values. Therefore, the number of algorithm combinations is 48 (4 x 2 x 3 x 2).

The nine new IKEv2 proposal algorithm combinations, 30 ECDH algorithms, four AES_GCM_128 algorithms, and 120 IKEv1 proposal algorithm combinations are obtained in the same way.

Considering the negotiation efficiency, the SHA256 and HMAC_SHA256 algorithms added to the IKEv2 proposal support only the nine combinations described in [Table 4-18](#).

To ensure algorithm security, DES and 3DES in the IKE encryption algorithms, MD5 in the IKE authentication algorithm, DH_GROUP1 and DH_GROUP2 in the DH groups, and HMAC_MD5 in the pseudo-random number algorithms will be deleted in later versions. In the current version, the interface supports configuration synchronization and delivery of these algorithms and the configured algorithms take effect. Therefore, avoid using these weak algorithms.

The UMPTb does not support the algorithm combinations listed in [Table 4-21](#), [Table 4-22](#), or [Table 4-23](#).

Table 4-17 Algorithms in the IKEv2 proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
3DES	SHA1	DH_GROUP2	HMAC_SHA1
AES128	AES_XCBC_96	DH_GROUP14	AES128_XCBC
AES192	-	DH_GROUP15	-
AES256	-	-	-

Table 4-18 New SHA256 algorithms in the IKEv2 proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES128	SHA256	DH_GROUP2	HMAC_SHA256
AES192		DH_GROUP14	

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES256		DH_GROUP15	

Table 4-19 New ECDH algorithms in the IKEv2 proposal (1)

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES128	SHA1	DH_GROUP19	HMAC_SHA1
AES192	AES_XCBC_96	DH_GROUP20	AES128_XCBC
AES256	-	-	-

Table 4-20 New ECDH algorithms in the IKEv2 proposal (2)

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES128	SHA256	DH_GROUP19	HMAC_SHA256
AES192	-	DH_GROUP20	-
AES256	-	-	-

Table 4-21 New AES_GCM_128 encryption algorithm in the IKEv2 proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES_GCM_128	-	DH_GROUP14	HMAC_SHA1
		DH_GROUP19	HMAC_SHA256

Table 4-22 New AES_GCM_256 encryption algorithm in the IKEv2 proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES_GCM_256	-	DH_GROUP20	HMAC_SHA384

Table 4-23 New SHA384 authentication and pseudorandom number algorithms in the IKEv2 proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm
AES256	SHA384	DH_GROUP20	HMAC_SHA384

Table 4-24 Algorithms in the IKEv1 proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	Authentication Method (Only IKEv1)
DES	MD5	DH_GROUP1	PSK
3DES	SHA1	DH_GROUP2	RSA-SIG
AES128	-	DH_GROUP14	DSS-SIG
AES192	-	DH_GROUP15	-
AES256	-	-	-

To improve the negotiation efficiency and compatibility with SeGWs, the base station first attempts to use a single proposal to carry multiple algorithm combinations (listed in [Table 4-25](#)) during IKEv2 negotiation. There are two proposals in total. If the negotiation fails, in order to achieve compatibility, the base station will use a single proposal to carry a single algorithm combination of IKEv2 and IKEv1 in [IKE SA Negotiation](#) in turn for negotiation. Use IKEv2 algorithm combinations as an example. [Table 4-26](#) lists the first five IKEv2 algorithm combinations supported by a single proposal carrying a single algorithm combination. If the negotiation still fails, the base station obtains transmission configuration from the public DHCP server again to set up a temporary IPsec tunnel and then initiates another IKE SA negotiation procedure.

During base station deployment by PnP, the base station does not have initial configuration and uses all supported algorithm combinations to negotiate with the peer end. Some SeGWs may support negotiation using only some of the algorithm combinations requested by the base station. As a result, the negotiation will fail. Therefore, ensure that the peer end can use the locally-planned algorithm combinations for negotiation. For example, the authentication algorithm is set to SHA256 or the PRF algorithm is set to HMAC_SHA256 for a certain type of SeGW, and one proposal can carry only one algorithm combination. During the base station deployment by PnP, the SeGW negotiates with only the first five algorithm combinations (as listed in [Table 4-26](#)) requested by the base station. The negotiation fails because the planned SHA256 (HMAC_SHA256), DH_GROUP19, and DH_GROUP20 are not among the first five algorithm combinations requested by the base station. As a result, the base station deployment by PnP fails.

Table 4-25 Multiple IKEv2 algorithm combinations in a single proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm (Only IKEv2)
AES_GCM_128 and AES_GCM_256	-	DH-GROUP14, DH-GROUP15, DH-GROUP19, DH-GROUP20, and DH-GROUP31	HMAC-SHA1, HMAC-SHA256, and HMAC-SHA384
3DES, AES128, AES192, and AES256	SHA1, AES_XCBC_96, SHA256, and SHA384	DH-GROUP2, DH-GROUP14, DH-GROUP15, DH-GROUP19, DH-GROUP20, and DH-GROUP31	HMAC-SHA1, AES128_XCBC, HMAC-SHA256, and HMAC-SHA384

Table 4-26 Single IKEv2 algorithm combination in a single proposal

Encryption Algorithm	Authentication Algorithm	Diffie-Hellman Group	PRF Algorithm (Only IKEv2)
AES128	SHA1	DH-GROUP2	HMAC-SHA1
3DES	SHA1	DH-GROUP2	HMAC-SHA1
AES256	AES_XCBC_96	DH_GROUP15	AES128_XCBC
AES192	SHA1	DH_GROUP14	HMAC_SHA1
AES128	SHA1	DH_GROUP14	HMAC_SHA1

NOTE

During base station deployment by PnP, the **IDTYPE (LTE eNodeB, 5G gNodeB)** parameter in the **IKEPEER** MO is set to **FQDN** by default and the base station uses *SubjectAltName* in the digital certificate as the local name of the base station for IKE negotiation.

Only the UMPTe, UMPTg, UMPTga, and UMPTHa boards support the following algorithm combinations listed in [Table 4-25](#).

- AES_GCM_128, DH_GROUP14, and HMAC_SHA256
- AES_GCM_256, DH_GROUP20, and HMAC_SHA384

During IKEv2 negotiation, the base station supports IKEv2 fragmentation negotiation by default for structured proposals. There are two possible situations:

1. The peer end supports IKEv2 fragmentation negotiation.
The base station attempts to send the negotiation packet in the original format. If the peer end does not respond, the base station attempts to send the negotiation packet in the fragment format.
2. The peer end does not support IKEv2 fragmentation negotiation.

The base station still sends negotiation packets in the original format.

For details about IKEv2 fragmentation, see *IPsec*.

IPsec SA Negotiation

During IPsec SA negotiation in the normal operation of the base station, the base station supports ESP and AH authentication in tunnel or transport mode. However, during base station deployment by PnP, the base station only supports ESP authentication in tunnel mode.

During IPsec SA negotiation in the normal operation of the base station, the base station supports multiple IPsec proposal algorithm combinations. To improve negotiation efficiency and compatibility with SeGWs, an IPsec proposal can carry multiple algorithm combinations (including PFS) during base station deployment by PnP. The four IPsec proposals listed in [Table 4-27](#) are preferentially used for negotiation. If the negotiation fails, the base station uses the IPsec proposal algorithm combinations (excluding PFS) listed in [Figure 4-30](#) for compatibility purposes. The base station performs IPsec SA negotiation in two steps as shown in [Figure 4-30](#). The sequence is as follows: {IKEv2, green and yellow algorithm groups in [Figure 4-30](#)}, {IKEv2, gray and blue algorithm groups in [Figure 4-30](#)}, {IKEv1, green algorithm groups in [Figure 4-30](#)}, {IKEv1, gray algorithm groups in [Figure 4-30](#)}.

Table 4-27 Multiple algorithm combinations in a single IPsec proposal

Encryption Algorithm	Authentication Algorithm	PFS
3DES, AES128, AES192, and AES256	SHA1, AES-XCBC-MAC-96, and SHA256	Disable
AES_GCM_128, AES_GCM_256, AES_GMAC_128, and AES_GMAC_256	-	Disable
3DES, AES128, AES192, and AES256	SHA1, AES-XCBC-MAC-96, and SHA256	PFS_GROUP1, PFS_GROUP2, PFS_GROUP14, PFS_GROUP15, PFS_GROUP19, PFS_GROUP20, and PFS_GROUP31
AES_GCM_128, AES_GCM_256, AES_GMAC_128, and AES_GMAC_256	-	PFS_GROUP1, PFS_GROUP2, PFS_GROUP14, PFS_GROUP15, PFS_GROUP19, PFS_GROUP20, and PFS_GROUP31

Figure 4-30 Encryption and authentication algorithms in the IPsec proposal

Encryption algorithm \ Authentication algorithm	SHA1	SHA256	AES-XCBC-MAC-96	-
3DES				
AES128				
AES_GCM_128				
AES_GCM_256				
AES192				
AES256				
AES_GMAC_128				
AES_GMAC_256				

NOTE

The base station supports a large number of security parameters. During base station deployment by PnP, the base station does not attempt to use all supported security parameters to establish an IPsec tunnel. Otherwise, the deployment may take a long time. For example, the base station will not try the supported DES algorithm during the PnP-based deployment due to limited security of the algorithm.

The MAE, BSC, DHCP server, and FTP server do not support IPsec. Therefore, during base station deployment by PnP, the base station must use the tunnel mode instead of the transport mode for encapsulation when establishing an IPsec tunnel.

To ensure algorithm security, 3DES will be deleted from the IPsec proposal encryption algorithms in later versions. Therefore, avoid using 3DES. This value is still supported in the current version.

During the negotiation of a temporary IPsec tunnel, the base station automatically configures the ACL rule "*temporary interface IP address* to any". The SeGW is required to assign a high priority to the Traffic Selector corresponding to the temporary interface IP address so that the negotiated Child SA allows the traffic between the base station and the built-in DHCP server of the MAE to pass.

If the IKE proposal and IPsec proposal algorithms of the base station or SeGW and their values are inconsistent with those used during base station deployment by PnP, OMCH establishment may fail. As a result, base station deployment by PnP may fail. Therefore, if PnP deployment is used, ensure that the values of these algorithms for the base station and SeGW meet the PnP deployment requirements.

4.3.3.6 Configuration Requirements for the MAE DHCP Server

The MAE DHCP server must be configured with the parameters (carried in DHCP packets) listed in [Table 4-28](#) as well as the parameters listed in [Table 4-8](#). These parameters are contained in subcodes of Option 43 in DHCP packets.

Table 4-28 Parameters specific to the MAE DHCP server

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
SeGW information	Serving SeGW IP	20	4	Mandatory. It is the IP address of the serving SeGW in IPsec networking scenarios.	DHCPOFFER DHCPACK
	Serving SeGW Local Name	32	1-32	Optional. It is the local name of the serving SeGW. It is provided by the base station to authenticate the serving SeGW in IPsec networking scenarios.	
	ACLRule Destination IP Address	81	4	Optional. During base station deployment by PnP, an ACL is specified for the base station to match the OM IP address segment when an IPsec tunnel is being established between the base station and the SeGW.	DHCPOFFER DHCPACK
	ACLRule Destination Subnet Mask	82	4	Optional. During base station deployment by PnP, an ACL is specified for the base station to match the OM IP address segment when an IPsec tunnel is being established between the base station and the SeGW.	DHCPOFFER DHCPACK

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
	Rsa Signature Hash Algorithm	70	1	Optional. It is the hash algorithm used by the RSA digital signature. • The value 3 indicates SHA1. • The value 14 indicates SHA2_256. • The value 15 indicates SHA2_384. This parameter is valid only when IKEPROPOSAL.AUTH METH (LTE eNodeB, 5G gNodeB) is set to IKE_CERT_SIG and the authentication method negotiation result is RSA digital signature. This parameter does not apply to IKEv1. This parameter is required when the SeGW does not support RFC 7427. If this parameter is not set, the base station uses the SHA1 algorithm. The SHA1 algorithm is considered insecure in the industry. If the algorithm is used, the transmitted data may be forged by attackers. The SHA256 or SHA384 algorithm is recommended.	DHCPOFFER DHCPACK

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
CA information	CA URL	44	1-128	Mandatory. In IPsec networking scenarios, the base station needs to obtain operator-issued certificates from this URL.	DHCPOFFER DHCPACK
	CA Name	38	1-127	Mandatory. It is the name of the preceding CA.	
	Certificate Key Type	45	1	Optional. It indicates the type of the certificate key. • The value 0 indicates RSA_2048. • The value 1 indicates RSA_3072. • The value 2 indicates RSA_4096. • The value 3 indicates ECDSA_P256. • The value 4 indicates ECDSA_P384. • The value 5 indicates ED25519. The default value is 0 .	

4.3.3.7 Obtaining Formal Transmission Configuration Information from the MAE DHCP Server

RFC 4306, the standard protocol for IKEv2, defines the MODE-CONFIG mode in which the base station uses the configuration payload (CP) to apply to the SeGW for certain configuration information. Using the MODE-CONFIG mode during IKE negotiation, the base station can obtain one temporary logical IP address used for accessing the trusted domain and one MAE DHCP server IP address. The base station can obtain a maximum of only one MAE DHCP server IP address.

 NOTE

In IKEv1, Configuration Payload is not standardized and is called MODE-CONFIG. The base station supports MODE-CONFIG only when the IKE SA is negotiated in aggressive mode. For details about MODE-CONFIG, see *RFC4306 Internet Key Exchange (IKEv2) Protocol*.

The base station follows procedures listed in [Table 4-29](#) to obtain formal transmission configuration information from the MAE DHCP server, depending on whether the logical IP address used for accessing the security domain and any MAE DHCP server IP address are obtained.

Table 4-29 Obtaining formal transmission configuration information from the MAE DHCP server

If...	Then...	Configuration Requirements for Network Equipment
The base station has obtained the interface IP address, logical IP address, and MAE DHCP server IP address NOTE The base station obtains the preceding IP addresses in different ways: • Interface IP address from the DHCP procedure • Logical IP address from MODE-CONFIG mode during IKE negotiation • MAE DHCP server IP address from the DHCP procedure or from MODE-CONFIG mode during IKE negotiation	• The base station uses the logical IP address as the source IP address and the IP address of the DHCP server as the destination IP address to unicast DHCP packets to all DHCP servers. Finally, only the DHCP server that has the identification information of the base station delivers the configuration to the base station. • The base station automatically configures an access control list (ACL) rule in Any to Any mode that allows DHCP packets to reach the base station.	See Table 4-30.

If...	Then...	Configuration Requirements for Network Equipment
The base station has obtained the interface IP address and MAE DHCP server IP address, but not the logical IP address	- The base station uses the interface IP address as the source IP address and the IP address of the DHCP server as the destination IP address to unicast DHCP packets to all DHCP servers. Finally, only the DHCP server that has the identification information of the base station delivers the configuration to the base station. - The base station automatically configures an ACL rule that allows DHCP packets to reach the base station. In the ACL rule, the source IP address is the interface IP address and the destination IP address is the IP address of the MAE DHCP server. If there are multiple internal DHCP servers, the corresponding ACL rule is generated when each DHCP server is connected.	See Table 4-31 .

Table 4-30 Configuration requirements for network equipment (1)

Network Equipment	Requirement
Public DHCP server	- Is configured with one to eight MAE DHCP server IP addresses only if the SeGW is not configured with any MAE DHCP server IP address. - If the SeGW is configured with the IP address of the MAE DHCP server, the preceding configuration is not required. - For detailed configurations, see 4.3.3.3 Configuration Requirements for the Public DHCP Server.

Network Equipment	Requirement
SeGW	- The MODE-CONFIG mode is supported to deliver temporary logical IP addresses and the IP address of the MAE DHCP server to the base station. Alternatively, only temporary logical IP addresses can be delivered. In this case, the public DHCP server needs to send the MAE DHCP server IP address. It is recommended that the operator plan all temporary logical IP addresses for accessing the trusted domain on the same network segment and on a different network segment from the OM IP address of the base station. - Automatically generates an ACL rule in Temporary Logical IP to Any mode after using the MODE-CONFIG mode to send the MAE DHCP server IP address. This eliminates the need to manually configure associated ACL rules. If an ACL rule is manually configured of which the source IP address is the temporary logical IP address for accessing the trusted domain, the IP addresses of all MAE DHCP servers must be on the network segment defined by this ACL rule.
All equipment between the base station and the MAE DHCP server	- Is configured with the firewall policy or the packet filtering policy to allow the transmission of packets with 67 or 68 as the source and destination UDP port number. - Is configured with a route of which the destination IP address is the logical IP address of the base station or the destination network segment is on the network segment of the base station. This enables the routing of related packets to the SeGW.
MAE DHCP server	Is configured with a route of which the destination IP address is the logical IP address of the base station.

Table 4-31 Configuration requirements for network equipment (2)

Network Equipment	Requirement
Public DHCP server	Is configured with the parameters listed in 4.3.3.3 Configuration Requirements for the Public DHCP Server. The public DHCP server must be configured with IP addresses of the MAE DHCP server. A maximum of eight IP addresses can be configured.
All equipment between the base station and the MAE DHCP server	- Is configured with the firewall policy or the packet filtering policy to allow the transmission of packets with 67 or 68 as the source and destination UDP port number. - Is configured with a route whose destination IP address is the interface IP address of the base station or the IP address of the network segment.

Network Equipment	Requirement
MAE DHCP server	Is configured with a route whose destination IP address is the interface IP address of the base station.

The base station obtains transmission configuration information in IPsec networking scenarios differently from non-IPsec networking scenarios:

- The DHCP server can only be deployed on the MAE, not the base station controller. That is, the MAE DHCP server is used.
- The base station may obtain IP addresses of multiple DHCP servers. Therefore, the base station attempts to interact with each DHCP server until it finds the DHCP server that manages the base station.
- IPsec secures OMCH data. Therefore, the MAE DHCP server must send the IP address of the SeGW to the base station. The local name of the SeGW is optional. The local name can be used to authenticate the SeGW.

4.3.3.8 Establishing a Formal IPsec Tunnel

The SeGW IP address obtained from the MAE DHCP server may or may not be the same as the SeGW IP address obtained from the public DHCP server. In either case, the base station must negotiate about an IKE SA and an IPsec SA with the SeGW before establishing a formal tunnel to the SeGW. The SeGW is identified by the SeGW IP address in the configuration information from the MAE DHCP server.

The procedure for establishing a formal IPsec tunnel differs from the procedure for establishing a temporary IPsec tunnel as follows:

- The base station uses the interface IP address delivered by the MAE DHCP server and SeGW IP address delivered by the MAE DHCP server for IKE SA and formal IPsec establishment negotiations between the base station and SeGW. During IPsec tunnel establishment, the base station automatically configures an ACL rule in OM IP to Any mode and the SeGW configures an ACL rule in Any to OM IP or Any to Any mode.
- The base station preferentially tries the IKE proposal algorithm and IPsec proposal algorithm with which the temporary IPsec tunnel was successfully established to establish the formal IPsec tunnel. If this fails, the base station follows the sequence described in [4.3.3.5 Establishing a Temporary IPsec Tunnel](#) to try other IKE proposal algorithms and IPsec proposal algorithms.

4.3.3.9 Establishing an OMCH

The procedure for establishing an OMCH in an IPsec networking scenario is similar to that in a non-IPsec networking scenario because the MAE does not need to know whether the base station uses the IPsec tunnel to access the MAE. The difference is that in an IPsec networking scenario, the MAE and base station must authenticate each other after the base station obtains operator-issued device certificates. The operator can choose SSL for authentication.

4.3.3.10 Configuration Requirements for Network Equipment

Table 4-32 lists the configuration requirements for NEs in IPsec networking scenario 1.

Table 4-32 Configuration requirements for NEs in IPsec networking scenario 1

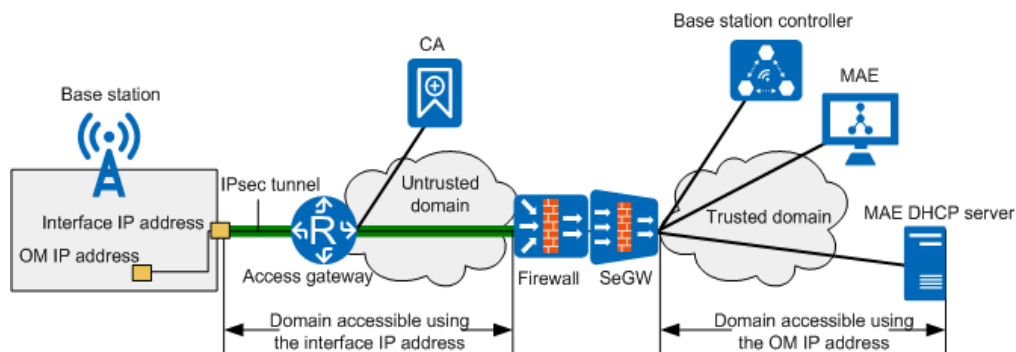
Network Equipment	Requirement
L2 device	- Allows the transmission of DHCP broadcast and unicast packets without filtering or modifying DHCP packets. - Is configured with correct VLAN information.
Next-hop gateway of the base station	- Is configured as the DHCP server or the DHCP relay agent and is configured with the IP address of the DHCP server. For the IP address requirements, see Table 4-45. - Is configured with a route with the destination IP address as the DHCP server IP address. - Is configured with routes with the destination addresses as the CA IP address and SeGW IP address.
L3 device	(NEs in the untrusted domain): Is configured with routes of which the destination addresses are the temporary and formal interface IP addresses of the base station, CA IP address, and SeGW IP address. (NEs in the trusted domain): Is configured with three routes of which the destination addresses are the OM IP address of the base station, MAE IP address, and FTP server IP address.
MAE	Is configured with a route of which the destination IP address is the OM IP address of the base station.
MAE DHCP server	Is configured with a route of which the destination IP address is that of the DHCP relay agent when the SeGW serves as the DHCP relay agent. If the SeGW does not serve as the DHCP relay agent, the MAE DHCP server is configured with a route of which the destination IP address is the temporary interface IP address of the base station.
FTP server	- Is configured with a route of which the destination IP address is the OM IP address of the base station. - Stores software and configuration file of the base station in a specified directory. - Provides access rights, such as the user name and password, for the base station.

Network Equipment	Requirement
SeGW	- Allows DHCP packets to be exchanged between the base station and the MAE. - Allows packets to be exchanged between the base station and the MAE over an OMCH and between the base station and the FTP server. - Is configured with security parameters listed in Table 4-16. - Is configured with ACL rules that allow the transmission of packets from the base station during a DHCP process. During the negotiation of a temporary IPsec tunnel, the base station automatically configures the ACL rule "<i>temporary interface IP address to any</i>". The SeGW is required to assign a high priority to the Traffic Selector corresponding to the temporary interface IP address so that the negotiated Child SA allows DHCPv4 traffic to pass. - Is configured with an "any to any" ACL rule or "any to base station OM IP" ACL rule. - Is configured with related IP address pool and assignment rules if the SeGW must assign an IP address for accessing the trusted domain or a DHCP server IP address to the base station. - Is configured with operator-issued CA certificates and the SeGW certificates.
CA	- Is configured with the IP address that can be accessed by NEs in the untrusted domain. - Is configured with Huawei-issued CA root certificates.

4.3.4 Automatic OMCH Establishment in IPsec Networking Scenario 2

4.3.4.1 Introduction to OMCH Networking

[Figure 4-31](#) shows OMCH networking in IPsec networking scenario 2. In this networking scenario, all packets except DHCP packets are protected by IPsec.

Figure 4-31 OMCH networking in scenario 2

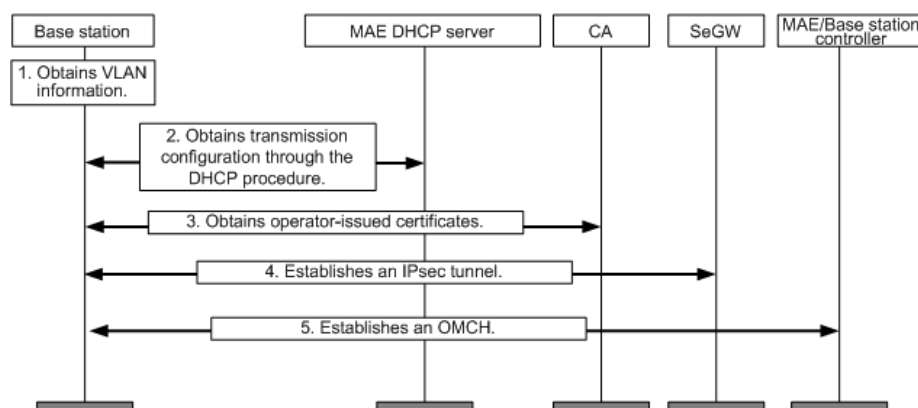
This networking has the following characteristics:

- An MAE DHCP server in the trusted domain is deployed. IPsec does not secure DHCP packets. Using a DHCP process in the untrusted domain, the base station obtains its OM channel information, CA information, and SeGW information. The base station in the untrusted domain cannot directly access NEs in the trusted domain. IP packets are encrypted over the IPsec tunnel between the base station in the untrusted domain and the SeGW before being transmitted to the MAE/BSC in the trusted domain.
- A CA is deployed and provides digital certificates for the base station. During PnP-based base station deployment, the CA can be accessed through IP addresses of NEs in the untrusted domain (for example, the interface IP address of the base station).
- After the base station starts, it must apply to the CA for operator-issued digital certificates before connecting to the SeGW. The base station then negotiates the IPsec tunnel with the SeGW.
- Only IPv4 OMCH networking is supported.

4.3.4.2 Automatic OMCH Establishment Process

In IPsec networking scenario 2, the base station must obtain the base station IP address and CA IP address from the MAE DHCP server, and then obtain digital certificates from the CA.

[Figure 4-32](#) shows the automatic OMCH establishment procedure in IPsec networking scenario 2.

Figure 4-32 Automatic OMCH establishment process in IPsec networking scenario 2

1. The base station obtains VLAN information. For details, see [4.2.7 Obtaining VLAN Information for DHCP Packets](#).
2. The base station obtains required configuration information from the MAE DHCP server. The information includes the OM IP address of the base station, the CA IP address, and the SeGW IP address.
3. By using the configuration information obtained from the MAE DHCP server, the base station applies to the CA for an operator-issued device certificate. (For details about the certificate application procedure, see [4.3.3.4 Obtaining an Operator-Issued Device Certificate](#).) The base station then adds the obtained certificate to the default trusted certificate list for subsequent IPsec tunnel establishment and SSL authentication.
4. By using the configuration information obtained from the MAE DHCP server, the base station establishes a formal IPsec tunnel to the SeGW.
5. After the formal IPsec tunnel is established, the base station waits for the OMCH establishment request from the MAE/BSC and then establishes an OMCH to the MAE/BSC. Since the base station has obtained the operator-issued certificate, SSL authentication is supported between the MAE and base station.

NOTE

If an IPsec tunnel or OMCH fails to be established, the base station automatically restarts the automatic OMCH establishment procedure.

IPsec Redundancy Among Multiple SeGWs is not supported during base station deployment by PnP when multiple SeGWs are configured. The active SeGW must function properly during base station deployment by PnP.

4.3.4.3 Configuration Requirements for the MAE DHCP Server

The MAE DHCP server must be configured with the parameters (carried in DHCP packets) listed in [Table 4-33](#) as well as the parameters listed in [Table 4-8](#). These parameters are contained in subcodes of Option 43 in DHCP packets.

Table 4-33 Parameters specific to the MAE DHCP server in IPsec networking scenario 2

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
SeGW information	Serving SeGW IP	20	4	Mandatory. It is the IP address of the SeGW in IPsec networking scenarios.	DHCPOFFER DHCPACK
	Serving SeGW Local Name	32	1–32	Mandatory. It is the local name of the serving SeGW. It is provided by the base station to authenticate the serving SeGW in IPsec networking scenarios.	
	ACLRule Destination IP Address	81	4	Optional. During base station deployment by PnP, an ACL is specified for the base station to match the OM IP address segment when an IPsec tunnel is being established between the base station and the SeGW.	DHCPOFFER DHCPACK
	ACLRule Destination Subnet Mask	82	4	Optional. During base station deployment by PnP, an ACL is specified for the base station to match the OM IP address segment when an IPsec tunnel is being established between the base station and the SeGW.	DHCPOFFER DHCPACK

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
	Rsa Signature Hash Algorithm	70	1	Optional. It is the hash algorithm used by the RSA digital signature. - The value 3 indicates SHA1. - The value 14 indicates SHA2_256. - The value 15 indicates SHA2_384. This parameter is valid only when IKEPROPOSAL.AUTHMETHOD (LTE eNodeB, 5G gNodeB) is set to IKE_CERT_SIG and the authentication method negotiation result is RSA digital signature. This parameter does not apply to IKEv1. This parameter is required when the SeGW does not support RFC 7427. If this parameter is not set, the base station uses the SHA256 algorithm. The SHA1 algorithm is considered insecure in the industry. If the algorithm is used, the transmitted data may be forged by attackers. The SHA256 or SHA384 algorithm is recommended.	DHCPOFFER DHCPACK
CA information	CA URL	44	1-128	Mandatory. In IPsec networking scenarios, the base station needs to obtain operator-issued certificates from this URL. This URL must be reachable in the untrusted domain.	DHCPOFFER DHCPACK

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
	CA Name	38	1-127	It is the name of the preceding CA.	
	Certificate Key Type	45	1	Optional. It indicates the type of the certificate key. - The value 0 indicates RSA_2048. - The value 1 indicates RSA_3072. - The value 2 indicates RSA_4096. - The value 3 indicates ECDSA_P256. - The value 4 indicates ECDSA_P384. - The value 5 indicates ED25519. The default value is 0.	

4.3.4.4 Configuration Requirements for Network Equipment

Table 4-34 lists the configuration requirements for network equipment in IPsec networking scenario 2.

Table 4-34 Configuration requirements for network equipment in IPsec networking scenario 2

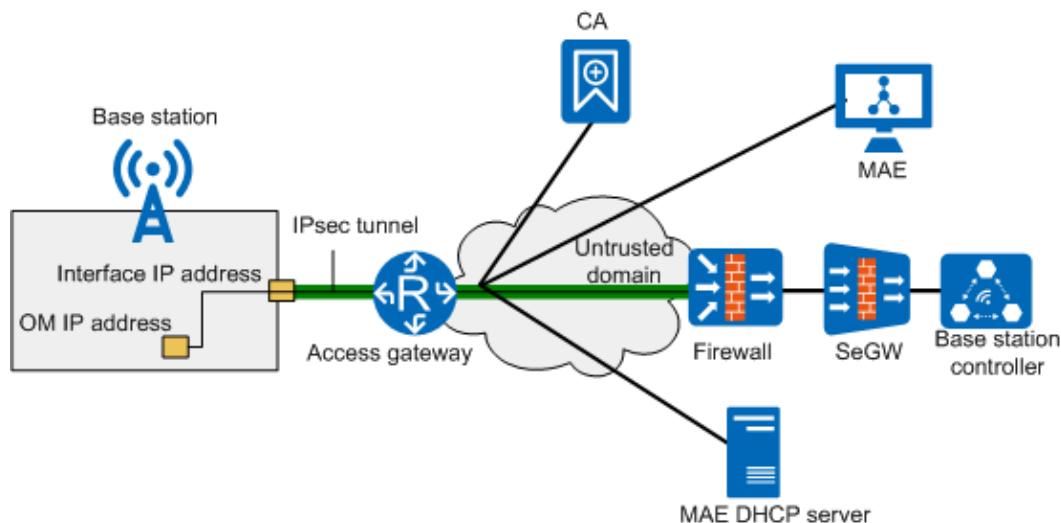
Network Equipment	Requirement
L2 devices	- Allow the transmission of DHCP broadcast and unicast packets and do not allow DHCP packet filtering or modification over the L2 network. - Are configured with correct VLAN information.
Next-hop gateway of the base station	- Is configured as the DHCP relay agent and is configured with the IP address of the DHCP server. For the IP address requirements, see Table 4-45. - Is configured with routes of which the destination IP addresses are the DHCP server IP address, CA IP address, and SeGW IP address.

Network Equipment	Requirement
L3 devices	• (NEs in the untrusted domain) Are configured with routes to the interface IP addresses of the base station and routes to the CA and the SeGW. • (NEs in the trusted domain) Are configured with routes of which the destination IP addresses are the OM IP address of the base station, MAE IP address, and FTP server IP address, respectively.
MAE	Is configured with a route of which the destination IP address is the OM IP address of the base station.
MAE DHCP server	Is configured with a route of which the destination IP address is the DHCP relay agent IP address.
SeGW	• Allows packets to be exchanged between the base station and the MAE over an OMCH and between the base station and the FTP server. • Is configured with security parameters for site deployment listed in Table 4-17, Table 4-24, and Table 4-34. • Is configured with an "any to any" or "any to base station OM IP" ACL rule. • Is configured with operator-issued CA certificates and the SeGW certificates.
CA	• Is configured with the following: An IP address that can be accessed by NEs in the untrusted domain. • Is configured with Huawei-issued CA root certificates.

4.3.5 Automatic OMCH Establishment in IPsec Networking Scenario 3

4.3.5.1 Introduction to IPsec Networking Scenario 3

[Figure 4-33](#) shows IPsec networking scenario 3 for the OM channel networking. In this networking scenario, IPsec secures service data and signaling data, but not DHCP data or OM data.

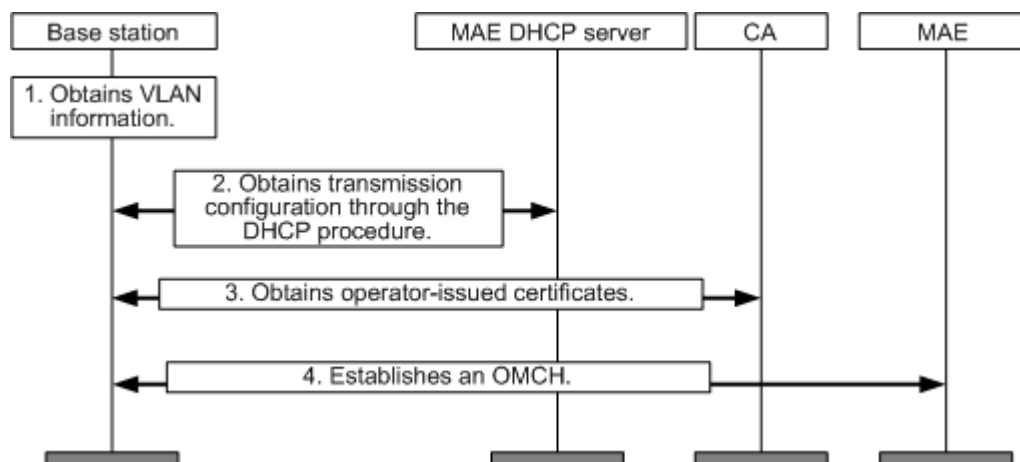
Figure 4-33 OMCH networking in scenario 3

This networking has the following characteristics:

- An MAE DHCP server is deployed as the only DHCP server in the network. The base station obtains the OMCH configuration data and CA configuration data from the MAE DHCP server. IPsec does not secure DHCP packets.
- IPsec does not secure OMCH data. The base station uses the OM IP address to access NEs in the untrusted domain. IPsec tunnels established between the base station and the SeGW are used to secure signaling and service data.
- Either party involved in IPsec negotiation uses digital certificates or PSK to authenticate the other party.
- A CA is required if digital certificates are used for authentication. After the base station starts, it must apply to the CA for operator-issued digital certificates before connecting to the SeGW. During base station deployment, the CA can be accessed by NEs in the untrusted domain.

4.3.5.2 Automatic OMCH Establishment Process

Figure 4-34 shows the automatic OMCH establishment procedure in IPsec networking scenario 3.

Figure 4-34 Automatic OMCH establishment process in IPsec networking scenario 3

1. The base station obtains VLAN information. For details, see [4.2.7 Obtaining VLAN Information for DHCP Packets](#).
2. The base station obtains the OMCH configuration data and CA configuration data from the DHCP server. If the base station uses the PSK for authentication, the base station does not need to obtain the CA configuration data. If the base station uses digital certificates for authentication, the base station must obtain CA configuration data.
3. The base station applies to the CA for an operator-issued device certificate if it has obtained CA information. (For details about the certificate application procedure, see [4.3.3.4 Obtaining an Operator-Issued Device Certificate](#).) The base station then adds the obtained certificate to the default trusted certificate list for subsequent IPsec tunnel establishment and SSL authentication.
4. Based on the configuration information obtained from the MAE DHCP server, the base station establishes an OMCH to the MAE/BSC. Since the base station has obtained the operator-issued certificate, SSL authentication is supported between the MAE and base station.

NOTE

After the OMCH is established, the base station obtains the formal configuration information and makes the configuration take effect. The base station is then restarted and establishes an IPsec tunnel to the SeGW to secure services and signaling.

4.3.5.3 Configuration Requirements for the MAE DHCP Server

If the base station uses digital certificates for authentication, the MAE DHCP server must be configured with the parameters (carried in DHCP packets) listed in [Table 4-35](#) as well as the parameters listed in [Table 4-8](#). These parameters are contained in subcodes of Option 43 in DHCP packets.

Table 4-35 Parameters specific to the MAE DHCP server in IPsec networking scenario 3

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
CA information	CA URL	44	1-128	Mandatory. In IPsec networking scenarios, the base station needs to obtain operator-issued certificates from this URL. This URL must be accessible by the NEs in the untrusted domain. That is, the OMCH local IP address that the base station obtains from the MAE DHCP server must be accessible.	DHCPOFFER DHCPACK
	CA Name	38	1-127	Mandatory. It is the name of the preceding CA.	

Parameter Category	Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
	Certificate Key Type	45	1	Optional. It indicates the type of the certificate key. • The value 0 indicates RSA_2048. • The value 1 indicates RSA_3072. • The value 2 indicates RSA_4096. • The value 3 indicates ECDSA_P256. • The value 4 indicates ECDSA_P384. • The value 5 indicates ED25519. The default value is 0.	

4.3.5.4 Configuration Requirements for Network Equipment

Table 4-36 lists the configuration requirements for network equipment in IPsec networking scenario 3.

Table 4-36 Configuration requirements for network equipment in IPsec networking scenario 3

Network Equipment	Requirement
L2 device	• Allows the transmission of DHCP broadcast and unicast packets without filtering or modifying DHCP packets. • Is configured with correct VLAN information.

Network Equipment	Requirement
Next-hop gateway of the base station	- Is enabled with the DHCP relay agent function and configured with the IP address of the DHCP server. For the IP address requirements, see Table 4-45. If an NAT server is deployed before the MAE, the IP address of the MAE must be converted by the NAT server. - Is configured with a route of which the destination IP address is the DHCP server IP address. - Is configured with a route of which the destination IP address is the OM IP address of the base station. This occurs if the OM IP address is not the same as the interface IP address of the base station. - Is configured with a route of which the destination IP address is the CA IP address.
L3 device	(NE in the untrusted domain) Is configured with routes of which the destination IP addresses are the interface IP address of the base station, OM IP address, MAE IP address, FTP server IP address, and CA IP address, respectively. (NE in the trusted domain) Is configured with routes of which the destination IP addresses are the OM IP address of the base station, MAE IP address, and FTP server IP address, respectively.
MAE	Is configured with a route of which the destination IP address is the OM IP address of the base station.
MAE DHCP server	Is configured with a route of which the destination IP address is the DHCP relay agent IP address.
CA	- Is configured with the IP address that can be accessed by NEs in the untrusted domain. - Is configured with Huawei-issued CA root certificates.

4.4 Automatic OMCH Establishment by the Separate-MPT Multimode Base Station

4.4.1 OMCH Networking

A separate-MPT multimode base station can use independent transmission or common transmission. When independent transmission is used, the OMCH establishment process is the same that for a single-mode base station. This section describes only the OMCH establishment process in common transmission scenarios. Only IPv4 co-transmission is supported.

Boards in a separate-MPT multimode base station can communicate with each other through panel interconnection or backplane interconnection. Generally, the

transmission board of a certain mode provides a shared transmission port for connecting to the transport network. The base station in this mode is called an upper-level base station, and base stations in the other modes are called lower-level base stations. The upper-level base station acts as the DHCP relay agent of lower-level base stations and needs to provide independent transmission through the panel.

Figure 4-35 shows the OMCH networking for the separate-MPT multimode base station that uses panel-based interconnection. The upper-level base station provides two transmission ports, one for panel-based interconnection (lower transmission port) and the other for connecting to the transport network (upper transmission port).

Figure 4-35 OMCH networking for the separate-MPT multimode base station that uses panel-based interconnection

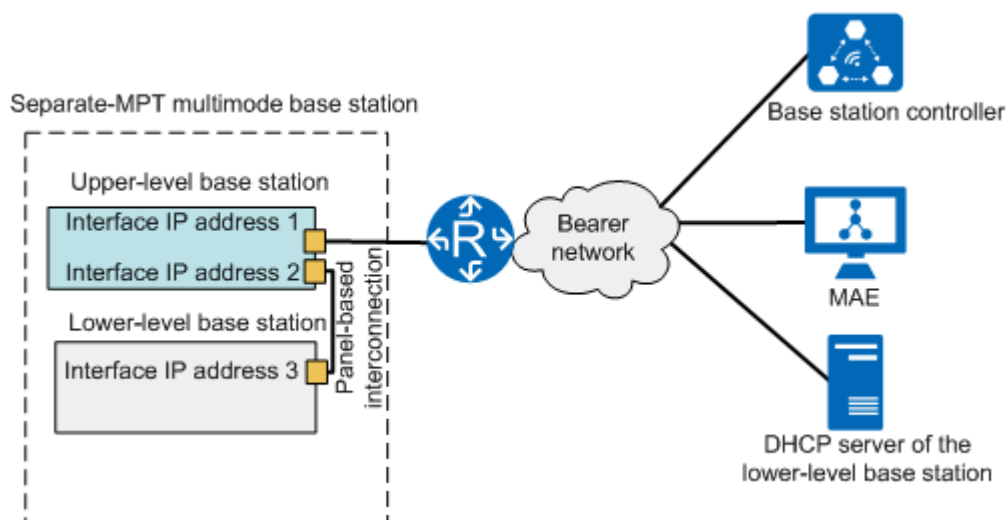
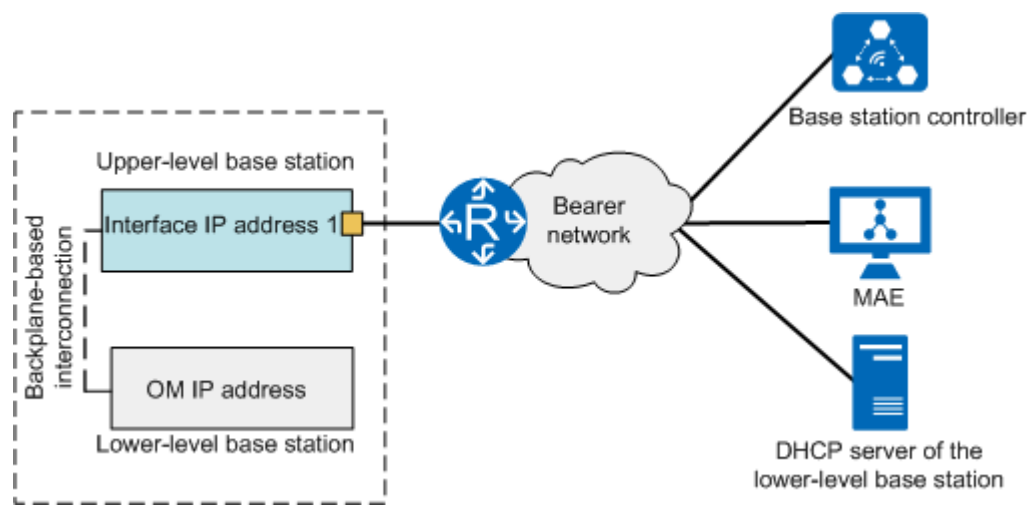


Figure 4-36 shows the OMCH networking for the separate-MPT multimode base station that uses backplane-based interconnection. For backplane-based interconnection, an end-to-end tunnel needs to be planned between the upper-level base station and each of its lower-level base station.

Figure 4-36 OMCH networking for the separate-MPT multimode base station that uses backplane-based interconnection

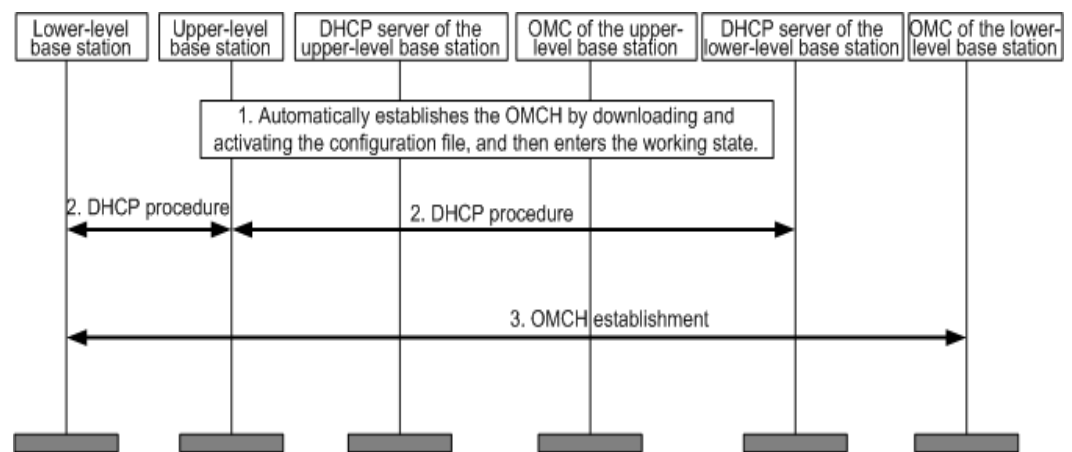


The automatic OMCH establishment procedure for the separate-MPT base station is similar to the respective automatic OMCH establishment procedure for each single-mode base station. Lower-level base stations can start the automatic OMCH establishment procedure only after the upper-level base station completes the procedure. This section describes the differences in the procedures between the separate-MPT base station and the single-mode base station.

4.4.2 Automatic OMCH Establishment Process

Figure 4-37 shows the automatic OMCH establishment process for the separate-MPT multimode base station.

Figure 4-37 Automatic OMCH establishment process



- 1. The upper-level base station has the same OMCH establishment process as a single-mode base station. Then the upper-level base station obtains the software and configuration file from the MAE/BSC over the established OMCH. The upper-level base station activates the software and configuration file and then enters the working state. For details about the automatic OMCH establishment for a single-mode base station, see [4.3 Automatic OMCH](#)

Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations.

2. Each lower-level base station exchanges DHCP packets with the DHCP relay agent (upper-level base station) and the DHCP server to obtain the transmission configuration.
3. Each lower-level base station establishes an OMCH to the MAE/BSC.

The DHCP servers of the upper-level base station and lower-level base stations can be deployed on the same NE or different NEs.

4.4.3 Configuration Requirements for the DHCP Server

Each mode in a separate-MPT multimode base station has almost the same configuration requirements for the DHCP server as a single-mode base station. For details about the configuration requirements for the DHCP server of a single-mode base station, see [4.3 Automatic OMCH Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations](#). This section describes only the differences in the configuration requirements for the DHCP server.

[Table 4-37](#) lists the additional parameter settings on DHCP servers of lower-level base stations. [Table 4-38](#) lists the additional parameter settings on DHCPv6 servers of lower-level base stations.

Table 4-37 Additional parameter settings on DHCP servers of lower-level base stations

Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
OM Bearing Board	250	1	Mandatory. Value: 0: An OMCH is established on the panel. 1: An OMCH is established on the backplane. Set this parameter to 0 for the lower-level base station when the separate-MPT multimode base station uses panel-based interconnection. Set this parameter to 1 for the lower-level base station when the separate-MPT multimode base station uses backplane-based interconnection.	DHCPOFFER DHCPACK

Table 4-38 Additional parameter settings on DHCPv6 servers of lower-level base stations

Parameter Name	Subcode	Length (Byte)	Description	DHCP Packet Involved
OM Bearing Board	250	1	Mandatory. Value: • 0: An OMCH is established on the panel. • 1: An OMCH is established on the backplane. Set this parameter to 0 for the lower-level base station when the separate-MPT multimode base station uses panel-based interconnection. Set this parameter to 1 for the lower-level base station when the separate-MPT multimode base station uses backplane-based interconnection.	ADVERTISE REPLY

 NOTE

If the source version of a lower-level base station is SRAN18.1 or later, the lower-level base station can enable SSL to establish an OM channel after applying for a certificate, without depending on certificate sharing. Certificate sharing is not recommended for these lower-level base stations. It is recommended that CA information be configured for these lower-level base stations during base station deployment so that the lower-level base stations can apply for certificates and then establish OM channels. For details about how to apply for an operator-issued certificate from the CA, see [4.3.3.4 Obtaining an Operator-Issued Device Certificate](#).

4.4.4 Configuration Requirements for Network Equipment

Each RAT in a separate-MPT multimode base station that has a route to the DHCP server has similar configuration requirements for network equipment to those of a single-mode base station. For details about these requirements, see [4.3 Automatic OMCH Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations](#). This section describes only the differences in configuration requirements.

The upper-level base station acts as the DHCP relay agent to forward DHCP packets and as a router to forward OMCH and service packets for lower-level base stations. The transport network for the upper-level base station must forward DHCP packets from the DHCP servers of lower-level base stations. The upper-level base station and its transport network must be configured with data listed as follows:

- Upper-level base station
 - Is enabled with the DHCP relay agent function and configured with IP addresses of the DHCP servers of lower-level base stations. For the IP address requirements, see [4.5.2 Impact of MAE Deployment on Base Station Deployment by PnP](#).
 - Is configured with the IP address of the transmission interface (used for panel-based interconnection) provided by the upper-level base station.
 - Is configured with uplink routes to the DHCP servers of lower-level base stations and to the peer IP addresses of lower-level base stations. If the lower-level base station is the GBTS/NodeB, uplink routes to the base station controller and MAE must be configured. If the lower-level base station is the eNodeB/gNodeB, uplink routes to the MAE and MME/S-GW must be configured.
 - Is configured with routes of which the source IP address is the IP address of the DHCP relay agent if source-based IP routing is configured for the upper-level base station.

 NOTE

In scenarios where backplane co-transmission is applied, the IP address of the DHCP relay agent must be configured. This applies if the IP address of the panel port connecting to the transport network is to be used as the IP address of the DHCP relay agent.

- Is configured with downlink routes to the OM IP address and service IP address of the lower-level base station.
 - Is configured with VLANs on the transmission interface connecting to the lower-level base station if VLANs are deployed between cascaded base stations. In this case, the network segment configured by **NEXTHOIP** (next-hop IP address) and **MASK** (subnet mask) must overlap with the network segment configured by the interconnection interface IP address. Single VLAN mode is recommended for both upper- and lower-level base stations.
 - If the DHCP packets and OM data of lower-level base stations are secured by the IPsec tunnel of the upper-level base station, security parameters must be configured on the upper-level base station for the passerby flows of lower-level base stations. The security parameters include the packet filtering rules, ACL rules, IPsec proposal, and IKE proposal.
- All devices on the transport network for the upper-level base station
 - Are configured with routes to the DHCP servers of lower-level base stations.
 - Are configured with routes to the IP address of the DHCP relay agent of the upper-level base station.
 - Are configured with routes to the OM IP address and service IP address of the lower-level base station.

- MAE/BSC
Is configured with routes to the OM IP address of the lower-level base station.
- DHCP servers of lower-level base stations
Are configured with routes to the IP address of the DHCP relay agent of the upper-level base station.
- Lower-level base stations
 - Are configured with routes destined for the MAE/BSC IP address.
 - Are configured with interface IP addresses that are on the same network segment as the IP address of the interface on the upper-level base station for interconnection.

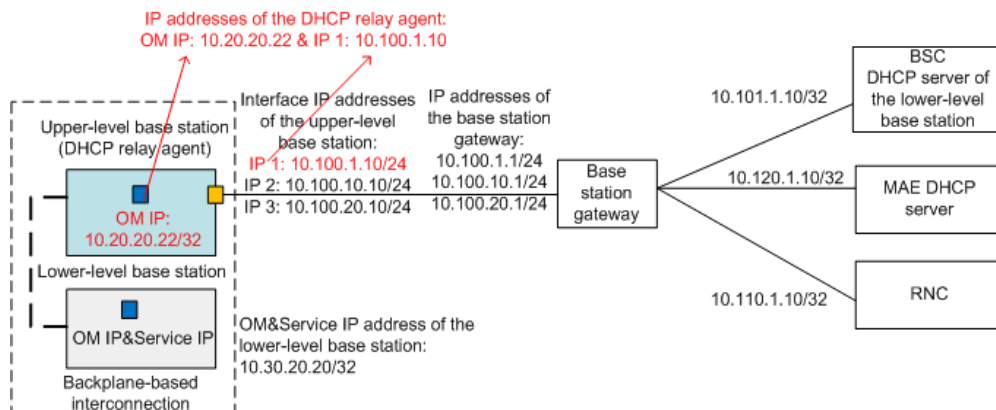
If ***DHCPRELAYIP (LTE eNodeB, 5G gNodeB)*** is not manually configured (for details about how to manually configure it, see [4.2.4.2.3 DHCPv4 Client and DHCPv4 Server](#) and [4.2.4.3.3 DHCPv6 Client and DHCPv6 Server](#)), the DHCP relay agent IP address of the upper-level base station must comply with the following rules:

- **Backplane-based Interconnection**
The DHCP relay agent IP addresses are the OM IP address of the upper-level base station and the IP address of the uplink port. If the uplink port of the upper-level base station has multiple interface IP addresses, the interface IP address used as the DHCP relay agent IP address is on the same network segment as the next-hop IP address of the route to the DHCP server IP address of the lower-level base station.
- **Panel-based Interconnection**
The DHCP relay agent IP addresses are the OM IP address of the upper-level base station and the IP address of the downlink port. If the downlink port of the upper-level base station has multiple interface IP addresses, the interface IP address used as the DHCP relay agent IP address varies with scenarios.
 - If VLANs are deployed for neither the OMCH or the service link on the lower-level base station, the interface IP addresses of the lower transmission port that is not configured with VLANs are used.
 - If VLANs are deployed for both the OMCH and the service link on the lower-level base station, the interface IP address that is used for deploying VLANs for the OMCH is used.
 - If VLANs are deployed for services but not for the OMCH on the lower-level base station, the interface IP addresses for which no VLAN is deployed are used.

In both backplane- and panel-based interconnection scenarios, if there are active and standby OMCHs on the upper-level base station, the OM IP address in use will be used as the IP address of the DHCP relay agent. For example, if the OM IP address of the standby OMCH is in use, it will be used as the IP address of the DHCP relay agent.

Backplane-based Interconnection

Figure 4-38 shows examples of DHCP relay agent's IP addresses and route deployment in backplane-based interconnection.

Figure 4-38 Examples of DHCP relay agent's IP addresses and route deployment in GBTS & NodeB backplane-based interconnection

When the old transmission configuration model is used

(**GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**), the configurations are as follows:

- IP addresses of the DHCP relay agent and route from the DHCP server to the IP address of the DHCP relay agent
 - IP addresses of the DHCP relay agent are 10.20.20.22 (OM IP address) and 10.100.1.10 (IP address 1).
 - The destination IP address of the route from the DHCP server to the IP address of the DHCP relay agent is 10.100.1.10 or 10.20.20.22.
- IP routes on the upper-level base station
 - Run the following command to configure a route to the DHCP server (BSC) of the lower-level base station:

```
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.101.1.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.1";
```
 - Run the following command to configure a route to the MAE IP address:

```
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.120.1.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.1";
```
 - Run the following command to configure a route to the RNC service IP address:

```
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.110.1.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.20.1";
```
 - Run the following command to configure a route to the OM IP address of the lower-level base station (the service IP address is the same as the OM IP address):

```
ADD IPRT: RTIDX=1, SN=6, SBT=BACK_BOARD, DSTIP="10.30.20.20",
DSTMASK="255.255.255.255", RTTYPE=IF, IFT=TUNNEL, IFNO=1;
```
- IP route on the lower-level base station

Run the following command to configure a route to the DHCP server:

```
ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.101.1.10", DSTMASK="255.255.255.255",
RTTYPE=OUTIF, ITFTYPE=TUNNEL, IFNO=1;
```
- IP route on the BSC

Run the following command to configure a route to the lower-level base station:

```
ADD IPRT: RTIDX=1, SRN=1, SN=7, DSTIP="10.30.20.20", DSTMASK="255.255.255.255",
NEXTHOPTYPE=Gateway, NEXTHOP="10.150.1.10", PRIORITY=HIGH;
```

When the new transmission configuration model is used

(**GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**), the configurations are as follows:

- IP addresses of the DHCP relay agent and route from the DHCP server to the IP address of the DHCP relay agent
 - IP addresses of the DHCP relay agent are 10.20.20.22 (OM IP address) and 10.100.1.10 (IP address 1).
 - The destination IP address of the route from the DHCP server to the IP address of the DHCP relay agent is 10.100.1.10 or 10.20.20.22.
- IP routes on the upper-level base station
 - Run the following command to configure a route to the DHCP server (BSC) of the lower-level base station:
`ADD IPROUTE4: RTIDX=1, DSTIP="10.101.1.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.1";`
 - Run the following command to configure a route to the MAE IP address:
`ADD IPROUTE4: RTIDX=1, DSTIP="10.120.1.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.10.1";`
 - Run the following command to configure a route to the RNC service IP address:
`ADD IPROUTE4: RTIDX=1, DSTIP="10.110.1.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.20.1";`
 - Run the following command to configure a route to the OM IP address of the lower-level base station (the service IP address is the same as the OM IP address):
`ADD IPROUTE4: RTIDX=1, DSTIP="10.30.20.20", DSTMASK="255.255.255.255", RTTYPE=IF, PT=TUNNEL, PORTID=1;`
- IP route on the lower-level base station

Run the following command to configure a route to the DHCP server:

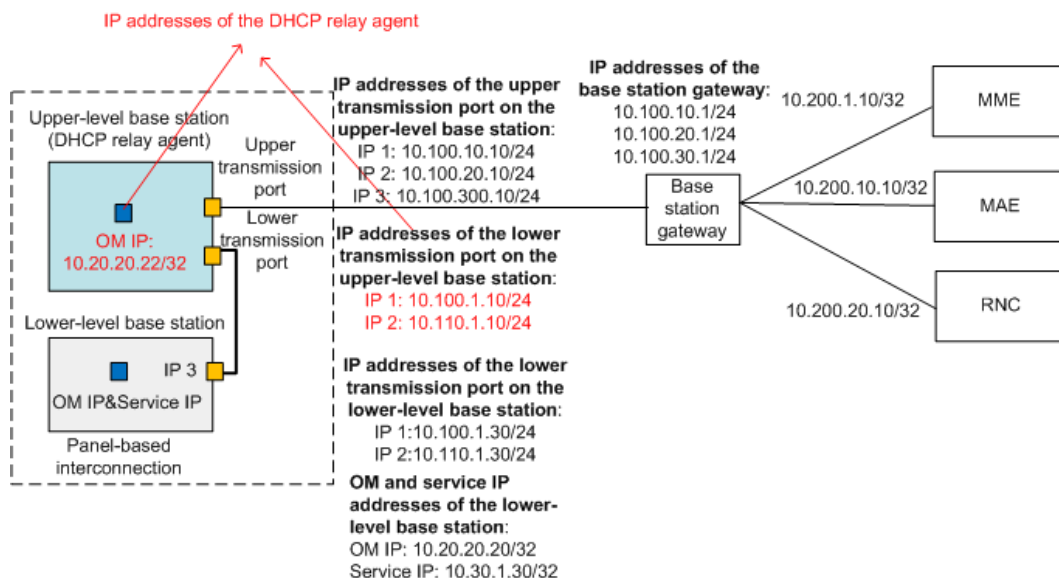
`ADD BTSIPRT: IDTYPE=BYID, BTSID=10, RTIDX=1, DSTIP="10.101.1.10", DSTMASK="255.255.255.255", RTTYPE=OUTIF, IFType=TUNNEL, IFNO=1;`
- IP route on the BSC

Run the following command to configure a route to the lower-level base station:

`ADD IPRT: RTIDX=1, SRN=1, SN=7, DSTIP="10.30.20.20", DSTMASK="255.255.255.255", NEXTHOPTYPE=Gateway, NEXTHOP="10.150.1.10", PRIORITY=HIGH;`

Panel-based Interconnection

Figure 4-39 shows examples of DHCP relay agent's IP addresses and route deployment in panel-based interconnection.

Figure 4-39 Examples of DHCP relay agent's IP addresses and route deployment in panel-based interconnection

When the old transmission configuration model is used (**GTRANS PARA. TRANSCFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**), the configurations are as follows:

- IP addresses of the DHCP relay agent and route from the DHCP server to the IP address of the DHCP relay agent
 - If VLANs have been deployed for neither the OMCH nor the service link on the lower-level base station:

IP addresses of the DHCP relay agent are 10.20.20.22 (OM IP address), 10.100.1.10 (IP address 1), and 10.110.1.10 (IP address 2).

The destination IP address of the IP route to the DHCP relay agent is 10.20.20.22, 10.100.1.10, or 10.110.1.10.
 - If VLANs are deployed for both the OMCH and the service link on the lower-level base station:

IP addresses of the DHCP relay agent are 10.20.20.22 (OM IP address) and 10.100.1.10 (IP address 1), either of which can be the destination IP address of the route to the IP address of the DHCP relay agent. To deploy VLANMAPs for the upper-level base station, perform the following operations accordingly:

```
//Configuring VLANs for the OMCH on the lower-level base station
ADD VLANMAP: VRFID=0, NEXTHOIP="10.100.1.30", MASK="255.255.255.0",
VLANMODE=SINGLEVLAN, VLANID=10, SETPRIO=DISABLE;
//Configuring VLANs for services on the lower-level base station
ADD VLANMAP: VRFID=0, NEXTHOIP="10.110.1.30", MASK="255.255.255.0",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
```

The destination IP address of the IP route to the DHCP relay agent is 10.20.20.22 or 10.100.1.10.
 - If VLANs have been deployed for services but not for the OMCH on the lower-level base station:

IP addresses of the DHCP relay agent are 10.20.20.22 (OM IP address) and 10.100.1.10 (IP address 1), either of which can be the destination IP

address of the route to the IP address of the DHCP relay agent. To deploy VLANMAPs for the upper-level base station, perform the following operations accordingly:

```
//Configuring VLANs for services on the lower-level base station
ADD VLANMAP: VRFIDX=0, NEXTHOPIP="10.110.1.30", MASK="255.255.255.0",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
```

The destination IP address of the IP route to the DHCP relay agent is 10.20.20.22 or 10.100.1.10.

- IP routes on the upper-level base station
 - Run the following command to configure a route to the MAE IP address:
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.200.10.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.1";
 - Run the following command to configure a route to the RNC service IP address:
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.200.20.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.20.1";
 - Run the following command to configure a route to the MME:
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.200.1.10",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.30.1";
 - Run the following command to configure a route to the OM IP address of the lower-level base station:
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.20.20.20",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.30";
 - Run the following command to configure a route to the service IP address of the lower-level base station:
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.30.1.30",
DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.110.1.30";
- IP route on the lower-level base station

Run the following command to configure a route to the MAE:
ADD IPRT: RTIDX=1, SN=6, SBT=BASE_BOARD, DSTIP="10.200.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.100.1.10";
- Route from the MAE to the OM IP address of the lower-level base station:

The destination IP address of the route is 10.20.20.20, the destination subnet mask is 255.255.255.255, and the next-hop IP address is 10.100.11.10.

When the new transmission configuration model is used

(**GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**), the configurations are as follows:

- IP addresses of the DHCP relay agent and route from the DHCP server to the IP address of the DHCP relay agent
 - If VLANs are deployed for neither the OMCH nor the service link on the lower-level base station:

IP addresses of the DHCP relay agent are 10.20.20.22 (OM IP address), 10.100.1.10 (IP address 1), and 10.110.1.10 (IP address 2).

Any of these IP addresses can be the destination IP address of the route to the IP address of the DHCP relay agent.
 - If VLANs are deployed for both the OMCH and the service link on the lower-level base station:

The IP addresses of the DHCP relay agent are the OM IP address 10.20.20.22 and IP1 10.100.1.10.

To deploy VLANs for the upper-level base station, perform the following operations accordingly:

- Set VLANs based on the interface as follows:

//Configuring VLANs for the OMCH on the lower-level base station

```
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=10;
ADD IPADDR4: ITFID=0, IP="10.100.1.10", MASK="255.255.255.0";
```

//Configuring VLANs for services on the lower-level base station

```
ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=20;
ADD IPADDR4: ITFID=1, IP="10.110.1.10", MASK="255.255.255.0";
```

- Set VLANs based on the VLANMAP as follows:

//Configuring VLANs for the OMCH on the lower-level base station

```
ADD VLANMAP: VRFID=0, NEXTHOPIP="10.100.1.30", MASK="255.255.255.0",
VLANMODE=SINGLEVLAN, VLANID=10, SETPRIO=DISABLE;
```

//Configuring VLANs for services on the lower-level base station

```
ADD VLANMAP: VRFID=0, NEXTHOPIP="10.110.1.30", MASK="255.255.255.0",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
```

The destination IP address of the IP route to the DHCP relay agent is 10.20.20.22 or 10.100.1.10.

- If VLANs have been deployed for services but not for the OMCH on the lower-level base station:

The IP addresses of the DHCP relay agent are the OM IP address 10.20.20.22 and IP1 10.100.1.10.

To deploy VLANs for the upper-level base station, perform the following operations accordingly:

- Set VLANs based on the interface as follows:

//Configuring VLANs for services on the lower-level base station

```
ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=20;
ADD IPADDR4: ITFID=1, IP="10.110.1.10", MASK="255.255.255.0";
```

- Set VLANs based on the VLANMAP as follows:

//Configuring VLANs for services on the lower-level base station

```
ADD VLANMAP: VRFID=0, NEXTHOPIP="10.110.1.30", MASK="255.255.255.0",
VLANMODE=SINGLEVLAN, VLANID=20, SETPRIO=DISABLE;
```

The destination IP address of the IP route to the DHCP relay agent is 10.20.20.22 or 10.100.1.10.

- IP routes on the upper-level base station

- Run the following command to configure a route to the MAE IP address:

```
ADD IPROUTE4: RTID=1, DSTIP="10.200.10.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.100.1.1";
```

- Run the following command to configure a route to the RNC service IP address:

```
ADD IPROUTE4: RTID=1, DSTIP="10.200.20.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.100.20.1";
```

- Run the following command to configure a route to the MME:

```
ADD IPROUTE4: RTID=1, DSTIP="10.200.1.10", DSTMASK="255.255.255.255",
RTTYPE=NEXTHOP, NEXTHOP="10.100.30.1";
```

- Run the following command to configure a route to the OM IP address of the lower-level base station:

- ADD IPROUTE4: RTIDX=1, DSTIP="10.20.20.20", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.30";
- Run the following command to configure a route to the service IP address of the lower-level base station:

ADD IPROUTE4: RTIDX=1, DSTIP="10.30.1.30", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.110.1.30";
 - IP route on the lower-level base station
Run the following command to configure a route to the MAE:

ADD IPROUTE4: RTIDX=1, DSTIP="10.200.10.10", DSTMASK="255.255.255.255", RTTYPE=NEXTHOP, NEXTHOP="10.100.1.10";
 - Route from the MAE to the OM IP address of the lower-level base station:
The destination IP address of the route is 10.20.20.20, the destination subnet mask is 255.255.255.255, and the next-hop IP address is 10.100.11.10.

4.5 Application Restrictions

4.5.1 Configuration Requirements for Base Stations and Other Network Equipment

When a base station is to be deployed by PnP, configuration requirements for the base station and related DHCP servers must be met to ensure successful automatic OMCH establishment. If configuration requirements are not met, automatic OMCH establishment may fail, leading to a deployment failure. The requirements in IPv4 transmission and IPv6 transmission scenarios are different.

4.5.1.1 IPv4 Transmission

Configuration Requirements for Base Stations (Old Model)

When the old transmission configuration model is used (**GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**), the configurations requirements are described in the following tables.

Table 4-39 Requirements for the configuration file of the base station in IPsec networking scenarios (old model)

MO	Requirement
OMCH	If only one of the OMCH and the service link is secured by IPsec, the OMCH and the service link must use different IP addresses. Otherwise, an error may occur in the parameters of the DHCP server.

MO	Requirement
ACLRULE	The ACL rules must meet one of the following conditions. Otherwise, errors may occur when SeGW parameters of the DHCP server are exported from the MAE-Deployment. As a result, PnP-based site deployment fails. The configured ACL rule meets either of the following requirements: - The SIP (LTE eNodeB, 5G gNodeB) and DIP (LTE eNodeB, 5G gNodeB) parameters are set to 0.0.0.0, and the SWC (LTE eNodeB, 5G gNodeB) and DWC (LTE eNodeB, 5G gNodeB) parameters are set to 255.255.255.255. That is, both the source and destination IP addresses can be any address. - The SIP (LTE eNodeB, 5G gNodeB) is set to the OM IP address. The DIP (LTE eNodeB, 5G gNodeB) parameter is set to the IP address of the MAE, the IP address of the MAE network segment, or 0.0.0.0. Note that if the ACTION (LTE eNodeB, 5G gNodeB) parameter is set to DENY(Deny) in an ACL rule, IPsec tunnels do not secure OMCHs that are established during base station deployment.
IKEPROPOSAL/ IPSECPROPOSAL	Parameter settings in the IPSECPROPOSAL MO must be consistent with those described in Figure 4-30. Parameter settings in the IKEPROPOSAL MO must be consistent with those described in Table 4-17, Table 4-18, and Table 4-24. If the base station uses the IPsec tunnel pair topology, only the active tunnel supports base station deployment by PnP.
BFDSESSION	If the base station uses the IPsec tunnel pair topology, the BFD session cannot be bound to a route during the BFD session configuration.
ETHTRK	Ethernet link aggregation groups cannot be manually configured on peer L2 devices of the base station.
CA	- The CA must be accessible to devices in the untrusted domain. - In the case of base station deployment by PnP, the base station does not support the polling mode. When the CA is in polling mode, the certificate application of the base station may fail due to timeout.

Configuration Requirements for Base Stations (New Model)

When the new transmission configuration model is used

(**GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**), the configurations requirements are described in the following tables.

Table 4-40 Requirements for the configuration file of the base station in IPsec networking scenarios (new model)

MO	Requirement
OMCH	If only one of the OMCH and the service link is secured by IPsec, the OMCH and the service link must use different IP addresses. Otherwise, an error may occur in the parameters of the DHCP server.
ACLRULE	The ACL rules must meet one of the following conditions. Otherwise, errors may occur when SeGW parameters of the DHCP server are exported from the MAE-Deployment. As a result, PnP-based site deployment fails. The configured ACL rule meets either of the following requirements: - The SIP (LTE eNodeB, 5G gNodeB) and DIP (LTE eNodeB, 5G gNodeB) parameters are set to 0.0.0.0, and the SWC (LTE eNodeB, 5G gNodeB) and DWC (LTE eNodeB, 5G gNodeB) parameters are set to 255.255.255.255. That is, both the source and destination IP addresses can be any address. - The SIP (LTE eNodeB, 5G gNodeB) is set to the OM IP address. The DIP (LTE eNodeB, 5G gNodeB) parameter is set to the IP address of the MAE, the IP address of the MAE network segment, or 0.0.0.0. Note that if the ACTION (LTE eNodeB, 5G gNodeB) parameter is set to DENY(Deny), IPsec tunnels do not secure OMCHs that are established during base station deployment. IPsec tunnels secure the OMCHs established in other cases.
IKEPROPOSAL/ IPSECPROPOSAL	Parameter settings in the IPSECPROPOSAL MO must be consistent with those described in Figure 4-30. Parameter settings in the IKERPROPOSAL MO must be consistent with those described in Table 4-17, Table 4-18, and Table 4-24. If the base station uses the IPsec tunnel pair topology, only the active tunnel supports base station deployment by PnP.
BFD	If the base station uses the IPsec tunnel pair topology, the BFD session cannot be bound to a route during the BFD session configuration.
ETHTRUNK	Ethernet link aggregation groups cannot be manually configured on peer L2 devices of the base station.
CA	- The CA must be accessible to devices in the untrusted domain. - In the case of base station deployment by PnP, the base station does not support the polling mode. When the CA is in polling mode, the certificate application of the base station may fail due to timeout.

Configuration Requirements for Network Equipment

NOTE

When configuring or modifying the information about the built-in DHCP server on the MAE, ensure that the destination IP address of the route for site deployment or the IP address of the network segment is correct.

Table 4-41 Requirements for the DHCP server

No.	Requirement
1	The public DHCP server can be configured with a maximum of eight MAE DHCP server IP addresses. If base stations of SRAN7.0, SRAN8.0, and later versions coexist on the network, the base stations of SRAN7.0 support only two built-in DHCP servers. If eight built-in DHCP servers are configured on the public DHCP server, base station deployment will fail. In this scenario, it is recommended that only two built-in DHCP servers be configured or base stations of SRAN7.0 be deployed in non-PnP mode.
2	If the main control board of the NodeB is a UMPT board, the base station ID configured on the DHCP server must be changed from being bound to the panel ESN (mapping subcode 43 in DHCP Option 43) to being bound to the backplane ESN (mapping subcode 1 in DHCP Option 43).

Table 4-42 Requirements for network equipment

No.	Requirement
1	VLAN isolation must be configured for the OM channel of the base station on the transport network. If VLAN aggregation is configured, base station deployment may fail.
2	In scenarios where Ethernet link aggregation is used, if LACP negotiation fails on the peer device, independent data forwarding in Ethernet port mode will be performed instead. As a result, base station deployment may fail.

4.5.1.2 IPv6 Transmission

Table 4-43 describes the configurations requirements.

Table 4-43 Requirements for the configuration file of the base station

MO	Requirement
OMCH	This MO is mandatory. If OMCHs of a base station use IPv6 transmission and the base station is configured with active and standby OMCHs, only the active OMCH can be configured for base station deployment by PnP. The active OMCH is the OMCH for which the OMCH.FLAG (LTE eNodeB, 5G gNodeB) parameter is set to MASTER(Master). The active OMCH must meet the following requirements: • If OMCH.PEERIP6 (LTE eNodeB, 5G gNodeB) is set to the MAE IPv6 address, the FTP server and MAE must be deployed on the same equipment or on the same network segment. The network segment specified by OMCH.PEERIP6PFXLEN (LTE eNodeB, 5G gNodeB) must cover the network segment where the IPv6 addresses of the MAE and FTP server reside. • The base station must be configured with a route whose destination IP address is on the same network segment as OMCH.PEERIP6 (LTE eNodeB, 5G gNodeB) or source-based routes from the next-hop IP address to the MAE and FTP server.
IROUTE6/ SRCIPROUT E6	If the OMCH is configured with active and standby routes, only the active route can be used for the base station deployment by PnP. The active route has a higher priority than the standby one. The smaller the number of the route priority, the higher the priority.
INTERFACE	If direct IPsec networking is used for services on the X2 or Xn/eXn interface, IPsec networking is not used for the OM channel, and the OMCH.IP6 (LTE eNodeB, 5G gNodeB) parameter references an IPv6 address that is an interface IP address, then the INTERFACE.VRFIDX (LTE eNodeB, 5G gNodeB) parameter referenced by the IP address must be different from the INTERFACE.VRFIDX (LTE eNodeB, 5G gNodeB) parameter referenced by the inner service IP address.
BFD	A BFD session is bound to a switchover route if the CATLOG parameter is set to RELIABILITY(Reliability) in active/standby route scenarios. If the base station uses a logical IP address as the OM IP address and the BFD session is bound to a switchover route, the base station cannot be deployed by PnP.

NOTE

When configuring or modifying the information about the built-in DHCPv6 server on the MAE, ensure that the destination IP address of the route for site deployment or the IP address of the network segment is correct.

Table 4-44 Requirements for network equipment

No.	Requirement
1	VLAN isolation must be configured for the OM channel of the base station on the transport network. If VLAN aggregation is configured, base station deployment may fail.
2	In scenarios where Ethernet link aggregation is used, if LACP negotiation fails on the peer device, independent data forwarding in Ethernet port mode will be performed instead. As a result, base station deployment may fail.

4.5.2 Impact of MAE Deployment on Base Station Deployment by PnP

During PnP deployment and subsequent commissioning, the base station needs to communicate with multiple application services of the MAE. These services include the DHCP service, FTP service, and OMCH management service.

The preceding three services can be deployed on different MAEs and use different IP addresses. Network planning and base station data configuration must ensure normal communication between the OM IP address of the base station and the IP addresses of these three services. The OMCH supports only one IP version at a time, either IPv4 or IPv6. The preceding services must run the same IP version as the OMCH.

Table 4-45 describes the impact of MAE deployment on automatic OMCH establishment.

Table 4-45 Impact of MAE deployment on automatic OMCH establishment

MAE Deployment	MAE Deployment Description	MAE Serving as the DHCP Server	MAE Serving as the OMCH	Requirement for the Base Station Deployment	Impact on the Network Configuration
Single-server system	- All application services are deployed on the same server. - The server (MAE) has only one IP address.	Single server	Single server	For details, see 4.3 Automatic OMCH Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations and 4.4 Automatic OMCH Establishment by the Separate-MPT Multimode Base Station .	For details, see 4.3 Automatic OMCH Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations and 4.4 Automatic OMCH Establishment by the Separate-MPT Multimode Base Station .
HA system	- The active and standby nodes have the same function and data on the two nodes are synchronized. - The active and standby nodes use the same IP address.	Active or standby node	Active or standby node	For details, see 4.3 Automatic OMCH Establishment for Single-mode Base Stations and Co-MPT Multimode Base Stations and 4.4 Automatic OMCH Establishment by the Separate-MPT Multimode Base Station .	

MAE Deployment	MAE Deployment Description	MAE Serving as the DHCP Server	MAE Serving as the OMCH	Requirement for the Base Station Deployment	Impact on the Network Configuration
SLS system/virtualization cluster	- The slave node only performs the NE management function. - The IP address of the master node is different from that of the slave node, and the IP addresses of the two nodes are in the same subnet.	Master node	Master or slave node	- The PEERIP parameter for the OMCH must be set to the IP address of the MAE that manages the base station. - If the OMCH is bound to a route, the route must be bound to the network segment of the MAE.	In IPsec networking scenario 1, the IP address of the built-in DHCP server configured on the public DHCP server must be the IP address of the master node. The SeGW must be configured with ACL rules related to the built-in DHCP server for the master node to allow DHCP packets from the built-in DHCP server to pass through. The SeGW must be configured with ACL rules which allow OM packets to pass for the MAE serving as the OMC. The DHCP server IP address configured on the DHCP relay must be the master node IP address of the MAE.

MAE Deployment	MAE Deployment Description	MAE Serving as the DHCP Server	MAE Serving as the OMCH	Requirement for the Base Station Deployment	Impact on the Network Configuration
Remote HA system	- The active and standby nodes are deployed in two locations. - The IP address of the active node is different from that of the standby node, and the IP addresses of the two nodes may not be in the same subnet.	Both the active and standby nodes	The MAE must serve as the DHCP server.	- The base station must be configured with routes to the two IP addresses or two network segments or source routes from the next-hop IP addresses to the two MAEs. - The PEERIP parameter for the OMCH of the base station must be set to the IP address of the MAE that serves as the DHCP server.	- In IPsec networking scenario 1, the IP address of the MAE DHCP server configured on the public DHCP server must be the IP address of the MAE that functions as the DHCP server. If the operator expects to use either of the active and standby MAE nodes as the DHCP server, the public DHCP server must be configured with the IP addresses of the active and standby MAE nodes. - The SeGW must be configured with ACL rules which allow DHCP packets to pass. If the operator expects to use either the active or standby MAE node as the DHCP server, the SeGW must be configured with ACL rules which allow packets of active and standby MAE nodes to pass. - The SeGW must be configured with

MAE Deployment	MAE Deployment Description	MAE Serving as the DHCP Server	MAE Serving as the OMCH	Requirement for the Base Station Deployment	Impact on the Network Configuration
					ACL rules which allow OM packets to pass for the MAE serving as the OMC. If the operator expects to use either the active or standby MAE node as the OMC, the SeGW must be configured with ACL rules which allow packets of active and standby MAE nodes to pass. The DHCP relay must be configured with the active and standby node IP addresses which serve as the DHCP server IP address.
Emergency system	The emergency system performs basic functions only and does not support PnP or DHCP.	Not supported	Not supported	Not supported	Not supported

NOTE

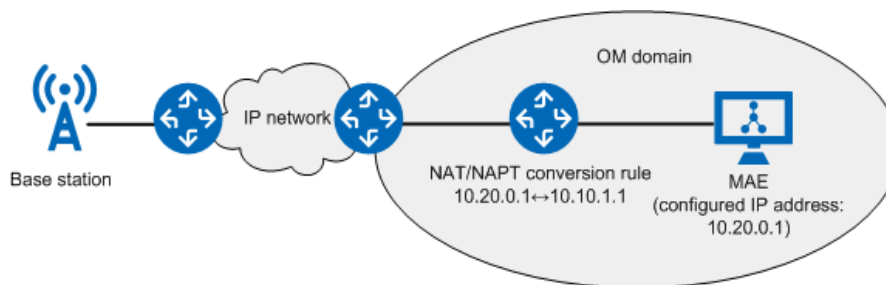
The active and standby MAE nodes in the preceding deployment mode must use the same IP version for a base station.

Below is an example. When the MAE uses the active/standby networking deployment mode, the DHCP service is deployed on the master server, whereas the FTP service and the OMCH management service can be deployed on either the master or slave server. When the FTP service and OMCH management service are deployed on different MAE servers and use different IP addresses, the route configuration on the base station and the transport network must be valid. This is

to ensure that the IP addresses of the two services are reachable using configured routes. If IPsec secures OMCH data, the IPsec SA's traffic selector (TS) successfully negotiated between the base station and the SeGW must cover the traffic between the OM IP address of the base station and the network-layer IP addresses of the two services.

IPv4 OMCH networking requires that the NAT server be deployed only on the MAE side, but not on the base station or BSC side. **Figure 4-40** shows OMCH networking when the NAT server is deployed on the MAE side.

Figure 4-40 OMCH networking when the NAT server is deployed on the MAE side



The IP address and port number of the MAE can only be unidirectionally converted by the NAT. When a route to the MAE IP address is configured on the base station side, the destination IP address must be the MAE IP address visible to the base station. As shown in **Figure 4-40**, the local IP address configured for the MAE is 10.20.0.1. That is, the source IP address of packets sent by the MAE is 10.20.0.1. After being translated by the NAT server, the source IP address of the TCP packets received by the base station is 10.10.1.1. Therefore, the route of which the destination IP address is 10.10.1.1 instead of 10.20.0.1 must be configured on the base station side.

NOTE

The IP address and port number on the base station side cannot be converted by the NAT server because the DHCP server uses the IP address of the DHCP relay agent (giaddr) or IP address of the DHCP client (ciaddr) as the destination IP address for responding to the DHCP message. The giaddr or ciaddr fields contained in the DHCP message cannot be converted by the NAT server.

5 ATM-based Automatic OMCH Establishment for Base Stations (UMTS)

5.1 Overview

ATM-based Automatic OMCH Establishment for Base Stations (corresponding to WRFD-031100 BOOTP) is used for the bootstrap of diskless workstations. It enables the diskless workstation to obtain the IP address from the server during startup. Compared with the Reverse Address Resolution Protocol (RARP) that implements the same function, BOOTP is more versatile and easier to use. BOOTP complies with the RFC 951 and RFC 1542 protocols.

BOOTP applies to ATM networks. It is used to establish an IPoA path between the MAE or LMT and a NodeB. The configuration information required for setting up an IPoA path includes the permanent virtual channel (PVC), transmission port carrying the PVC, and IP address.

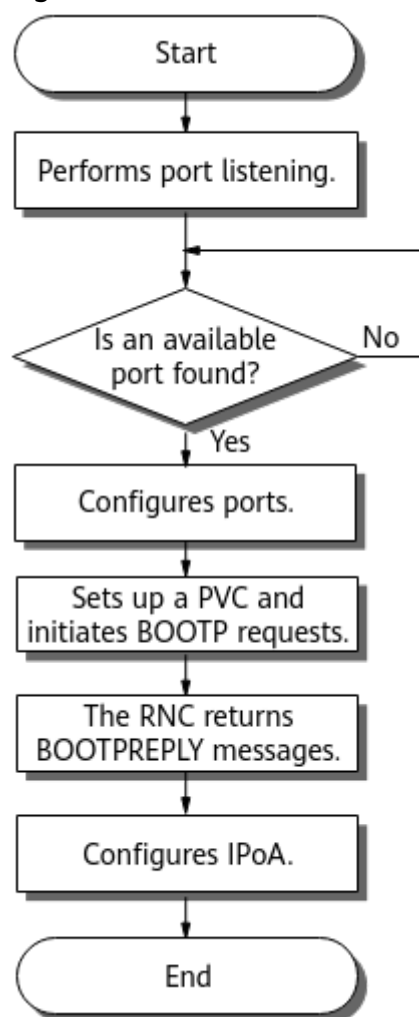
The NodeB configuration data contains the data of the IPoA path. If the configuration data is correct, you can remotely manage and maintain the NodeB. If the data is incorrect, BOOTP helps the NodeB to establish a correct IPoA path so that the NodeB can be remotely maintained.

After the BOOTP is applied in the RAN system, the NodeB can establish an IPoA path with the MAE or LMT based on the obtained IP address and the default PVC. In this manner, the OMCH is established.

If the MAE of SRAN18.1 or later is newly deployed or the certificate system of the MAE is switched to the operator's certificate system, this feature is not supported in ATM transmission scenarios.

5.2 Principles

The procedure of BOOTP establishment consists of port listening, port configuration, PVC setup and BOOTP request initiation, RNC returning the BOOTPREPLY message, and IPoA configuration, as shown in [Figure 5-1](#).

Figure 5-1 Procedure of BOOTP establishment

5.2.1 Port Listening

Overview

Port listening enables the NodeB to listen to the configuration data of peer ports so that the NodeB transport ports that carry PVCs can be correctly configured.

Port listening requires that the physical links must be connected properly. The transmission ports on the transmission device between the RNC and the NodeB must also be correctly configured.

The port types applied to ATM networking are as follows:

- Inverse Multiplexing over ATM (IMA)
- User Network Interface (UNI)
- Fractional ATM
- Unchannelized STM-1/OC-3

The procedure of BOOTP establishment is different in the case of different port types. For the unchannelized STM-1/OC-3 ports, the PVC can be set up without port listening as interconnection is not involved.

Port Listening in the Case of IMA/UNI

Through IMA/UNI ports, the NodeB can obtain the configuration data from peer ports by listening to the IMA Control Protocol (ICP) cells of the peer end. According to the obtained configuration data, the NodeB sets up an IMA group that carries the PVC (including the IMA links in the IMA group) or UNI links.

The NodeB cannot determine whether the IMA/UNI ports or fractional ATM ports are used and first listens to the IMA/UNI ports. If the listening task fails, the NodeB listens to the fractional ATM ports.

Port Listening in the Case of Fractional ATM

The fractional ATM link requires a bitmap of all types of timeslots contained in the link. If the timeslots are inconsistent at two ends, the setup of a fractional ATM link will fail.

Each E1 link consists of 32 timeslots and each T1 link contains 24 timeslots. Each timeslot occupies 64 kbit/s. The exhaustive method is applied to these typical timeslot bitmaps, which is a way to configure the fractional ATM links. If the links function properly, the listening is successful. However, if the links function abnormally, it indicates that the timeslot bitmap does not match the configuration at the peer end, and the NodeB must try other timeslot bitmaps.

Listening to the timeslots by using the exhaustive method will be time-consuming because the combinations of timeslots are countless. To avoid this issue, the range of timeslot combinations must be minimized. The combinations must contain only the typical timeslot bitmaps commonly used by telecom operators.

The NodeB cannot determine whether the physical links connected to the NodeB are E1s or T1s and first uses the E1 timeslot bitmaps to listen to the ports. If the listening task fails, the NodeB uses the T1 timeslot bitmaps to listen to the ports.

After the listening is successful, the PVC can be set up.

5.2.2 Port Configuration

The NodeB configures its IMA or UNI ports based on the configuration data of the ports at the peer end. The configuration parameters of the peer ports, obtained through port listening, include the protocol version and IMA frame length.

NOTE

For an IMA port, when configuring an IMA group on the RNC or ATM network equipment, you must set the minimum number of activated links (specified by the **IMAGRP.MINLNKNUM** parameter) to 1.

5.2.3 PVC Setup and BOOTP Request Initiation

The PVC used by BOOTP is permanently 1/33. That is, its virtual path identifier (VPI) is set to 1 and virtual channel identifier (VCI) is set to 33. Such a PVC must be configured on the RNC or ATM network equipment. The BOOTP process is implemented on this PVC.

After the PVC is set up, the NodeB sends a BOOTREQUEST message on this PVC to the RNC and requires the assignment of an IP address. The IP address will be used

as the OM address of the NodeB. This IP address can be used for logging in to the NodeB and for maintenance purposes.

5.2.4 RNC Returning the BOOTREPLY Message

The RNC responding to a BOOTREQUEST message must be configured with a PVC (fixed to 1/33) for the related NodeB and have obtained the corresponding IP addresses.

On reception of the BOOTREQUEST message, the RNC replies with a BOOTREPLY message containing the assigned IP address.

5.2.5 IPoA Configuration

After receiving the BOOTREPLY message from the RNC, the NodeB configures an IPoA path, which finalizes the BOOTP implementation process.

5.3 Configuration Guidelines

In the IP network:

- For details about data to prepare before a base station starts the automatic OMCH establishment procedure, see *3900 & 5900 Series Base Station Initial Configuration Guide*.
- For details about software and configuration file downloading, activation, and commissioning on a base station after the automatic OMCH establishment procedure is complete, see *3900 & 5900 Series Base Station Commissioning Guide*.

The following describes how to configure BOOTP on an ATM network.

On the RNC Side

On the RNC side, run the **ADD IPOAPVC** command to configure the PVC. When using BOOTP, the PVC is to be configured with VPI = 1 and VCI = 33. The main parameters of this command are as follows:

- **CARRYVPI**: This parameter specifies the VPI value of the PVC. It is set to **1**.
- **CARRYVCI**: This parameter specifies the VCI value of the PVC. It is set to **33**.
- **IPADDR**: This parameter specifies the local IP address.
- **PEERIPADDR**: This parameter specifies the IP address of the peer end, that is, IP address of the NodeB.

On the RNC side, run the **ADD UNODEBIP** command to configure the IP address of the OM channel. The main parameters of this command are as follows:

- **NBATMOAMIP**: This parameter specifies the OM IP address of the NodeB in ATM networking.
- **NBCTRLSN**: This parameter specifies the main control board slot number of the NodeB.

When there are multiple main control boards in a base station, the RNC compares the slot number of a main control board reported in the BOOTP process with the slot number specified by users. If the reported and specified

slot numbers are the same, the RNC returns a BOOTPREPLY message to the base station.

On the NodeB Side

The BOOTP process can be implemented without any NodeB configuration data, and therefore it is unnecessary to configure BOOTP on the NodeB side.

6 TDM-based Base Station Automatic OMCH Establishment (GSM)

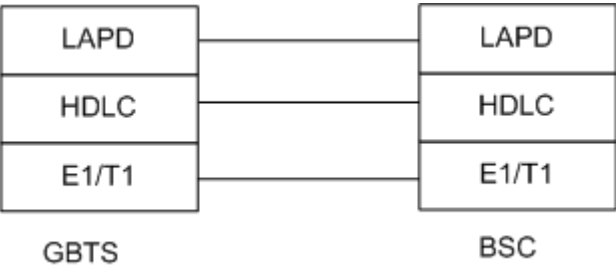
6.1 Overview

In TDM networking, the protocol stack on the Abis interface is as follows:

- Physical layer: Data is carried over E1/T1 links.
- Data link layer: High-Level Data Link Control (HDLC) is used.
- Application layer: Link access procedure on the D channel (LAPD) is used. LAPD includes layer 2 management link (L2ML), OML, radio signaling link (RSL), and extended signaling link (ESL).

Figure 6-1 shows the protocol stack on the Abis interface in TDM networking.

Figure 6-1 Protocol stack on the Abis interface in TDM networking

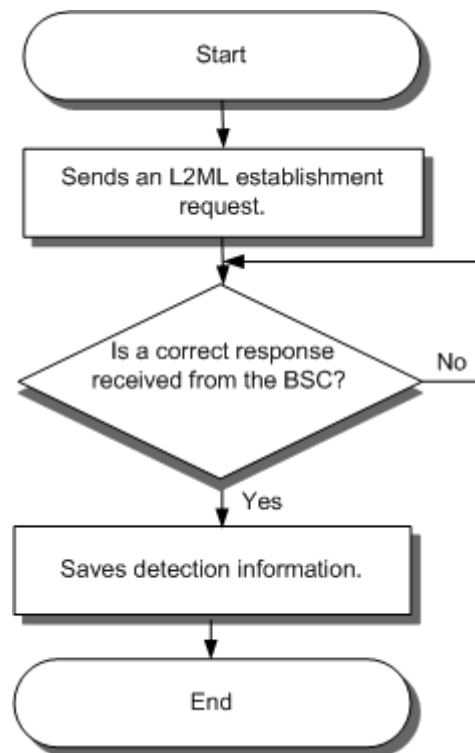


OML timeslot detection in TDM networking applies to the GBTS in Abis over TDM mode. This function is used to establish an OMCH (that is, an OML) between the GBTS and BSC.

6.2 Process

As shown in Figure 6-2, the process of OML timeslot detection in TDM networking consists of two procedures: sending L2ML establishment requests and saving detection information.

Figure 6-2 Process of OML timeslot detection in TDM networking



6.2.1 Sending L2ML Establishment Requests

The procedure for sending L2ML establishment requests is as follows:

1. The GBTS determines whether an E1 or T1 link is used for OML timeslot detection based on the DIP switch of the main control board.
2. To establish an OML to the BSC, the GBTS attempts to send L2ML establishment requests based on certain combinations of bandwidths and E1/T1 ports that support OML timeslot detection.

OML timeslot detection in TDM networking requires 64 kbit/s or 16 kbit/s bandwidth and can be implemented on E1/T1 ports 0 and 1 of the main control board. The GBTS uses four possible combinations in the following order:

1. E1/T1 port 0, 64 kbit/s bandwidth
2. E1/T1 port 0, 16 kbit/s bandwidth
3. E1/T1 port 1, 64 kbit/s bandwidth
4. E1/T1 port 1, 16 kbit/s bandwidth

If the 64 kbit/s bandwidth is used:

- For an E1 link, the GBTS sends L2ML establishment requests over 64 kbit/s timeslots 1 through 31.
- For a T1 link, the GBTS sends L2ML establishment requests over 64 kbit/s timeslots 1 through 24.

If the 16 kbit/s bandwidth is used:

- For an E1 link, the GBTS sends L2ML establishment requests over the third 16 kbit/s sub-timeslots of 64 kbit/s timeslots 1 through 31.

- For a T1 link, the GBTS sends L2ML establishment requests over the third 16 kbit/s sub-timeslots of 64 kbit/s timeslots 1 through 24.

Upon receiving an L2ML establishment request, the BSC selects a 64 kbit/s timeslot or a 16 kbit/s sub-timeslot based on base station configurations, and responds to the request. By default, the BSC selects the last 64 kbit/s timeslot of an E1/T1 link, or the third 16 kbit/s sub-timeslot of the last 64 kbit/s timeslot. The last 64 kbit/s timeslot is timeslot 31 for an E1 link and timeslot 24 for a T1 link.

If the last 64 kbit/s timeslot or the third 16 kbit/s sub-timeslot of the last 64 kbit/s timeslot cannot carry an OML, run the **SET BTSOMLTS** command on the BSC LMT to set the timeslot that is used to carry the OML, and run the **SET BTSOMLDETECT** command to set the OML timeslot detection function.

Upon receiving a correct response over a timeslot, the GBTS uses the timeslot to carry the OML. Otherwise, the GBTS attempts to establish an OML on other ports or timeslots.

6.2.2 Saving Detection Information

The GBTS saves the combination of the bandwidth and E1/T1 port number that was used for the previous successful L2ML establishment. Upon the next startup, the GBTS preferentially uses the saved combination for OML establishment, which reduces startup time.

7 Related Features

Prerequisite Features

None

Mutually Exclusive Features

None

Impacted Features

None

8 Network Impact

8.1 Benefits

With the Automatic OMCH Establishment feature, a base station can establish OMCHs by network communication (not requiring local end operations). This enables remote base station deployment by PnP, thereby reducing site visits and deployment cost and time.

8.2 Impacts

None

9 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

10 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

11 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

12 Reference Documents

- *SRAN Networking and Evolution Overview*
- *IPsec Feature Parameter Description*
- *PKI Feature Parameter Description*
- *SSL Feature Parameter Description*
- *IPv4 Transmission Feature Parameter Description*
- Documents in *3900 & 5900 Series Base Station Product Documentation*:
 - *3900 & 5900 Series Base Station Commissioning Guide*
 - *3900 & 5900 Series Base Station Communication Matrix*
 - *3900 & 5900 Series Base Station Initial Configuration Guide*
- *BSC6900/6910 GSM Initial Configuration Guide* in *BSC6900/6910 GSM Product Documentation*
- *BSC6900/6910 UMTS Initial Configuration Guide* in *BSC6900/6910 UMTS Product Documentation*